

Multipliers of Spherical Harmonic Expansions

by
Jacob Denson

A dissertation submitted in partial fulfillment
of the requirements for the degree of
Doctor of Philosophy
(Mathematics)

at the
University of Wisconsin-Madison
2025

Date of Final Oral Exam: July 24th, 2025

The dissertation is approved by the following members of the Final Oral Committee:

Sean Paul, Professor, Mathematics
Andreas Seeger, Professor, Mathematics
Brian Street, Professor, Mathematics
Betsy Stovall, Professor, Mathematics

Multipliers of Spherical Harmonic Expansions

Jacob Denson

Abstract

Let T be a Schwartz operator on S^d commuting with rotations. Then there exists a function $a : \mathbb{Z} \rightarrow \mathbb{C}$, the symbol of T , such that for any spherical harmonic $f : S^d \rightarrow \mathbb{C}$ of degree k , $Tf = a(k)f$. This thesis discusses the work of the author on obtaining new results about the boundedness of such operators on $L^p(S^d)$ for $p \neq 2$ and $d \geq 4$.

Given any regulated function $a : [0, \infty) \rightarrow \mathbb{C}$, define operators T and $\{T_\rho : \rho > 0\}$ on S^d , commuting with rotations, with symbols $a(k)$ and $a_\rho(k) = a(\rho(k + \frac{d-1}{2}))$ respectively. For $d \geq 4$, and $1 < p < 2(d-1)/(d+1)$, we obtain new sufficient conditions on a in order for T to be bounded on $L^p(S^d)$, and necessary and sufficient conditions for the operators $\{T_\rho\}$ to be uniformly bounded on $L^p(S^d)$, the first such conditions for any $p \neq 2$. One consequence of these results are new transplantation principles, which control the L^p boundedness of the multiplier operators $\{T_\rho\}$ in terms of the radial Fourier multiplier operator on $\mathbb{R}^d$ with symbol $a(|\cdot|) : \mathbb{R}^d \rightarrow \mathbb{C}$. This is the first nontrivial transplantation result of it's kind.

Our methods do not take advantage of the rotational symmetries of S^d , and can thus be applied in greater generality, to spectral multipliers of a self-adjoint elliptic pseudo-differential operator P , under assumptions related to the curvature of level sets of the principal symbol of P , and the periodicity of a Hamiltonian flow associated with P .

In order to prove these results, we obtain new quasi-orthogonality estimates for averages of solutions to the half-wave equation $\partial_t - iP = 0$, via a connection between pseudo-differential operators satisfying an appropriate curvature condition and Finsler geometry. We also combine estimates for different eigenfunction scales via arguments based on a theory of atomic decompositions in L^p for $p > 1$.

The characterizations of L^p boundedness for compactly supported and regulated functions are adapted from a paper of the author, with the methods of combining frequency scales taken from forthcoming work.

Acknowledgements

First and foremost, I am deeply indebted to my advisor Andreas Seeger for his patience, guidance, and generosity with both his time and mathematical insight over the past five years. I am very grateful for all the mathematical and professional insight he has gifted me. His encouragement helped me persevere through the slow and often frustrating process of mathematical research.

I would also like to express my gratitude to the many teachers and mentors who have guided me along the way to my PhD. To Shaoming Guo, who provided much advice, especially during my initial years in Madison, and for broadening my perspective with the organization of the Harmonic Analysis summer schools in Madison during my time here. To Betsy Stovall, for organizing many of the student seminars which enriched my semesters here at Madison. To Christoph Thiele, for his summer schools in Bonn, which provided a space for me to meet the European community of harmonic analysts. To David Beltran, who first introduced me to the field of analysis on manifolds. And To Sergey Denisov, Mihaela Ifrim, and Brian Street, for their courses and insights over the years.

I sincerely thank my committee members, Sean Paul, Andreas Seeger, Brian Street, and Betsy Stovall, for their time and feedback, especially for accommodating my defense, despite the complications of summer schedules.

I owe an immeasurable debt of gratitude to my parents, whose support, and belief in me laid the foundation for my academic work. I would also like to thank for grandfather. Without his influence and conversations through my undergraduate and postgraduate

years, it is unlikely I would have found my passion for higher mathematics.

Thank you to my partner Annabel, for being there for me every step of the way, and for bringing me so much joy and balance to my life during my PhD years.

Thank you to all my fellow students in the harmonic analysis group, for making our group so supportive and collegial during my time here.

Thank you to all the friends I have had the fortune of meeting during my PhD, for your friendship and support through the years. Because of you, the city of Madison will always hold a special place in my heart.

Finally, I thank the National Science Foundation, for supporting my research through their grants DMS-2037851, DMS-2054220, and DMS-2348797.

Contents

Notation	vi
Chapter 1 Introduction	1
Chapter 2 Statement of Main Results	3
2.1 Multipliers of Spherical Harmonic Expansions	3
2.2 Spectral Multipliers of an Elliptic Operator	8
Chapter 3 Radial Multipliers on Euclidean Space	16
3.1 Convolution Kernels of Fourier Multipliers	18
3.2 The Radial Multiplier Conjecture	19
3.3 Radial-Multiplier Bounds by Density Decompositions	22
3.4 Combining Scales with Atomic Decompositions	35
3.5 Quasi-Radial Multipliers and Local Smoothing	40
Chapter 4 Wave Equations on Compact Manifolds	43
4.1 An Analogue of the Radial Multiplier Conjecture on Compact Manifolds .	43
4.2 Geometries Induced by Elliptic Operators	45
4.3 Fourier Integral Operator Techniques	55
4.4 Periodic Geodesics and Assumption B	69

Chapter 5	The Boundedness of Spectral Multipliers with Compact Support	72
5.1	Preliminary Setup	73
5.2	Quasi-Orthogonality Estimates For Wave-Packets	79
5.3	Analysis of Regime I via Density Methods	93
5.4	Analysis of Regime II via Local Smoothing	109
Chapter 6	Combining Scales Via Atomic Decompositions	111
6.1	Short Range Bounds	117
6.2	Long Range Bounds	118
6.3	Wave Bounds on the Sphere	121
Appendix 7	Elementary Theorems	125
Appendix 8	Distributions Defined by Oscillatory Distributions	130
Appendix 9	Pseudo-Differential Operators	134
Bibliography		137

Notation

- We use the normalization of the Fourier transform given by the formula

$$\widehat{f}(\xi) = \int f(x) e^{-2\pi i \xi \cdot x} dx.$$

Overloading notation, for a function $f : [0, \infty) \rightarrow \mathbb{C}$ and $\lambda \geq 0$ we write the cosine transform of f as $\widehat{f}(\lambda) = \int_0^\infty f(t) \cos(2\pi\lambda t) dt$.

- For $p \in [1, \infty]$, we let $p' \in [1, \infty]$ be its dual exponent, satisfying $1/p + 1/p' = 1$.
- We use the translation and L^∞ -normalized dilation operators

$$\text{Trans}_y f(x) = f(x - y) \quad \text{and} \quad \text{Dil}_t f(x) = f(x/t),$$

defined so that if a function f is supported on a closed set A , then $\text{Trans}_y f$ is supported on $A + y$, and $\text{Dil}_t f$ is supported on tA .

- On $\mathbb{R}^d$, we let ∂_i denote the usual partial derivative operators in the i th coordinate direction (if we are using a variable $x = (x_1, \dots, x_n)$ to denote these coordinates, we might also write this derivative as ∂_{x_i}), and use D_j to denote the self-adjoint normalization $D_j f = (2\pi i)^{-1} \partial_j$. This normalization has the convenience that for a

d -variate polynomial $P : \mathbb{R}^d \rightarrow \mathbb{C}$,

$$P(D_j)\{f\} = \int P(\xi) \widehat{f}(\xi) e^{2\pi i \xi \cdot x} d\xi,$$

and simplifies many formulas associated with Fourier integral operators. We also consider the compositions of these operators ∂^α and D^α given by multi-indices α .

- For a differentiable function $F : \mathbb{R}^m \rightarrow \mathbb{R}^n$ and $x \in \mathbb{R}^m$, we let $F_*(x) = \{\partial_j F_k(x)\}$ denote the $n \times m$ matrix of partial derivatives of the components of F . If $F : X \rightarrow Y$ is a smooth map between two manifolds, we also let $F_* : TX \rightarrow TY$ denote the map, which is a linear map from $T_x X$ to $T_{f(x)} Y$ for each $x \in X$, given by the matrix of partial derivatives of F in any particular pair of coordinate systems.
- We will often use the Japanese bracket $\langle x \rangle = (1 + |x|^2)^{1/2}$ for $x \in \mathbb{R}^d$.
- Given a topological space X and two sets U and V , we write $U \Subset V$ if the closure of U is a compact subset of V .
- Let $\Omega \subset \mathbb{R}^d$ be a bounded open set. For each affine transformation $T : \mathbb{R}^d \rightarrow \mathbb{R}^d$, consider a smooth function ϕ_T supported on $T(\Omega)$. We say these functions are *adapted* if $|\partial_x^\alpha \{\phi_T \circ T\}| \lesssim_\alpha 1$, uniformly in T . We will apply this definition when Ω is a cube (to obtain functions adapted to certain rectangles), and when Ω is an annulus (to obtain functions adapted to various annuli).
- We let $\dot{\mathbb{R}}^p = \mathbb{R}^p - \{0\}$. A smooth function $f : \mathbb{R}^n \times \dot{\mathbb{R}}^p \rightarrow \mathbb{C}$ is a *symbol of order s* if it satisfies bounds of the form $|\partial_x^\alpha \partial_\theta^\lambda f(x, \theta)| \lesssim_{\alpha, \beta} \langle \theta \rangle^{s-|\lambda|}$ for all multi-indices α and λ , and all $x \in \mathbb{R}^n$ and $\theta \in \dot{\mathbb{R}}^p$, where $\langle \cdot \rangle$ is the Japanese bracket defined above. More generally, symbols $s : \Gamma \rightarrow \mathbb{C}$ can be defined for *conical subsets* Γ of $\mathbb{R}^n \times \dot{\mathbb{R}}^p$, i.e. sets Γ with the property that if $(x, \theta) \in \Gamma$ and $\rho > 0$, then $(x, \rho\theta) \in \Gamma$. For a sequence

of symbols $\{a_k : k \geq 0\}$, we write

$$a \sim \sum_k a_k$$

if, for any $l > 0$, there exists N_0 such that for $N \geq N_0$, $a - \sum_{k=0}^N a_k$ is a symbol of order $s - l$. We view this relation as an asymptotic expansion of the function a

- We let $\mathcal{S}(\mathbb{R}^d)$ denote the Schwartz space of rapidly decreasing smooth functions on $\mathbb{R}^d$, i.e. the functions $f : \mathbb{R}^d \rightarrow \mathbb{C}$ such that for any multi-index α and any $N \geq 0$, the inequality $|\partial^\alpha f(x)| \lesssim_{\alpha,N} \langle x \rangle^{-N}$ holds for all $x \in \mathbb{R}^d$.
- For a measure space X , $L^\infty(X)$ is the Banach space of essentially-bounded functions, defined almost everywhere. For a set X , $l^\infty(X)$ is the Banach space of bounded functions on X , defined everywhere. We also used mixed norm spaces; for a function $f(x, y)$ defined on the Cartesian product of two measure spaces X and Y , we define

$$\|f\|_{L^p(X)L^q(Y)} = \left\| \|f\|_{L^q(Y)} \right\|_{L^p(X)} = \left(\int_X \left(\int_Y |f(x, y)|^q dy \right)^{p/q} dx \right)^{1/p}.$$

Similarly, we can define the interchanged norm $L^q(Y)L^p(X)$ by swapping the order of application of the norms. It follows from Minkowski's inequality that if $p \geq q$ then $\|f\|_{L^p(X)L^q(Y)} \leq \|f\|_{L^q(Y)L^p(X)}$ and if $p \leq q$, then $\|f\|_{L^p(X)L^q(Y)} \geq \|f\|_{L^q(Y)L^p(X)}$.

- If T is a Schwartz operator from a manifold Y to a manifold X , each equipped with some smooth density, we let K_T denote the Schwartz distribution on $X \times Y$ which is the integral kernel for T , i.e. such that for any $f \in C_c^\infty(X)$ and $g \in C_c^\infty(Y)$,

$$\int_X f(x)(Tg)(x) dx = \int_{X \times Y} K_T(x, y)f(x)g(y) dx dy.$$

We define the *support* of a Schwartz operator T from a manifold Y to a manifold

X to be the smallest closed set $\text{supp}(T) \subset X \times Y$ such that $\langle f, Tg \rangle = 0$ whenever $f \in C_c^\infty(X)$ and $g \in C_c^\infty(Y)$ are such that $\text{supp}(f) \times \text{supp}(g)$ is disjoint from $\text{supp}(T)$.

- For $1 \leq p \leq \infty$ and $s \in \mathbb{R}$, we consider the Sobolev spaces $W^{s,p}(\mathbb{R}^d)$ with norm

$$\|f\|_{W^{s,p}(\mathbb{R}^d)} = \|(1 - \Delta)^{s/2} f\|_{L^p(\mathbb{R}^d)},$$

and for $1 \leq r \leq \infty$, we consider the (homogeneous) Besov spaces

$$\|f\|_{\dot{B}_r^{s,p}(\mathbb{R}^d)} = \left(\sum_{k \in \mathbb{Z}} \left[2^{ks} \|P_k f\|_{L^p(\mathbb{R}^d)} \right]^r \right)^{1/r}.$$

Here $(1 - \Delta)^{s/2}$ is the Fourier multiplier operator with symbol $(1 + |\xi|^2)^{s/2}$, and the operators $\{P_k\}$ are Littlewood-Paley projection operators, i.e. Fourier multiplier operators with compactly supported symbols $\chi(|\cdot|/2^k)$, where χ is compactly supported and $\sum \chi(t/2^k) = 1$ for all $t > 0$. By working in coordinate systems, these definitions can also be used to define Sobolev spaces $W^{s,p}(X)$ of functions on a compact manifold X . We write $H^s(X)$ for $W^{s,2}(X)$.

- If X is a compact manifold equipped with a smooth volume density, we let $\text{COP}_s^+(X)$ denote the family of all classical, elliptic pseudo-differential operators of order s on X which are positive-semidefinite in the sense that $\langle Pf, g \rangle = \langle f, Pg \rangle$ and $\langle Pf, f \rangle \geq 0$ hold for all $f, g \in C^\infty(X)$.

Chapter 1

Introduction

Let T be a Schwartz operator on S^d commuting with rotations. Then T is diagonalized by the spherical harmonics; we can find a complex-valued function a known as the *symbol* of T , such that for any degree k spherical harmonic f on S^d , $Tf = a(k)f$. For a function $a : [0, \infty) \rightarrow \mathbb{C}$, we can consider a family of operators $\{T_\rho : \rho > 0\}$ on S^d , where T_ρ commutes with rotations and has a symbol obtained by dilating a by a factor ρ . In this thesis, we study the following question:

What conditions on a function a ensure that the operators $\{T_\rho\}$ are uniformly bounded.

Aside from testing our ability to understand interactions between spherical harmonics, this question is closely related to smoothing phenomena for waves on S^d , and naturally arises when studying the convergence of spherical harmonic expansions of a function. For $d \geq 4$ and $1 < p < 2(d-1)/(d+1)$, this thesis obtains new sufficient conditions for the boundedness of the operator T on $L^p(S^d)$. Moreover, the thesis obtains new necessary and sufficient conditions for uniform boundedness of the operators $\{T_\rho\}$ on $L^p(S^d)$, the first of their kind aside from the relatively simple study of uniform boundedness on $L^2(S^d)$.

The sphere S^d lacks a family of dilation symmetries, which makes uniform control

on the operators $\{T_\rho\}$ difficult when compared to the study of analogous operators on $\mathbb{R}^d$. From this perspective, the study of such operators provides a useful setting with which to test and develop methods of harmonic analysis in a ‘variable-coefficient setting’, where a lack of certain symmetries forces us to introduce more robust methods than are required in the Euclidean setting. In fact, the methods of this thesis not only get around the lack of dilation symmetry on S^d , but also do not exploit the rotational symmetry of S^d in any explicit way. We thus obtain results for analogous spectral multipliers operators on more general compact manifolds lacking any symmetries appropriate to the study of such operators.

In Chapter 2, we describe the problem of the thesis in more detail, and relate it to the study of spectral multipliers of an elliptic operator on a compact manifold. In Chapter 3, we discuss results known for radial and quasi-radial multipliers which are analogue to the endpoint bounds we will establish for spectral multipliers on manifolds. In Chapter 4, we return to the study of spectral multipliers on manifolds, introducing the background on Finsler geometry and Fourier integral operators we will need in the proofs to come. In Chapters 5 and 6, we provide the proofs of the new results on the uniform boundedness of the operators $\{T_\rho\}$.

Chapter 2

Statement of Main Results

2.1 Multipliers of Spherical Harmonic Expansions

Consider a bounded operator T on $L^2(S^d)$ commuting with rotations, i.e. such that

$$T \circ R^* = R^* \circ T \quad \text{for all } R \in \mathrm{O}(d+1), \quad (2.1.1)$$

where $(R^*f)(x) = f(Rx)$. There is an orthogonal decomposition $L^2(S^d) = \bigoplus \mathcal{H}_k$, where $\mathcal{H}_k$ is the space of *spherical harmonics of degree k* , i.e. the restrictions of harmonic, homogeneous polynomials of degree k on $\mathbb{R}^{d+1}$ to S^d . All operators commuting with rotations are diagonalized by this orthogonal decomposition, i.e. there exists a bounded function $a : \mathbb{N} \rightarrow \mathbb{C}$ such that $Tf = a(k)f$ for all $f \in \mathcal{H}_k(S^d)^*$. Conversely, any bounded function a defines an operator T_a on $L^2(S^d)$ commuting with rotations. We will call such an operator a *multiplier operator for spherical harmonic expansions*.

To study the regularity of such operators, we introduce the space $M^p(S^d)$, consisting

*This follows from Schur's Lemma, and that the vector spaces $\mathcal{H}_k$ are non-isomorphic irreducible representations of $\mathrm{O}(d+1)$. See Theorem 2.33 of [40] for a proof of irreducibility, and Theorem 3.5 of [10] for a discussion of Schur's Lemma for unitary representations.

of all functions $a : \mathbb{N} \rightarrow \mathbb{C}$ for which the operator T_a is bounded on $L^p(S^d)$. The space $M^p(S^d)$ is then equipped with the norm

$$\|a\|_{M^p(S^d)} = \sup \left\{ \frac{\|T_a f\|_{L^p(S^d)}}{\|f\|_{L^p(S^d)}} : f \in C^\infty(S^d) \right\}. \quad (2.1.2)$$

$M^2(S^d)$ is isometric to $l^\infty(\mathbb{N})$ (as the calculations below in Lemma 2.1 show), but the structure of $M^p(S^d)$ for $p \neq 2$ is unclear. Duality implies that $M^p(S^d)$ is isometric to $M^{p'}(S^d)$, so attention may be restricted to the case $1 \leq p \leq 2$.

We wish to study operators commuting with rotations analogously to how we might study Fourier integral operators on $\mathbb{R}^d$, i.e. using oscillatory integrals. However, the discrete decomposition of $L^2(S^d)$ into spherical harmonics makes it difficult to apply methods of oscillatory integrals to the problem. Fortunately, certain ‘semi-classical’ heuristics tell us that these problems disappear when we restrict the problem to ‘high frequency inputs’. In order to take advantage of this fact, rather than studying what conditions ensure the operators T_a are bounded, we study conditions that ensure the operators T_{a_ρ} are uniformly bounded on $L^p(X)$, where $a_\rho(\lambda) = a(\rho(\lambda + \frac{d-1}{2}))$. As $\rho \rightarrow 0$, the function becomes ‘concentrated at high frequency’. To prevent pathological examples from arising, we restrict ourselves to *regulated functions* a , i.e. functions such that

$$a(\lambda_0) = \lim_{\delta \rightarrow 0} \int_{|\lambda - \lambda_0| \leq \delta} a(\lambda) d\lambda \quad \text{for all } \lambda_0 \in [0, \infty). \quad (2.1.3)$$

Define the set $M_{\text{Dil}}^p(S^d)$ of all regulated functions a such that $\|a\|_{M_{\text{Dil}}^p(S^d)} = \sup_\rho \|a_\rho\|_{M^p(S^d)}$ is finite. With notation introduced, we now state precisely the problem studied in this thesis:

*Can one find simple conditions on a function a ,
necessary and sufficient to be contained in $M_{\text{Dil}}^p(S^d)$ for $p \neq 2$.*

For $p = 2$, orthogonality gives the isometric equivalence $M_{\text{Dil}}^2(X) = L^\infty[0, \infty)$.

Lemma 2.1. *For any regulated a , $\|a\|_{M_{\text{Dil}}^2(S^d)} = \|a\|_{L^\infty[0,\infty)}$.*

Proof. Given $f \in L^2(S^d)$, if we write $f = \sum f_k$ with $f_k \in \mathcal{H}_k$, then for any $R > 0$,

$$\|T_{a,\rho}f\|_{L^2(S^d)} = \left\| \sum_k a_\rho(k) f_k \right\|_{L^2(S^d)} = \left(\sum_k |a_\rho(k)|^2 \|f_k\|_{L^2(S^d)}^2 \right)^{1/2}. \quad (2.1.4)$$

Conversely,

$$\|f\|_{L^2(S^d)} = \left(\sum_k \|f_k\|_{L^2(S^d)}^2 \right)^{1/2}. \quad (2.1.5)$$

Setting $c_k = \|f_k\|_{L^2(S^d)}$, (2.1.4) and (2.1.5) show $\|a_\rho\|_{M^2(S^d)}$ is the smallest constant so that

$$\left(\sum_k |a_\rho(k)|^2 c_k^2 \right)^{1/2} \leq \|a_\rho\|_{M^2(S^d)} \left(\sum_k c_k^2 \right)^{1/2}. \quad (2.1.6)$$

for all sequences $\{c_k\}$. It is then clear that $\|a_\rho\|_{M^2(S^d)} = \|a_\rho\|_{l^\infty(\mathbb{N})}$, and thus (now using the fact that a is regulated),

$$\|a\|_{M_{\text{Dil}}^2(S^d)} = \sup_\rho \|a_\rho\|_{l^\infty(\mathbb{N})} = \|a\|_{L^\infty[0,\infty)}, \quad (2.1.7)$$

which completes the proof. $\square$

The main result of this thesis, proved in Chapters 5 and 6, are new sufficient conditions to be contained in $M^p(S^d)$, and new necessary and sufficient conditions for a function to be contained in $M_{\text{Dil}}^p(S^d)$, for $d \geq 4$ and $1 < p < (d-1)/2(d+1)$. To introduce the results, fix a non-zero $\chi \in C_c^\infty(0, \infty)$. For $s \geq 0$ and $1 \leq p \leq \infty$, define the norm

$$\|a\|_{R^{s,p}[0,\infty)} = \sup_{R>0} \left(\int_0^\infty |\widehat{a}_R(t) \langle t \rangle^s|^p dt \right)^{1/p} \quad \text{where} \quad a_R(\lambda) = \chi(\lambda) a(R\lambda). \quad (2.1.8)$$

Here $\widehat{a}_R(t) = \int_0^\infty a_R(\lambda) \cos(2\pi\lambda t) d\lambda$ is the cosine transform of a_R , i.e. the Fourier transform of the even extension of a to a function on $\mathbb{R}$. The particular choice of χ is irrelevant, as

the resulting norms will all be equivalent.

Theorem 2.2. *Suppose $1 < p < 2(d-1)/(d+1)$. If $s = (d-1)(1/p - 1/2)$, and $a : [0, \infty) \rightarrow \mathbb{C}$ is a regulated function, then*

$$\|a\|_{M^p(S^d)} \lesssim \|a\|_{R^{s,p}[0,\infty)} \quad \text{and} \quad \|a\|_{M_{\text{Dir}}^p(S^d)} \sim \|a\|_{R^{s,p}[0,\infty)}.$$

One may view control on the $R^{s,p}$ norm as a smoothness condition on the multiplier a , and so the theorem fits the natural heuristic that one can control the regularity of a multiplier operator through control on the smoothness of its symbol; indeed the Hausdorff-Young inequality implies a Sobolev space estimate

$$\|a_R\|_{W^{s,p'}[0,\infty)} \lesssim \|a\|_{R^{s,p}[0,\infty)}. \quad (2.1.9)$$

The space $R^{s,p}[0, \infty)$ cannot be characterized by Sobolev, Besov, or Triebel-Lizorkin estimates, though we do have the Besov space estimate

$$\|a\|_{R^{s,p}[0,\infty)} \lesssim \sup_R \|a_R\|_{\dot{B}_p^{s+1/p-1/2,2}[0,\infty)}. \quad (2.1.10)$$

The result is somewhat intuitive. Inequality (2.1.9) tells us that elements of $R^{s,p}$ have s derivatives in $L^{p'}$, though with ‘some extra control’ occurring on the other side of the Fourier transform. The Sobolev embedding theorem (see Theorem 7.2) tell us that having $s+1/p-1/2$ derivatives in L^2 is sufficient to have s derivatives in $L^{p'}$. And the presence of L^2 type control on the right hand side of (2.1.10) allows us, not only to obtain control on s derivatives of a using Sobolev embedding, but also to capture the appropriate information on the Fourier transform of a to obtain (2.1.10), since L^2 type control is not lost under taking the Fourier transform because of the unitary nature of the Fourier transform.

Interest in bounding multiplier operators for spherical harmonics has a long history

throughout the 20th century. Classical methods involving the analysis of special functions and orthogonal polynomials culminated in Bonami and Clerc's extension of the Marcinkiewicz multiplier theorem to multipliers for spherical harmonic expansions [5]. The methods used to attack the problem changed in the 1960s, when Hörmander introduced the powerful method of Fourier integral operators, which he used to obtain L^p bounds for Bochner-Riesz multiplier expansions for spherical harmonics [19], with later developments on S^d and for spectral multipliers on other compact manifolds by Christopher D. Sogge [41, 44, 45], Michael Christ and Sogge [6], Andreas Seeger and Sogge [35], Seeger [36], Terence Tao [49], and Jongchon Kim [24]. None of these results were able to *completely* characterize the uniform L^p boundedness of dilated multipliers, making Theorem 2.2 the first result of that kind. The state of art results of [24] obtain boundedness under the weakest possible Besov assumptions on the function a , i.e. proving that

$$\|a\|_{M_{\text{Dil}}^p(S^d)} \lesssim \|a\|_{\dot{B}_p^{s+1/p-1/2,2}[0,\infty)}. \quad (2.1.11)$$

Theorem 2.4 needs $1/p - 1/2$ fewer derivatives than required for (2.1.11) to apply. The methods we use to prove Theorem 2.2 also follow the line of attack originating in the work of Hörmander, but are also heavily inspired by recent results characterizing L^p boundedness for radial Fourier multiplier operators on $\mathbb{R}^d$ due to Yaryong Heo, Fedor Nazarov, and Andreas Seeger [15], Laura Cladek [7], Gustavo Garrigos and Seeger [11], and Jongchon Kim [25]. This might be expected since transplantation methods of Boris Mitjagin [31], which we will discuss in more detail in Section 4.1, show any results uniformly which give necessary conditions for containment in $M_{\text{Dil}}^p(S^d)$ also give necessary conditions for the boundedness of associated radial multipliers. In particular, the main results of [16] are a corollary of Theorem 2.2.

To prove Theorem 2.2, we will now switch our point of view to a more general setting,

which is essential for us to apply the Fourier integral operator methods mentioned above.

2.2 Spectral Multipliers of an Elliptic Operator

We now restate our results in the more general setting of the study of spectral multipliers of an elliptic pseudo-differential operator. Appendix 9 gives a very brief summary of the relevant theory of pseudo-differential operators for our purposes. We suppose X is a compact manifold, equipped with a smooth volume density. We then let $\text{COP}_s^+(X)$ denote the family of all classical, elliptic pseudo-differential operators of order s on a compact manifold X which are formally positive-semidefinite, in the sense that $\langle Pf, g \rangle = \langle f, Pg \rangle$ and $\langle Pf, f \rangle \geq 0$ hold for all $f, g \in C^\infty(X)$. Consider $P \in \text{COP}_s^+(X)$ for $s > 0$. By general properties of elliptic operators*, $1 + P$ is an isomorphism between the Sobolev space $H^s(X)$ and $L^2(X)$, and so the inverse $(1 + P)^{-1}$, viewed as a map from $L^2(X)$ to itself, is compact by a form of the Rellich-Kondrachov embedding theorem†. By the spectral theorem for bounded, self-adjoint compact operators, there exists a discrete set $\Lambda \subset [0, \infty)$, and a decomposition $L^2(X) = \bigoplus_{\lambda \in \Lambda} \mathcal{V}_\lambda$, where $\mathcal{V}_\lambda$ is a finite dimensional subspace of $C^\infty(X)$, such that $Pf = \lambda f$ for all $f \in \mathcal{V}_\lambda$. We use this decomposition to define a functional calculus for the operator P ; given a bounded function $a : \Lambda \rightarrow \mathbb{C}$, we can define an operator $a(P)$ on $L^2(X)$ so that $a(P)f = a(\lambda)f$ for all $f \in \mathcal{V}_\lambda$. Our goal is to study the regularity of $a(P)$, in terms of properties of the function a .

In what follows, we will assume the operator P is an elliptic operator *of order one*. This does not restrict the scope of our analysis; given a formally positive operator $P \in \text{COP}_s^+(X)$, a standard argument‡ shows that the operator $P^{1/s}$ defined by the functional

*See Theorem 18.1.29 of [21] for a proof that $1 + P$ is an isomorphism from $H^s(X)$ to $L^2(X)$.

†See Proposition 4.4 of [50] for the statement of the Rellich-Kondrachov theorem.

‡See Theorem 3.3.1 of [42] for a proof that $P^{1/s}$ is in $\text{COP}_1^+(X)$, based on a technique of [39] initially developed for the study of elliptic differential operators, but which can be straightforwardly adapted to elliptic pseudo-differential operators.

calculus above is in $\text{COP}_1^+(X)$, and the spectral theory of P is identical with the spectral theory of $P^{1/s}$. Fixing the order of our operator reflects the fact that the theory of Fourier integral operators we will eventually employ is most elegant when the resulting oscillatory integrals have homogeneous phases of order one.

The main example of elements of $\text{COP}_s^+(X)$ for our purposes are obtained from a Laplace-Beltrami operator Δ on a compact Riemannian manifold X . The operator $-\Delta$ is classical (since it is a differential operator), and formally positive-semidefinite with respect to the volume form dV on X , because of the ‘integration by parts’ identity

$$\int_X (\Delta f_1) f_2 \, dV = - \int_X g(\nabla_g f_1(x), \nabla_g f_2) \, dV.$$

Since we restrict our attention to elliptic operators of order one, the archetypal object of study is then the Dirac operator $P = \sqrt{-\Delta}$.

We claim that the study of spectral multipliers for eigenfunctions is a generalization of the study of multipliers for spherical harmonic expansions introduced in the previous section. Indeed, define an operator P_{SH} on S^d by setting $P_{\text{SH}}f = (k + \frac{d-1}{2})f$ for $f \in \mathcal{H}_k$. Then $T_{a_\rho} = a(\rho P_{\text{SH}})$. In Lemma 2.3 we will see that P_{SH} is an element of $\text{COP}_s^+(S^d)$ with respect to the Riemannian volume density on S^d . Thus uniform bounds on $\{T_{a_\rho}\}$ are equivalent to uniform bounds on the operators $\{a(\rho P_{\text{SH}})\}$, i.e. $M_{\text{Dil}}^p(S^d) = M_{\text{Dil}}^p(S^d, P_{\text{SH}})$.

To study the regularity of the operators $a(P)$, we introduce the spaces $M^p(X, P)$, consisting of all functions $a : \Lambda \rightarrow \mathbb{C}$ for which the operator $a(P)$ is bounded on $L^p(X)$. The space $M^p(X, P)$ is then equipped with a norm

$$\|a\|_{M^p(X, P)} = \sup \left\{ \frac{\|a(P)f\|_{L^p(X)}}{\|f\|_{L^p(X)}} : f \in C^\infty(X) \right\}. \quad (2.2.1)$$

$M^2(X, P)$ is isometrically equal to $l^\infty(\Lambda)$, and duality implies $M^p(X, P)$ is isometric to $M^{p'}(X, P)$. We then define $\|a\|_{M_{\text{Dil}}^p(X, P)} = \sup \|a_\rho\|_{M^p(X, P)}$, where $a_\rho(\lambda) = a(\rho\lambda)$.

Before the results of this thesis, no simple conditions had been proved necessary and sufficient for inclusion in $M_{\text{Dil}}^p(X)$ for $p \neq 2$ and $X \neq \mathbb{T}^d$. The main result of this thesis is that under suitable assumptions on X and the operator P , we can find and prove the sufficiency of such a necessary condition. The first assumption is a curvature condition on the principal symbol of the P , and the second is an assumption about the operator's eigenvalues.

Assumption A: For each $x_0 \in M$, the co-sphere

$$S_{x_0} = \{\xi \in T_{x_0}^*M : p(x_0, \xi) = 1\}$$

is a hypersurface in T_x^*M with non-vanishing Gauss curvature, where $p : T^*X \rightarrow [0, \infty)$ is the principal symbol of the operator P .

Assumption B: The spectrum of the operator P is contained in an arithmetic progression.

Assumption C: The operators $\cos(2\pi tP)$ on X defined by the functional calculus above have ‘finite propagation speed’, in the sense that for one (and thus all) Riemannian metrics on X , there exists $C > 0$ such that for all $t \in \mathbb{R}$, the kernel of the spectral multiplier operators $\cos(2\pi tP)$ is supported on $\{(x, y) : d(x, y) \leq Ct\}$.

Lemma 2.3. *The operator P_{SH} is an element of $\text{COP}_1^+(S^d)$ with respect to the Riemannian volume form on S^d , and satisfies Assumptions A, B, and C.*

Proof. Let Δ denote the Laplace-Beltrami operator on S^d . Then $P = \sqrt{-\Delta}$ is an element of $\text{COP}_1^+(S^d)$ satisfying Assumption A. If f is a spherical harmonic of degree k , then

$$\Delta f = -\sqrt{k(k+d-1)}f. \quad (2.2.2)$$

Thus we can write $P_{\text{SH}} = \alpha(P)$, where $\alpha(\lambda) = \sqrt{\lambda^2 + c^2}$ with $c = \frac{d-1}{2}$. Since α is a symbol*

*See the notation section for a definition of classes of symbols of a certain order.

of order 1 with principal symbol $|\lambda|$, it follows that $P_{\text{SH}} \in \text{COP}_1^+(S^d)$, and P_{SH} and P share the same principal symbol*. Thus P_{SH} satisfies Assumption A. The operator also satisfies Assumption B by definition since P_{SH} is diagonalized by the spherical harmonics, and its eigenvalues form the arithmetic progression $\mathbb{N} + \frac{d-1}{2}$.

Now we prove Assumption C. Fix $a \in \mathbb{R}$, and let $Lu = (\Delta + c)u$. Then as t varies, for each input f the functions $u = \cos(tP_{\text{SH}})f$ solve the differential equation $\partial_t^2 u = Lu$ with $c = (d-1)^2/4$, and so it suffices to show solutions to this partial differential equation have finite propagation speed. This follows if we can find $t_0 > 0$ so that for $0 < t < t_0$, if $u(x, 0) = 0$ for $d(x, x_0) \leq t$, then $u(x, s) = 0$ for $d(x, x_0) \leq t - s$ and $s \leq t$, since the result for $|t| \leq t_0$ then follows by time inversion symmetry, and then the result for all t then follows by repeatedly composing the result with itself, using the time invariance of the equation. We prove this result by using standard energy methods. See Theorem 2.4.2 of [43] for another example of this approach.

Fix $x_0 \in X$ and $0 < t < 1/2$. Define a geodesic normal coordinate ball[†] $B_t(x_0)$ in S^d , and set $e = u^2 + (\partial_t u)^2 + |\nabla_g u|_g^2$. Then define $E(s) = \int_{B_{t-s}(x_0)} e \, dV$. For solutions to $\partial_t^2 u = Lu$, we have $\partial_t e = 2[(c+1)u\partial_t u + \text{div}_g\{\partial_t u \nabla_g u\}]$. Substituting this identity into the equation

$$E'(s) = \int_{B_{t-s}(x_0)} \partial_t e \, dV - \int_{\partial B_{t-s}} e \, dS, \quad (2.2.3)$$

we find that

$$E'(s) = 2 \int_{B_{t-s}(x_0)} [(c+1)u\partial_t u + \text{div}_g\{\partial_t u \nabla_g u\}] \, dV - \int_{\partial B_{t-s}(x_0)} e \, dS. \quad (2.2.4)$$

*See Theorem 4.3.1 of [42] for a proof that $P_{\text{SH}} = \alpha(P)$ is a pseudo-differential operator with the same principal symbol as P .

[†]See Theorem 31 of [46] for a proof that the exponential map from $T_{x_0}S^d$ to S^d is an embedding when restricted to a ball of radius t at the origin in $T_{x_0}S^d$, provided t is smaller than the cut time of S^d , which is π .

The divergence theorem* implies that if ν is the outward pointing unit normal vector field to ∂B_{t-s} , then

$$\begin{aligned} 2 \left| \int_{B_{t-s}(x_0)} \operatorname{div}_g \{ \partial_t u \nabla_g u \} dV \right| &= 2 \left| \int_{\partial B_{t-s}(x_0)} \partial_t u \langle \nabla_g u, \nu \rangle_g dS \right| \\ &\leq \int_{\partial B_{t-s}(x_0)} [|\partial_t u|^2 + |\nabla_g u|_g^2] dS \\ &\leq \int_{\partial B_{t-s}(x_0)} e dS, \end{aligned} \quad (2.2.5)$$

so

$$E'(s) \leq \int_{B_{t-s}(x_0)} 2(c+1)u \partial_t u \leq |c+1|E(s). \quad (2.2.6)$$

Thus by Gronwall's inequality[†], $E(s) \leq E(0)e^{|c+1|s}$. Since $E(0) = 0$, we have $E(s) = 0$ for $0 \leq s \leq t$, which implies that $u(x, s) = 0$ for $d(x, x_0) \leq t - s$. $\square$

To obtain more general examples of operators satisfying Assumptions A and B, we will see in Section 4.2 that if an operator P satisfies Assumption A then it gives X a certain geometry, and Assumption B is closely related to this geometry. Assumption B implies that geodesics on X with respect to this geometry are all closed and have commensurable lengths. Conversely, for most naturally occurring operators in $\operatorname{COP}_1^+(X)$ with this geometrical property, including the Riemannian case where $P = \sqrt{-\Delta}$, there exists[‡] there exists $\tilde{P} \in \operatorname{COP}_1^+(X)$ satisfying Assumptions A and B, and such that $P - \tilde{P}$ is a pseudo-differential operator of order 0, so we can view $\tilde{P}$ as a perturbation of P . We can study the regularity of a subfamily of multipliers of P by representing them as multipliers of $\tilde{P}$. Thus we can study multipliers of an appropriate perturbation of the Laplacian on any compact rank

*See Proposition 59 of [46] for a proof of the form of the divergence theorem used here.

†See Theorem 1.10 of [48] for a proof of Gronwall's inequality as it is used here.

‡See Lemma 29.2.1 of [22] for a proof of the existence of $\tilde{P}$. We discuss this result in slightly more detail in Section 4.4.

one symmetric space, or any Zoll manifold*, since the geodesics on these manifolds are all closed and have commensurable lengths.

In order for Assumption C to hold, it is necessary that the operator P^2 be *local* rather than merely *pseudo-local*, i.e. it must be true that $\text{supp}(P^2u) \subset \text{supp}(u)$ for all inputs u rather than that P^2u is smooth and rapidly decaying away from the support of u . A result of Peetre [33] implies that P^2 must be a *differential operator*, and thus that P be a Dirac operator. Thus a use of Assumption C restricts our assumption to Dirac operators, and thus explains the presence of the shift $\frac{d-1}{2}$ that occurs in our definition of dilations in our discussion of multipliers of spherical harmonic expansions in the previous section (the only Dirac operators that are also multiplier operators for spherical harmonic expansions are of the form $\sqrt{c - \Delta}$ for $c \geq 0$, which has symbol $a(k) = \sqrt{k^2 + (d-1)k + c}$, and it is only when $c = (d-1)/2$ that a simplifies to a linear function of k). This assumption is only needed when combining frequency scales together in our theorem, and thus does not occur when we restrict our attention to the study of spectral multipliers of the form $a(P)$, where a is compactly supported away from the origin. We hope to remove this assumption in further work.

Under Assumption A and Assumption B, in Chapter 5 we prove necessary and sufficient conditions hold for a compactly supported function to be contained in $M_{\text{Dil}}^p(X)$ for $1 < p < (d-1)/2(d+1)$.

Theorem 2.4. *Suppose X is a compact d -dimensional manifold, and $P \in \text{COP}_1^+(X)$ satisfies Assumptions A and B. Then for $1 < p < 2(d-1)/(d+1)$, if $s = (d-1)(1/p - 1/2)$,*

*The rank one d -dimensional symmetric spaces are S^d , $\mathbb{RP}^d$, $\mathbb{CP}^{d/2}$ (if d is even), $\mathbb{HP}^{d/4}$ (if d is a multiple of 4), and the Cayley plane $\mathbb{OP}^2$ (if $d = 16$). Zoll manifolds are Riemannian manifolds diffeomorphic to S^d whose geodesics are all closed with commensurable period. See Chapter 3 of [4] for a definition and classification of the rank one symmetric spaces, and Chapter 4 for an analysis of Zoll manifolds.

then for any regulated a with $\text{supp}(a) \subset [1/2, 2]$,

$$\|a\|_{M_{\text{Dil}}^p(X,P)} \sim \|a\|_{R^{s,p}[0,\infty)} \sim \left(\int_0^\infty |\widehat{a}(t)|^p \langle t \rangle^s dt \right)^{1/p}.$$

For most manifolds X and P , elements of $\mathcal{V}_\lambda$ are difficult to describe explicitly; even for the relatively simple case of spherical harmonics on the sphere S^d , many questions about the geometric behavior of eigenfunctions remain open. Many arguments in harmonic analysis involve an interplay between spatial and frequential control, and without explicit descriptions of eigenfunctions, spatial control becomes difficult. Nonetheless, we will find we can obtain some spatial control by utilizing the wave equation on M , which carries geometric information via the behavior of wave propagation. Under the curvature assumptions we make, wave propagation has sufficient smoothing properties to match certain necessary conditions that multipliers need in order to be bounded. And Assumption B implies that the wave equation associated with the operator P is periodic, which simplifies the large time analysis of the wave equation.

Under Assumption C, and the additional assumption of certain wave estimates associated with P at each frequency scale, in Chapter 6 we are able to characterize all elements of $M_{\text{Dil}}^p(X)$ rather than just the compactly supported ones.

The statement of these estimates is somewhat technical, but relates to control over discrete averages of a wave equation on X . Fix $k \geq 0$, and consider a pair of maximal 2^{-k} separated subsets $\mathcal{X}_k$ and $\mathcal{T}_k$ of X and of $[0, 1]$ respectively*. Fix a family of L^1 -normalized functions $\mathbf{b} = \{b_{t_0}\}$ and $\mathbf{u} = \{u_{x_0}\}$ for $t_0 \in \mathcal{T}_k$ and $x_0 \in \mathcal{X}_k$, with u_{x_0} supported on a 2^{1-k} neighborhood of x_0 , and b_{t_0} on a 2^{1-k} interval centered at t_0 . Also fix a bump function $q \in C_c^\infty(\mathbb{R})$ with $\text{supp}(q) \subset [1/4, 4]$, and with $q(\lambda) = 1$ for $\lambda \in [1/2, 2]$, and define $Q_k = q(P/2^k)$, which we view as ‘frequency localizations’ at a scale 2^k . For each

*It is natural to use the metric on X induced by P here to define 2^{-k} separated subsets of X , though any Finsler or Riemannian metric will suffice.

$(x_0, t_0) \in \mathcal{X}_k \times \mathcal{T}_k$, define $f_{x_0, t_0} = \int b_{t_0}(t)(\cos(2\pi itP) \circ Q_k)\{u_{x_0}\}$, a time average of a frequency localized solution to the wave equation $\partial_t^2 u = -P^2 u$. Finally, define an operator A_k from functions on $\mathcal{X}_k \times \mathcal{T}_k$ to functions on X by $A_k\{c\} = \sum_{(x_0, t_0) \in \mathcal{X}_k \times \mathcal{T}_k} c(x_0, t_0) f_{x_0, t_0}$.

Assumption Wave-Bound_p(P): There exists a constant $C_0 > 0$ such that

$$\|A_k\{c\}\|_{L^p(X)} \leq C_0 2^{kd/p'} \left(\sum_{(x_0, t_0) \in \mathcal{X}_k \times \mathcal{T}_k} [|c(x_0, t_0)| \langle 2^k t_0 \rangle^s]^p \right)^{1/p}.$$

uniformly in k , b , and u , where $s = (d-1)(1/q - 1/2)$.

Theorem 2.5. *Let X be a compact manifold, and let $P \in COP_1^+(X)$ satisfy Assumptions A, B, and C. Fix $1 < p < q < 2$ with $q < 2d/(d+1)$, and suppose that Wave-Bound_q(P) holds. Then for any regulated function a , if $s = (d-1)(1/p - 1/2)$,*

$$\|a\|_{M_{Dil}^p(X, P)} \sim \|a\|_{R^{s, p}[0, \infty)}.$$

In Chapter 6, we verify that Wave-Bound_p(P) holds for $1/p - 1/2 > 1/(d-1)$ when $P = P_{SH}$. Theorem 2.2 then follows from Theorem 2.5.

In the next chapter, we will begin our study by describing what is currently known for the boundedness problem on the compact manifold $\mathbb{T}^d$, where eigenfunctions are explicit, and, using a simple transplantation theorem, one can study the behavior of spectral multipliers via the Fourier transform and radial convolution operators on $\mathbb{R}^d$. Our analysis here will help gain intuition and motivate potential hypotheses in the more general setting.

Chapter 3

Radial Multipliers on Euclidean Space

Consider the elliptic operator $P = \sqrt{-\Delta}$ on $\mathbb{T}^d$, where $\Delta = \partial_1^2 + \cdots + \partial_d^2$ is the usual Laplacian. Unlike for other compact manifolds, $\mathbb{T}^d$ has a natural explicit basis of eigenfunctions of Δ : for a given eigenvalue $\lambda > 0$, the space $\mathcal{V}_\lambda$ has an orthonormal basis consisting of the exponentials $e^{2\pi i n \cdot x}$, where $n \in \mathbb{Z}^d$ and $|n| = \lambda$. Since the Fourier series of a function gives the expansion of the function in this basis, it follows that we can expand the spectral multiplier operator $T = a(P)$ using a Fourier series, i.e. writing

$$Tf(x) = \sum_{n \in \mathbb{Z}^d} a(|n|) \widehat{f}(n) e^{2\pi i n \cdot x}. \quad (3.0.1)$$

Thus a multiplier of Δ on $\mathbb{T}^d$ is nothing more than a Fourier multiplier operator on $\mathbb{T}^d$ whose symbol is radial. Methods of transplantation* show that

$$\|a\|_{M^p(\mathbb{R}^d)} \sim \|a\|_{M_{\text{dil}}^p(\mathbb{T}^d)}, \quad (3.0.2)$$

*See Section 3.6.2 of Grafakos [12] for a proof of this result, based on methods of de Leeuw [28].

where $\|a\|_{MP(\mathbb{R}^d)}$ is the operator norm of the Fourier multiplier on $\mathbb{R}^d$ given by

$$Tf(x) = \int_{\mathbb{R}^d} a(|\xi|) \widehat{f}(\xi) e^{2\pi i \xi \cdot x} d\xi \quad (3.0.3)$$

on $L^p(\mathbb{R}^d)$. In this section we will focus on operators of this type, which we call *radial Fourier multiplier operators*.

The study of the regularity of Fourier multiplier operators has proved central to the development of modern harmonic analysis and the theory of linear partial differential operators. This is because essentially any operator T on $\mathbb{R}^d$ commuting with translations is a Fourier multiplier operator*, i.e. we can find a tempered distribution m on $\mathbb{R}^d$, the *symbol* of T , such that for any Schwartz function f ,

$$Tf(x) = \int_{\mathbb{R}^d} m(\xi) \widehat{f}(\xi) e^{2\pi i \xi \cdot x} d\xi. \quad (3.0.4)$$

The study of translation invariant operators emerges from classical questions in analysis, such as the convergence of Fourier series, and problems in mathematical physics related to the study of the heat, wave, and Schrödinger equations. These physical equations also often have *rotational* symmetry, so it is natural to restrict our attention to translation-invariant operators which are also rotation-invariant. These operators are precisely the family of radial multiplier operators. If $m(\xi) = a(|\xi|)$ for a tempered distribution on $[0, \infty)$ we will write $T = a(P)$, where $P = \sqrt{-\Delta}$, since $Tf = a(\lambda)f$ whenever $\Delta f = -\lambda^2 f$.

*See Theorem 6.33 of [34] for a proof that every continuous operator from $C_c^\infty(\mathbb{R}^d)$ to $C(\mathbb{R}^d)$ commuting with translations is given by convolution against a distribution u . Provided that this operator extends to a continuous operator from $\mathcal{S}(\mathbb{R}^d)$ to $C(\mathbb{R}^d)$, this distribution is tempered, and so the operator is a Fourier multiplier operator whose symbol is the Fourier transform of u . See Theorem 2.5.2 of [12] for a proof that any operator bounded from $L^p(\mathbb{R}^d)$ to $L^q(\mathbb{R}^d)$ for some $p, q \in [1, \infty]$ is a Fourier multiplier operator.

3.1 Convolution Kernels of Fourier Multipliers

It is often useful to study spatial representations of these operators, since one can often exploit certain geometric information about the behavior of operators. Given any translation invariant operator T on $\mathbb{R}^d$, we can associate a tempered distribution k , the *convolution kernel* of T , such that

$$Tf(y) = \int_{\mathbb{R}^d} k(x)f(y-x) dx \quad \text{for all } f \in \mathcal{S}(\mathbb{R}^d). \quad (3.1.1)$$

If T is radial, then so is k , and so we can write $k(x) = b(|x|)$ for some distribution b on $[0, \infty)$, and then we have a representation

$$T = \int_0^\infty b(r)S_r dr, \quad \text{where} \quad S_rf(x) = \int_{|y|=r} f(x+y) dy \quad (3.1.2)$$

are the *spherical averaging operators*.

With the notation as above, the function k is the Fourier transform of m , and the function b is a Bessel transform of a , i.e. $b = \mathcal{B}_d a$, where

$$\mathcal{B}_d a(r) = \int_0^\infty s^{\frac{d-1}{2}} J_{\frac{d-1}{2}}(2\pi s) a(s) ds, \quad (3.1.3)$$

and where

$$J_\alpha(\lambda) = \frac{(\lambda/2)^\alpha}{\Gamma(\alpha + 1/2)} \int_{-1}^1 e^{i\lambda s} (1-s^2)^{\alpha-1/2} ds. \quad (3.1.4)$$

Using the theory of stationary phase, for each d we can write

$$J_d(\lambda) = e^{2\pi i \lambda} s_1(\lambda) + e^{-2\pi i \lambda} s_2(\lambda). \quad (3.1.5)$$

for symbols s_1 and s_2 of order $-1/2$. The presence of $e^{2\pi i \lambda}$ and $e^{-2\pi i \lambda}$ allows one to

relate the Bessel transform of a function to its Fourier transform, to a certain extent. In particular, we record the following result of Garrigos and Seeger [11].

Theorem 3.1. *Suppose $d > 1$ and $1/2d < 1/p - 1/2 < 1/2$, and suppose $a : [0, \infty) \rightarrow \mathbb{C}$ has compact support away from the origin. Then, with implicit constants depending on the support of a ,*

$$\left(\int_0^\infty |\mathcal{B}_d a(t)|^p t^{d-1} dt \right)^{1/p} \sim_{p,d,\phi} \left(\int_0^\infty |\widehat{a}(t)|^p \langle t \rangle^{(d-1)(1-p/2)} dt \right)^{1/p},$$

where $\widehat{a}(t) = \int_0^\infty a(\lambda) \cos(2\pi\lambda t) d\lambda$ is the cosine transform of a .

3.2 The Radial Multiplier Conjecture

The general study of the boundedness properties of Fourier multiplier operators in multiple variables was initiated in the 1950s, as connections of the theory to partial differential equations became more fully realized*. It was quickly realized that the most fundamental estimates were obtained by studying the boundedness of such operators on Lebesgue spaces. It is therefore natural to introduce the space $M^p(\mathbb{R}^d)$, consisting of all symbols m which induce a Fourier multiplier operator T bounded on $L^p(\mathbb{R}^d)$. Duality implies that $M^p(\mathbb{R}^d)$ is isometric to $M^{p'}(\mathbb{R}^d)$, where p and p' are conjugates, so it suffices to study the spaces $M^p(\mathbb{R}^d)$ where $1 \leq p \leq 2$ or when $2 \leq p \leq \infty$. We only know simple characterizations of $M^p(\mathbb{R}^d)$ for very particular p :

- The spaces $M^1(\mathbb{R}^d) = M^\infty(\mathbb{R}^d)$ can be characterized, by virtue of the fact that the boundedness of operators with domain $L^1(\mathbb{R}^d)$ or range $L^\infty(\mathbb{R}^d)$ is often simple; we have $M^1(\mathbb{R}^d) = \widehat{M}(\mathbb{R}^d)$, where $M(\mathbb{R}^d)$ is the space of all finite signed Borel measures, equipped with the total variation norm. The proof follows from Schur's

*See [18] for a survey of what was known about such operators at the time.

test for integral kernels, which often gives tight estimates to bound operators with domain L^1 or range L^∞ .

- The unitary nature of the Fourier transform also allows for the characterization $M^2(\mathbb{R}^d) = L^\infty(\mathbb{R}^d)$. The proof follows from Parseval's identity.

It is perhaps surprising that these are the *only* known characterizations of the spaces $M^p(\mathbb{R}^d)$. No necessary and simple conditions for boundedness are known for any other values of p , and perhaps no simple characterization exists.

Despite the lack of a characterization of the classes $M^p(\mathbb{R}^d)$, it is surprising that we *can* conjecture a characterization of the subspace of $M^p(\mathbb{R}^d)$ consisting of *radial symbols*, for an appropriate range of exponents. The conjectured range of estimates was first suggested by a result of [11], concerning the boundedness of a radial Fourier multiplier T with symbol $a(|\cdot|)$ *restricted to radial functions*, i.e. such that the norm

$$\|a\|_{M_{\text{Rad}}^p(\mathbb{R}^d)} = \sup \left\{ \frac{\|a(P)f\|_{L^q(\mathbb{R}^d)}}{\|f\|_{L^p(\mathbb{R}^d)}} : f \text{ is radial} \right\} \quad (3.2.1)$$

is finite. For any $\chi \in C_c^\infty(\mathbb{R})$, if we define k_R to be the Fourier transform of $\chi(|\cdot|)a(R|\cdot|)$, then the identity $k_R = a(RP)\{\widehat{\chi}\}$ and dilation symmetry imply that

$$\|k_R\|_{L^p(\mathbb{R}^d)} \lesssim \|a\|_{M_{\text{Rad}}^p(\mathbb{R}^d)}, \quad (3.2.2)$$

with implicit constants depending on χ . For $d > 1$, and for $1 < p < 2d/(d+1)$, Garrigos and Seeger proved [11] the converse bound

$$\|a\|_{M_{\text{Rad}}^p(\mathbb{R}^d)} \lesssim \sup_{R>0} \|k_R\|_{L^p(\mathbb{R}^d)}, \quad (3.2.3)$$

Theorem 3.1 implies that

$$\sup_{R>0} \|k_R\|_{L^p(\mathbb{R}^d)} \sim \|a\|_{R^{s,p}[0,\infty)}, \quad (3.2.4)$$

where $s = (d-1)(1/p - 1/2)$. Thus Garrigos and Seeger have proved that the isomorphism $M_{\text{Rad}}^p(\mathbb{R}^d) = R^{s,p}[0, \infty)$ holds in the range above.

For any other value of p , $M_{\text{Rad}}^p(\mathbb{R}^d)$ is a proper subset of $R^{s,p}[0, \infty)$:

- For $p = 1$, the inclusion ‘fails by a logarithm’, as we can see by the characterization of $M^1(\mathbb{R}^d)$ in the last section; if a is supported on an interval I , the inequality $\|a\|_{M^1(\mathbb{R}^d)} \lesssim \log |I| \|a\|_{R^{s,1}[0,\infty)}$ holds with $s = (d-1)/2$, and where the dependence on I is in general sharp, and cannot be removed.
- Suppose $p \geq 2d/(d+1)$. Then $R^{s,p}[0, \infty)$ contains all elements of the Besov space $\dot{B}_p^{1/2,2}[0, \infty)$ supported on $[1/2, 2]$. Thus, in particular, $R^{s,p}[0, \infty)$ must contain unbounded functions for $p > 1$, essentially because the Sobolev embedding theorem fails when used at the endpoint to embed into L^∞ . Since $M^p(\mathbb{R}^d) \subset M^2(\mathbb{R}^d)$, and $M^2(\mathbb{R}^d) = L^\infty[0, \infty)$, $M^p(\mathbb{R}^d)$ cannot contain unbounded functions, and thus $M^p(\mathbb{R}^d)$ is a proper subset of $R^{s,p}(\mathbb{R}^d)$.

It is natural to conjecture the same bounds hold when we remove the constraint that the inputs are radial, i.e so that for a radial function $m(\xi) = a(|\xi|)$ and $1 < p < 2d/(d+1)$,

$$\|m\|_{M^p(\mathbb{R}^d)} \sim_{p,d} \|a\|_{R^{p,s}[0,\infty)} \quad \text{where } s = (d-1)(1/p - 1/2). \quad (3.2.5)$$

We call this the *radial multiplier conjecture* on $\mathbb{R}^d$.

We now know, by results of Yaryong Heo, Fedor Nazarov, and Andreas Seeger [16] that the radial multiplier conjecture is true when $d \geq 4$ and when $1 < p < 2(d-1)/(d+1)$. A summary of the proof strategies of this argument is provided in the following two

sections, and a major part of our bounds for spectral multipliers follow by adapting this argument to the non-Euclidean setting. Partial improvements were obtained by Laura Cladek [7] for symbols m compactly supported away from the origin when $d = 4$ and when $1 < p < 36/29$, and establishing *restricted weak type* bounds when $d = 3$ and $1 < p < 13/12$. But the radial multiplier conjecture has not been resolved fully in any dimension d , we do not have any strong type L^p bounds when $d = 3$, and no bounds whatsoever are known when $d = 2$.

3.3 Radial-Multiplier Bounds by Density Decompositions

In this section and the following, we give an overview of the proof of the radial multiplier bounds obtained by Heo, Nazarov, and Seeger in [16].

Theorem 3.2. *Suppose $1 < p < 2(d - 1)/(d + 1)$, and $m(\xi) = a(|\xi|)$ is radial. Then*

$$\|m\|_{M^p(\mathbb{R}^d)} \lesssim \|a\|_{R^{s,p}[0,\infty)},$$

where $s = (d - 1)(1/p - 1/2)$.

The main tool we will take away for application to the proof of Theorem 2.4 is the method of *density decompositions*, and a method of *atomic decompositions* used to combine frequency scales in the problem. We give an overview of the density decomposition argument in this section to establish a single scale version of Theorem 3.2, as described in the following lemma, and discuss the particular atomic decomposition method in the following section.

Proposition 3.3. *Suppose $1 < p < 2(d - 1)/(d + 1)$, and k is a radial function with Fourier*

transform supported on $1/2 \leq |\xi| \leq 2$. Then

$$\|k * f\|_{L^p(\mathbb{R}^d)} \lesssim \|k\|_{L^p(\mathbb{R}^d)} \|f\|_{L^p(\mathbb{R}^d)}$$

We reduce Proposition 3.3 to an inequality for sums of functions oscillating on spheres. Let σ_r be the surface measure for the sphere of radius r centered at the origin in $\mathbb{R}^d$. Also fix a nonzero, radial, compactly supported function $\psi \in \mathcal{S}(\mathbb{R}^d)$ whose Fourier transform is non-negative, and vanishes to high order at the origin. Given $x \in \mathbb{R}^d$ and $r \geq 1$, define $\chi_{x,r} = \text{Trans}_x(\sigma_r * \psi)$, where $\text{Trans}_x g(y) = g(y-x)$. Then $\chi_{x,r}$ is a smooth function adapted* to a thickness $O(1)$ annulus of radius r centered at x , which is *slightly oscillating*. We will verify the following proposition.

Proposition 3.4. *For any $a : \mathbb{R}^d \times [1, \infty) \rightarrow \mathbb{C}$, if $1 < p < 2(d-1)/(d+1)$,*

$$\left\| \int_{\mathbb{R}^d} \int_1^\infty a(x, r) \chi_{x,r} dx dr \right\|_{L^p(\mathbb{R}^d)} \lesssim \left(\int_{\mathbb{R}^d} \int_1^\infty |a(x, r)|^p r^{d-1} dr dx \right)^{1/p}.$$

The implicit constant here depends on p , d , and ψ .

Proof of Proposition 3.3 from Proposition 3.4. Suppose $k(\cdot) = b(|\cdot|)$ for $b : [0, \infty) \rightarrow \mathbb{C}$.

If we set $a(x, r) = f(x)b(r)$ for any function $f : \mathbb{R}^d \rightarrow \mathbb{C}$, then

$$(k * \psi * f)(y) = \int_{\mathbb{R}^d} \int_1^\infty a(x, r) \chi_{x,r}(y) dx dr, \quad (3.3.1)$$

Proposition 3.4, applied to (3.3.1) tells us that

$$\|k * \psi * f\|_{L^p(\mathbb{R}^d)} \lesssim \|k\|_{L^p(\mathbb{R}^d)} \|f\|_{L^p(\mathbb{R}^d)}. \quad (3.3.2)$$

If we choose ψ so that $\widehat{\psi}$ is non-vanishing on the support of $\widehat{k}$, then the function $1/\widehat{\psi}(\cdot)$ is

*As defined in the appendix.

smooth on the support of $\widehat{k}$; if T is a Fourier multiplier operator with a smooth, compactly supported symbol agreeing with $1/\widehat{\psi}(\cdot)$ on the support of $\widehat{k}$, then the convolution kernel of T is Schwartz, and so T is bounded on $L^p(\mathbb{R}^d)$ by Schur's test. Since $T(k * \psi * f) = k * f$, we conclude

$$\|k * f\|_{L^p(\mathbb{R}^d)} = \|T(k * \psi * f)\|_{L^p(\mathbb{R}^d)} \lesssim \|k * \psi * f\|_{L^p(\mathbb{R}^d)} \lesssim \|k\|_{L^p(\mathbb{R}^d)} \|f\|_{L^p(\mathbb{R}^d)}, \quad (3.3.3)$$

which completes the argument. $\square$

Next, we consider a discretization of Proposition 3.4.

Proposition 3.5. *Fix a 1-separated set $\mathcal{E} \subset \mathbb{R}^d \times [1, \infty)$. Then for any $a : \mathcal{E} \rightarrow \mathbb{C}$, and for $1 < p < 2(d-1)/(d+1)$,*

$$\left\| \sum_{(x,r) \in \mathcal{E}} a(x,r) \chi_{x,r} \right\|_{L^p(\mathbb{R}^d)} \lesssim \left(\sum_{(x,r) \in \mathcal{E}} |a(x,r)|^p r^{d-1} \right)^{1/p},$$

where the implicit constant is independent of $\mathcal{E}$.

Proof of Proposition 3.4 from Proposition 3.5. For any $a : \mathbb{R}^d \times [1, \infty) \rightarrow \mathbb{C}$, if we consider the vector-valued function $A(x, r) = a(x, r) \chi_{x,r}$, then

$$\int_{\mathbb{R}^d} \int_1^\infty A(x, r) \, dr \, dx = \int_{[0,1]^d} \int_0^1 \sum_{n \in \mathbb{Z}^d} \sum_{m>0} A(x+n, r+m) \, dr \, dx \quad (3.3.4)$$

The triangle inequality and the increasing property of norms on $[0, 1)^d \times [0, 1]$ imply that

$$\begin{aligned}
& \left\| \int_{\mathbb{R}^d} \int_1^\infty A(x, r) dr dx \right\|_{L^p(\mathbb{R}^d)} \\
& \leq \int_{[0,1)^d} \int_0^1 \left\| \sum_{n \in \mathbb{Z}^d} \sum_{m>0} A(x+n, r+m) \right\|_{L^p(\mathbb{R}^d)} dr dx \\
& \lesssim \int_{[0,1)^d} \int_0^1 \left(\sum_{n \in \mathbb{Z}^d} \sum_{m>0} |a(x+n, r+m)|^p r^{d-1} \right)^{1/p} dr dx \quad (3.3.5) \\
& \leq \left(\int_{[0,1)^d} \int_0^1 \sum_{n \in \mathbb{Z}^d} \sum_{m>0} |a(x+n, r+m)|^p r^{d-1} dr dx \right)^{1/p} \\
& = \left(\int_{\mathbb{R}^d} \int_1^\infty |a(x, r)|^p r^{d-1} dr dx \right)^{1/p},
\end{aligned}$$

which completes the proof. $\square$

Proposition 3.5 can be further reduced by considering it as a bound on the operator

$$a \mapsto \sum_{(x,r) \in \mathcal{E}} a(x, r) \chi_{x,r}. \quad (3.3.6)$$

In particular, since Proposition 3.5 is an estimate for an open interval of L^p spaces, by the Marcinkiewicz interpolation theorem (see Theorem 7.3) it suffices to prove a restricted strong type bound L^p bound for $1 < p < 2(d-1)/(d+1)$. So Proposition 3.5 is implied by the following proposition.

Proposition 3.6. *For $1 < p < 2(d-1)/(d+1)$, if $\mathcal{E}$ is a 1-separated set, and $\mathcal{E}_k$ is the set of $(x, r) \in \mathcal{E}$ with $2^k \leq r < 2^{k+1}$, then*

$$\left\| \sum_{(x,r) \in \mathcal{E}} \chi_{x,r} \right\|_{L^p(\mathbb{R}^d)} \lesssim \left(\sum_{k \geq 1} 2^{k(d-1)\#(\mathcal{E}_k)} \right)^{1/p}.$$

Remark. Note that if $r \sim 2^k$, then $\|\chi_{x,r}\|_{L^p(\mathbb{R}^d)} \sim 2^{k(d-1)/p}$, and so Proposition 3.6 says

$$\left\| \sum_{(x,r) \in \mathcal{E}} \chi_{x,r} \right\|_{L^p(\mathbb{R}^d)} \lesssim_p \left(\sum_{(x,r) \in \mathcal{E}} \|\chi_{x,r}\|_{L^p(\mathbb{R}^d)}^p \right)^{1/p}. \quad (3.3.7)$$

Thus we are proving a p th root cancellation bound for partial sums of the functions $\{\chi_{x,r}\}$.

To control these sums, we apply a ‘density decomposition’, which splits a discrete set into different parts which are either spread out or clustered on small sets. The density decomposition will enable us to obtain L^p bounds from L^2 bounds. We say a 1-separated set $\mathcal{E}$ in a metric space X is of *density type* (u, r) if $\#(B \cap \mathcal{E}) \leq u \cdot \text{diam}(B)$ for each ball B in X with diameter at most r .

Lemma 3.7. *For any family of 1-separated sets $\mathcal{E}_k \subset \mathbb{R}^d \times [2^k, 2^{k+1})$, there exists a decomposition $\mathcal{E}_k = \bigcup_{m=1}^{\infty} \mathcal{E}_k(2^m)$ with the following properties:*

- *For each m , $\mathcal{E}_k(2^m)$ has density type $(2^m, 2^k)$.*
- *If B is a ball in $\mathbb{R}^{d+1}$ of radius $r \leq 2^k$ containing at least $2^m r$ points of $\mathcal{E}_k$, then*

$$B \cap \mathcal{E}_k \subset \bigcup_{m' \geq m} \mathcal{E}_k(2^{m'}).$$

- *For each m , there are disjoint balls $\{B_i\}$ in $\mathbb{R}^{d+1}$ with radii $\{r_i\}$, such that*

$$\sum_i r_i \leq 2^{-m} \# \mathcal{E}_k,$$

such that $r_i \leq 2^k$ for all i , and such that $\bigcup B_i^$ covers $\bigcup_{m' \geq m} \mathcal{E}_k(2^{m'})$, where B_i^* denotes the ball with the same center as B_i but 5 times the radius.*

Proof. Define a function $M : \mathcal{E}_k \rightarrow [0, \infty)$ by setting

$$M(x, r) = \sup \left\{ \frac{\#(\mathcal{E}_k \cap B)}{\text{rad}(B)} : (x, r) \in B \text{ and } \text{rad}(B) \leq 2^k \right\}. \quad (3.3.8)$$

We can establish a kind of weak L^1 estimate for M using a Vitali type argument. Let

$$\widehat{\mathcal{E}}_k(2^m) = \{(x, r) \in \mathcal{E}_k : M(x, r) \geq 2^m\}. \quad (3.3.9)$$

We can therefore cover $\widehat{\mathcal{E}}_k(2^m)$ by a family of balls $\{B\}$ such that $\#(\mathcal{E}_k \cap B) \geq 2^m \text{rad}(B)$. The Vitali covering lemma allows us to find a disjoint subcollection of balls $B_1, \dots, B_N$ such that $B_1^*, \dots, B_N^*$ covers $\widehat{\mathcal{E}}_k(2^m)$. We find that

$$\#(\mathcal{E}_k) \geq \sum_i \#(B_i \cap \mathcal{E}_k) \geq 2^m \sum_i \text{rad}(B_i), \quad (3.3.10)$$

Setting $\mathcal{E}_k = \widehat{\mathcal{E}}_k(2^m) - \bigcup_{k' > k} \widehat{\mathcal{E}}_{k'}(2^m)$ thus gives the required result. $\square$

Remark. We will later apply the Lemma with $\mathbb{R}^d \times [0, \infty)$ replaced by $X \times [0, \infty)$, where X is a compact, d -dimensional manifold equipped with a Finsler metric. A version of the Vitali covering lemma also holds for this metric space, so that the same proof allows one to perform a density decomposition in this setting.

To prove Proposition 3.6, we perform a decomposition of $\mathcal{E}_k$ for each k into the sets $\mathcal{E}_k(2^m)$, and then define $\mathcal{E}^m = \bigcup_{k \geq 1} \mathcal{E}_k^m$. For appropriate exponents, we prove L^p bounds on the functions $F^m = \sum_{(x,r) \in \mathcal{E}^m} \chi_{x,r}$ which are exponentially decaying in m , i.e. so that

$$\|F^m\|_{L^p(\mathbb{R}^d)} \lesssim 2^{-m((1/p-1/2)-(d-1)^{-1})} \left(\sum_k 2^{k(d-1)} \#(\mathcal{E}_k) \right)^{1/p}, \quad (3.3.11)$$

with implicit constants depending polynomially on m . Thus, in the range $1/p - 1/2 > (d-1)^{-1}$, i.e. for $1 < p < 2(d-1)/(d+1)$ there is geometric decay in m , decaying much quicker than the polynomial implicit constant, and so we may sum in m using the triangle inequality to conclude that

$$\|F\|_{L^p(\mathbb{R}^d)} \lesssim \left(\sum_k 2^{k(d-1)} \#(\mathcal{E}_k) \right)^{1/p}, \quad (3.3.12)$$

proving Proposition 3.6. To get the bound on F^m , we interpolate between an L^2 bound for F^m , and an ' L^0 bound' (a bound on the measure of the support of F^m).

Lemma 3.8. *For each k ,*

$$|\text{supp}(F_k^m)| \lesssim 2^{-m} 2^{k(d-1)} \#\mathcal{E}_k.$$

Thus we have

$$|\text{supp}(F^m)| \leq \sum_k |\text{supp}(F_k^m)| \lesssim \sum_k 2^{-m} 2^{k(d-1)} \#\mathcal{E}_k.$$

Proof. We recall that for each k and m , we can find disjoint balls $B_1, \dots, B_N$ with radii $r_1, \dots, r_N \leq 2^k$ such that $\mathcal{E}_k(2^m)$ is covered by the expanded balls $B_1^* \cup \dots \cup B_N^*$, and

$$\sum_i r_i \leq 2^{-m} \#\mathcal{E}_k. \quad (3.3.13)$$

If we write

$$F_{k,i}^m = \sum_{(x,r) \in \mathcal{E}_k(2^m) \cap B_i^*} \chi_{x,r}, \quad (3.3.14)$$

then $\text{supp}(F_k^m) \subset \bigcup_i \text{supp}(F_{k,i}^m)$. For each $(x, r) \in B_i^* \cap \mathcal{E}_k(2^m)$, the support of $\chi_{x,r}$, an annulus of thickness $O(1)$ and radius r , is contained in an annulus of thickness $O(r_i)$ and radius $O(2^k)$ with the same center as B_i . Thus we conclude that

$$|\text{supp}(F_{k,i}^m)| \lesssim r_i 2^{k(d-1)}, \quad (3.3.15)$$

and it follows that

$$|\text{supp}(F_k^m)| \leq \sum_i r_i 2^{k(d-1)} \leq 2^{-m} 2^{k(d-1)} \#\mathcal{E}_k, \quad (3.3.16)$$

which completes the proof. $\square$

Interpolating, it suffices to prove the following L^2 estimate on the function F^m .

Lemma 3.9. *Suppose $\mathcal{E} = \bigcup_k \mathcal{E}_k$ is a 1-separated set, where $\mathcal{E}_k \subset \mathbb{R}^d \times [2^k, 2^{k+1})$ is a set*

of density type $(2^m, 2^k)$. Then

$$\left\| \sum_{(x,r) \in \mathcal{E}} \chi_{x,r} \right\|_{L^2(\mathbb{R}^d)} \lesssim \sqrt{m} \cdot 2^{\frac{m}{d-1}} \left(\sum_k 2^{k(d-1)} \#(\mathcal{E}_k) \right)^{1/2}.$$

The L^2 bound in Lemma 3.9 gets worse and worse as m grows, whereas the L^0 bound in Lemma 3.8 gets better and better, since annuli are concentrating in a small set, which is problematic from the perspective of constructive interference, but inert from the perspective of a support bound. To prove the L^2 bound, we require an analysis of the interference patterns of pairs of the functions $\chi_{x,r}$, as provided by the following lemma.

Lemma 3.10. *For any $N > 0$, $x_1, x_2 \in \mathbb{R}^d$ and $r_1, r_2 \geq 1$,*

$$|\langle \chi_{x_1, r_1}, \chi_{x_2, r_2} \rangle| \lesssim_N \left(\frac{r_1 r_2}{\langle (x_1, r_1) - (x_2, r_2) \rangle} \right)^{\frac{d-1}{2}} \sum_{\pm, \pm} \langle |x_1 - x_2| \pm r_1 \pm r_2 \rangle^{-N}.$$

Remark. Suppose $r_1 \leq r_2$. Then Lemma 3.10 implies that χ_{x_1, r_1} and χ_{x_2, r_2} are roughly uncorrelated, except when they are supported on annuli that roughly have the same radii and centers, and in addition, one of the following two properties hold:

- $r_1 + r_2 \approx |x_1 - x_2|$, which holds when the two annuli are ‘approximately’ externally tangent to one another.
- $r_2 - r_1 \approx |x_1 - x_2|$, which holds when the two annuli are ‘approximately’ internally tangent to one another.

Heo, Nazarov, and Seeger do not exploit the tangency information, though utilizing the tangencies seems important to improve the results they obtain. Cladek exploits this tangency information further, to obtain improved results.

Proof. We write

$$\begin{aligned}
\langle \chi_{x_1 r_1}, \chi_{x_2 r_2} \rangle &= \langle \widehat{\chi}_{x_1 r_1}, \widehat{\chi}_{x_2 r_2} \rangle \\
&= \int_{\mathbb{R}^d} \widehat{\sigma_{r_1} * \psi}(\xi) \overline{\widehat{\sigma_{r_2} * \psi}(\xi)} e^{2\pi i(x_2 - x_1) \cdot \xi} d\xi \\
&= (r_1 r_2)^{d-1} \int_{\mathbb{R}^d} \widehat{\sigma}(r_1 \xi) \overline{\widehat{\sigma}(r_2 \xi)} |\widehat{\psi}(\xi)|^2 e^{2\pi i(x_2 - x_1) \cdot \xi} d\xi.
\end{aligned} \tag{3.3.17}$$

Define functions A and B such that $B(|\xi|) = \widehat{\sigma}(\xi)$, and $A(|\xi|) = |\widehat{\psi}(\xi)|^2$. Then

$$\langle \chi_{x_1, r_1}, \chi_{x_2, r_2} \rangle = C_d (r_1 r_2)^{d-1} \int_0^\infty s^{d-1} A(s) B(r_1 s) B(r_2 s) B(|x_2 - x_1| s) ds. \tag{3.3.18}$$

Using well known asymptotics for $\widehat{\sigma}$, we have, for any $N > 0$,

$$B(s) = s^{-(d-1)/2} \sum_{n=0}^{N-1} (c_{n,+} e^{2\pi i s} + c_{n,-} e^{-2\pi i s}) s^{-n} + O_N(s^{-N}). \tag{3.3.19}$$

Write

$$a_{n,\tau} = \int_0^\infty A(s) s^{-\frac{d-1}{2} - n_1 - n_2 - n_3} e^{2\pi i(\tau_1 r_1 + \tau_2 r_2 + \tau_3 |x_2 - x_1|)s} ds. \tag{3.3.20}$$

Assuming $A(s)$ vanishes to suitably high order at the origin, depending on N , we find that

$$\begin{aligned}
\langle \chi_{x_1 r_1}, \chi_{x_2 r_2} \rangle &= C_d \left(\frac{r_1 r_2}{|x_1 - x_2|} \right)^{(d-1)/2} \sum_{n,\tau} a_{n,\tau} c_{n,\tau} r_1^{-n_1} r_2^{-n_2} |x_2 - x_1|^{-n_3} \\
&\lesssim \left(\frac{r_1 r_2}{|x_1 - x_2|} \right)^{\frac{d-1}{2}} \left(1 + \frac{1}{|x_1 - x_2|^N} \right) \sum_{\tau} \langle \tau_1 r_1 + \tau_2 r_2 + \tau_3 |x_2 - x_1| \rangle^{-N}.
\end{aligned} \tag{3.3.21}$$

This gives the result provided that $1 + |x_1 - x_2| \geq |r_1 - r_2|/10$ and $|x_1 - x_2| \geq 1$. If $1 + |x_1 - x_2| \leq |r_1 - r_2|/10$, then the supports of χ_{x_1, r_1} and χ_{x_2, r_2} are disjoint, so the inequality is trivial. On the other hand, if $|x_1 - x_2| \leq 1$, then the bound is trivial by the last sentence unless $|r_1 - r_2| \leq 10$, and in this case the inequality reduces to the simple inequality

$$\langle \chi_{x_1, r_1}, \chi_{x_2, r_2} \rangle \lesssim_N (r_1 r_2)^{(d-1)/2}. \tag{3.3.22}$$

But this follows immediately from the Cauchy-Schwarz inequality. $\square$

The exponent $\frac{d-1}{2}$ in Lemma 3.10 is too weak to apply the general theory of almost orthogonality directly to obtain L^2 bounds on $\sum_{(x,r) \in \mathcal{E}_k} \chi_{x,r}$ without exploiting further geometric information about the functions $\chi_{x,r}$. But together with the density decomposition assumption we will be able to obtain Lemma 3.9.

Proof of Lemma 3.9. Without loss of generality, we may assume that the set of k such that $\mathcal{E}_k \neq \emptyset$ is 10-separated. Write

$$F = \sum_{(x,r) \in \mathcal{E}} \chi_{x,r} \quad (3.3.23)$$

and $F_k = \sum_{(x,r) \in \mathcal{E}_k} \chi_{x,r}$. First, we deal with $F_{\leq m} = \sum_{k \leq 10m} F_k$ trivially, i.e. writing

$$\|F\|_{L^2(\mathbb{R}^d)} \lesssim m^{1/2} \left(\sum_{k \leq 10m} \|F_k\|_{L^2(\mathbb{R}^d)}^2 + \left\| \sum_{k > 10m} F_k \right\|_{L^2(\mathbb{R}^d)} \right)^{1/2}. \quad (3.3.24)$$

We then decompose

$$\left\| \sum_{k > 10m} F_k \right\|_{L^2(\mathbb{R}^d)}^2 \leq \sum_{k > 10m} \|F_k\|_{L^2(\mathbb{R}^d)}^2 + 2 \sum_{k' > k > 10m} |\langle F_k, F_{k'} \rangle|. \quad (3.3.25)$$

Let us analyze $\langle F_k, F_{k'} \rangle$. The term will become a sum of the form $\langle \chi_{x,r}, \chi_{y,s} \rangle$, where $r \sim 2^k$ and $s \sim 2^{k'}$. Because of our assumption of being 10-separated, we have $r \leq s/2^{10}$. If $\langle \chi_{x,r}, \chi_{y,s} \rangle \neq 0$, then since the support of $\chi_{y,s}$ is an annulus of radius s centered at y , with thickness $O(1)$, and $\chi_{x,r}$ has support on an annulus of radius r centered at x , with thickness $O(1)$, the fact that r is comparatively smaller than s implies that (x, r) must be contained in the annulus of radius s centered at y , with thickness $O(2^k)$. Such an annulus is covered by $O(2^{(k'-k)(d-1)})$ balls of radius 2^k . Each ball can only contain 2^{k+m} points (x, r) , and so there can be at most $O(2^{k'(d-1)-k(d-1)+k+m}) = O(2^{k'(d-1)-k(d-2)+m})$ pairs $(x, r) \in \mathcal{E}_k$ for which

$\langle \chi_{x,r}, \chi_{y,s} \rangle \neq 0$. For such pairs we have

$$|\langle \chi_{x,r}, \chi_{y,s} \rangle| \lesssim \left(\frac{2^k 2^{k'}}{2^{k'}} \right)^{\frac{d-1}{2}} = 2^{\frac{k(d-1)}{2}}. \quad (3.3.26)$$

Thus we conclude that

$$|\langle F_k, \chi_{y,s} \rangle| \lesssim 2^{-k(\frac{d-3}{2}) + k'(d-1) + m}. \quad (3.3.27)$$

Summing over $10m < k < k'$, we conclude that since $d \geq 4$,

$$\sum_{10m < k < k'} |\langle F_k, \chi_{y,s} \rangle| \lesssim 2^{k'(d-1) + m} \sum_{10m < k < k'} 2^{-k\frac{d-3}{2}} \lesssim 2^{k'(d-1) + m} 2^{-5m} \lesssim 2^{k'(d-1)}. \quad (3.3.28)$$

But this means that

$$\sum_{10m < k < k'} |\langle F_k, F_{k'} \rangle| \lesssim 2^{k'(d-1)} \cdot \#(\mathcal{E}_{k'}). \quad (3.3.29)$$

This means that

$$\left\| \sum_{k > 10m} F_k \right\|_{L^2(\mathbb{R}^d)}^2 \lesssim \sum_{k > 10m} \|F_k\|_{L^2(\mathbb{R}^d)}^2 + \sum_{k'} 2^{k'(d-1)} \#(\mathcal{E}_{k'}), \quad (3.3.30)$$

and it now suffices to deal with estimates the $\|F_k\|_{L^2(\mathbb{R}^d)}$, i.e. the interactions of functions supported on radii of comparable magnitude. To deal with these, we further decompose the radii, writing $[2^k, 2^{k+1})$ as the disjoint union of intervals $I_{k,\mu} = [2^k + (\mu-1)2^{um}, 2^k + \mu 2^{um}]$, for some u to be chosen later. These interval induces a decomposition $\mathcal{E}_k = \bigcup_{\mu} \mathcal{E}_{k,\mu}$. Again, incurring a constant loss at most, we may assume that the μ such that $\mathcal{E}_{k,\mu} \neq \emptyset$ are 10 separated. We write $F_k = \sum F_{k,\mu}$, and we have

$$\|F_k\|_{L^2(\mathbb{R}^d)}^2 = \sum_{\mu} \|F_{k,\mu}\|_{L^2(\mathbb{R}^d)}^2 + \sum_{\mu < \mu'} |\langle F_{k,\mu}, F_{k,\mu'} \rangle|. \quad (3.3.31)$$

We now consider $\chi_{x,r}$ and $\chi_{y,s}$ with $r \in I_{k,\mu}$ and $s \in I_{k,\mu'}$. Then we must have $|x - y| \lesssim 2^k$

and $2^{am} \leq |r - s| \lesssim 2^k$, and so we have

$$\begin{aligned} \left| \sum_{\mu < \mu'} \langle F_{k,\mu}, \chi_{y,s} \rangle \right| &\lesssim 2^{k(d-1)} \sum_{\substack{(x,r) \in \mathcal{E}_k \\ 2^{um} \leq |(x,r)-(y,s)| \lesssim 2^k}} |(x,r) - (y,s)|^{-\frac{d-1}{2}} \\ &\lesssim 2^{k(d-1)} \sum_{um \leq l \leq k} 2^{-l(d-1)/2} \#\{(x,r) \in \mathcal{E}_k : |(x,r) - (y,s)| \sim 2^l\} \end{aligned} \quad (3.3.32)$$

Using the density assumption,

$$\#\{(x,r) \in \mathcal{E}_k : |(x,r) - (y,s)| \sim 2^l\} \lesssim 2^{l+m} \quad (3.3.33)$$

and so we obtain that, again using the assumption that $d \geq 4$,

$$\left| \sum_{\mu < \mu'} \langle F_{k,\mu}, \chi_{y,s} \rangle \right| \lesssim 2^{k(d-1)} 2^{m(1-u(d-3)/2)}. \quad (3.3.34)$$

Now summing over all (y,s) , we obtain that

$$\left| \sum_{\mu < \mu'} \langle F_{k,\mu}, F_{k,\mu'} \rangle \right| \lesssim 2^{k(d-1)} 2^{m(1-u(d-3)/2)} \#\mathcal{E}_{k,\mu'}. \quad (3.3.35)$$

and now summing over μ' gives that

$$\|F_k\|_{L^2(\mathbb{R}^d)}^2 \lesssim \sum_{\mu} \|F_{k,\mu}\|_{L^2(\mathbb{R}^d)}^2 + 2^{k(d-1)} 2^{m(1-u(d-3)/2)} \#\mathcal{E}_k, \quad (3.3.36)$$

Now we are left to analyze $\|F_{k,\mu}\|_{L^2(\mathbb{R}^d)}$, i.e. analyzing interactions between annuli which have radii differing from one another by at most $O(2^{am})$. Since the family of all possible radii are discrete, the set $\mathcal{R}_{k,\mu}$ of all possible radii has cardinality $O(2^{um})$. We do not really have any orthogonality to play with here, so we just apply the Cauchy-Schwarz inequality, writing $F_{k,\mu} = \sum_{r \in \mathcal{R}_{k,\mu}} F_{k,\mu,r}$, to write

$$\|F_{k,\mu}\|_{L^2(\mathbb{R}^d)}^2 \lesssim 2^{am} \sum_r \|F_{k,\mu,r}\|_{L^2(\mathbb{R}^d)}^2. \quad (3.3.37)$$

Recall that $\chi_{x,r} = \text{Trans}_x(\sigma_r * \psi)$, where ψ is a compactly supported function whose Fourier transform is non-negative and vanishes to high order at the origin, and $\text{Trans}_x f(y) = f(y - x)$ is the translation operator by x . In particular, we now make the additional assumption that $\psi = \psi_\circ * \psi_\circ$ for some other compactly function $\psi_\circ$ whose Fourier transform is non-negative and vanishes to high order at the origin. Then we find that $F_{k,\mu,r}$ is equal to the convolution of the function

$$G_r = \sum_{(x,r) \in \mathcal{E}} \text{Trans}_x \psi_\circ \quad (3.3.38)$$

with the function $\sigma_r * \psi_\circ$. Using the standard asymptotics for the Fourier transform of σ_r , i.e. that for $|\xi| \geq 1$,

$$|\widehat{\sigma_r}(\xi)| \lesssim r^{d-1} (1 + r|\xi|)^{-\frac{d-1}{2}}, \quad (3.3.39)$$

and since $|\widehat{\psi_\circ}(\xi)| \lesssim_N |\xi|^N$, we get that if $r \geq 1$, then for $|\xi| \leq 1/r$,

$$|\widehat{\sigma_r}(\xi) \widehat{\psi_\circ}(\xi)| \lesssim_N r^{d-1-N} \quad (3.3.40)$$

and for $|\xi| \geq 1/r$,

$$|\widehat{\sigma_r}(\xi) \widehat{\psi_\circ}(\xi)| \lesssim_N r^{\frac{d-1}{2}} |\xi|^{-N}. \quad (3.3.41)$$

Thus in particular, the L^∞ norm of the Fourier transform of $\sigma_r * \psi_\circ$ is $O(r^{(d-1)/2})$. Now the functions $\psi_\circ$ are compactly supported, so since the set of x such that $(x, r) \in \mathcal{E}$ is one-separated, we find that

$$\|G_r\|_{L^2(\mathbb{R}^d)} \lesssim \#\{x : (x, r) \in \mathcal{E}\}^{1/2}. \quad (3.3.42)$$

But this means that

$$\|F_{k,\mu,r}\|_{L^2(\mathbb{R}^d)} = \|G_r * (\sigma_r * \psi_\circ)\|_{L^2(\mathbb{R}^d)} \lesssim r^{\frac{d-1}{2}} \#\{x : (x, r) \in \mathcal{E}\}^{1/2}. \quad (3.3.43)$$

Thus we have that

$$\|F_{k,\mu}\|_{L^2(\mathbb{R}^d)}^2 = 2^{um} \cdot \#\mathcal{E}_{k,\mu} \cdot 2^{k(d-1)}. \quad (3.3.44)$$

Summing over μ gives that

$$\|F_k\|_{L^2(\mathbb{R}^d)}^2 = 2^{k(d-1)} \#\mathcal{E}_k (2^{um} + 2^{m(1-u(d-3)/2)}). \quad (3.3.45)$$

Picking $u = 2/(d-1)$ optimizes this bound, giving

$$\|F_k\|_{L^2(\mathbb{R}^d)} \lesssim 2^{m/(d-1)} 2^{k(d-1)/2} (\#\mathcal{E}_k)^{1/2}. \quad (3.3.46)$$

Combining (3.3.30) and (3.3.46) completes the proof. $\square$

This completes a proof of the single scale estimates of the paper. The paper then uses an atomic decomposition method to combine frequency scales and thus complete the proof of Theorem 3.2. We discuss these methods in the following section.

3.4 Combining Scales with Atomic Decompositions

We now sketch a proof as to how we can complete the proof of Theorem 3.2. Our arguments are deliberately vague, as our goal is to build intuition about the techniques involved, in order to carry out analogous methods more rigorously in the compact manifold setting later in Chapter 6. If $T = m(P)$, then Proposition 3.3, appropriately rescaled, implies that if we write $T = \sum T_j$, where T_j is the radial Fourier multiplier operator with convolution kernel $k_j(\cdot/2^j)$, then

$$\|T_j\|_{L^p(\mathbb{R}^d) \rightarrow L^p(\mathbb{R}^d)} \lesssim \|k_j\|_{L^p(\mathbb{R}^d)}. \quad (3.4.1)$$

The goal of this section is to prove that we can bound the sum of the operators T_j by $\sup_j \|k_j\|_{L^q(\mathbb{R}^d)}$, which morally speaking, is a bound of the form

$$\left\| \sum_j T_j \right\| \lesssim \sup_j \|T_j\|. \quad (3.4.2)$$

We must thus show that the operators $\{T_j\}$ do not constructively interfere with one another to a significant extent.

Atomic decompositions are a powerful way to control interactions between operators. The method involves decomposing functions into more elementary components, which we call *atoms*, that have controlled size, spatial support, and oscillatory properties. Such atoms are chosen via careful, non-linear selection processes, often related to stopping times, to ensure certain cancellation conditions hold. We use a variant of this method, involving a use of a Whitney decomposition rather than a stopping time argument, to obtain an atomic decomposition where atoms do not cluster, in an appropriate sense. The method employed was first introduced in [37].

Since we are controlling parts of an operator supported on different dyadic frequency ranges on functions in $L^p(\mathbb{R}^d)$ for $1 < p < \infty$, it is natural to consider the Littlewood-Paley inequality $\|Sf\|_{L^p(\mathbb{R}^d)} \lesssim \|f\|_{L^p(\mathbb{R}^d)}$, where

$$Sf(x) = \left(\sum_j |P_j f(x)|^2 \right)^{1/2}, \quad (3.4.3)$$

and the operators P_j are Littlewood-Paley projections, Fourier multiplier operators with symbol $\psi(\cdot/2^j)$, where $\psi \in C_c^\infty(\mathbb{R}^d)$ is equal to one on the support of the function χ used to decompose T into the operators T_j above, so that $T_j f = T_j f_j$, where $f_j = P_j f$. Roughly speaking, our atomic decomposition will be of the following form: for each dyadic number H , we consider a family of dyadic cubes $\mathcal{W}_H$, whose union is the set $\{x : |Sf(x)| \sim H\}$, and whose doubles have the bounded overlap property. We will then

obtain a decomposition $f_j = \sum A_{j,H,W}$, where $A_{j,H,W}$ has Fourier support on $|\xi| \sim 2^j$, is supported on the cube W , and for each fixed H ,

$$\left(\sum_j \left| \sum_W |A_{j,H,W}(x)| \right|^2 \right)^{1/2} \sim H \quad \text{if } |Sf(x)| \sim H. \quad (3.4.4)$$

The advantage of this decomposition is that it controls how many *local interactions* the atoms $\{A_{j,H,W}\}$ have. To see why this might be useful, suppose we were considering a Fourier multiplier operator T with the property that $TA_{j,H,W}$ is supported on the cube W^* obtained by doubling the side lengths of the cube W , but maintaining the same center. The bounded overlap property of the dyadic cubes $\mathcal{W}_H$ would then imply an almost orthogonality bound for each H , that

$$\begin{aligned} \left\| \sum_{j,W} T_j A_{j,H,W} \right\|_{L^2(\mathbb{R}^d)} &\lesssim \left(\sum_{j,W} \|T_j A_{j,H,W}\|_{L^2(\mathbb{R}^d)}^2 \right)^{1/2} \\ &\lesssim \left(\sum_{j,W} \|A_{j,H,W}\|_{L^2(\mathbb{R}^d)}^2 \right)^{1/2} \\ &= \left\| \left(\sum_{j,W} |A_{j,H,W}|^2 \right)^{1/2} \right\|_{L^2(\mathbb{R}^d)} \lesssim H |\Omega_H|^{1/2} \end{aligned} \quad (3.4.5)$$

Since the sum is supported on Ω_H , Hölder's inequality then justifies that

$$\left\| \sum_{j,W} T_j A_{j,H,W} \right\|_{L^1(\mathbb{R}^d)} \lesssim |\Omega_H|^{1/2} \left\| \sum_{j,W} T_j A_{j,H,W} \right\|_{L^2(\mathbb{R}^d)} \lesssim H |\Omega_H|. \quad (3.4.6)$$

Real interpolation (Lemma 7.6) allows us to sum in H , so that for $1 < p < 2$,

$$\|Tf\|_{L^p(\mathbb{R}^d)} \lesssim \left(\sum [H |\Omega_H|^{1/p}]^p \right)^{1/p} \lesssim \|Sf\|_{L^p(\mathbb{R}^d)} \lesssim \|f\|_{L^p(\mathbb{R}^d)}. \quad (3.4.7)$$

and thus we have proven the boundedness of such operators.

Such a simple proof will not suffice for our analysis, since the class of multipliers we are considering is not pseudo-local at all; the kernels k_j are only assumed to be uniformly

bounded in $L^p(\mathbb{R}^d)$, and need not satisfy any decay bound as $|x| \rightarrow \infty$. Nonetheless, the argument above can be used to prove bounds for other more pseudo-local multipliers, for instance, obtaining an alternate proof of the endpoint results of [38], and we will be able to exploit the above calculations to control *close range interactions* of general multipliers. Given T and f , write

$$Tf = \sum T_j \{A_{j,H,W}\} = \sum T_{j,W,\text{Short}} \{A_{j,H,W}\} + T_{j,W,\text{Long}} \{A_{j,H,W}\}, \quad (3.4.8)$$

where $T_{j,W,\text{Short}}$ and $T_{j,W,\text{Long}}$ are Schwartz operators with kernels

$$K_{j,W,\text{Short}}(x, y) = \chi_{W^*}(x) K_{T_j}(x, y) \quad \text{and} \quad K_{j,W,\text{Long}}(x, y) = \chi_{(W^*)^c}(x) K_{T_j}(x, y), \quad (3.4.9)$$

and where χ_{W^*} is a smooth function adapted to W^* , and $\chi_{(W^*)^c} = 1 - \chi_{W^*}$. The functions $T_{j,W,\text{Short}} \{A_{j,H,W}\}$ remain almost orthogonal as j and W vary, and so the calculations of the previous paragraph can be adapted to show that

$$\left\| \sum_{j,W,H} T_{j,W,\text{Short}} \{A_{j,H,W}\} \right\|_{L^p(\mathbb{R}^d)} \lesssim \|f\|_{L^p(\mathbb{R}^d)}, \quad (3.4.10)$$

It remains to find a way to control the long range interactions between atoms.

There are several related techniques to control the long range interactions, the most elegant formulation provided in the paper [15]. The idea is to take the single scale estimate we obtained in 3.3, and upgrade this to an estimate with a geometrically decaying constant term when our inputs have amplitudes that are locally constant on a much larger scale than required by the uncertainty principle and the frequency support of the inputs, i.e. that

$$\left\| \sum_H \sum_{W \in \mathcal{W}_{H,l-j}} T_{j,W,\text{Long}} \{A_{j,H,W}\} \right\|_{L^p(\mathbb{R}^d)} \lesssim 2^{-l\varepsilon} \left(\sum_H \left\| \sum_{W \in \mathcal{W}_{H,l-j}} A_{j,H,W} \right\|_{L^\infty(\mathbb{R}^d)}^p |W| \right)^{1/p}, \quad (3.4.11)$$

where $\mathcal{W}_{H,a}$ are the set of all cubes of side-length 2^a . The functions $T_{j,W,\text{Long}}\{A_{j,H,W}\}$ are almost orthogonal as j varies, and so using Lemma 7.5 one can show

$$\begin{aligned} & \left\| \sum_j \sum_H \sum_{W \in \mathcal{W}_{H,l-j}} T_{j,W,\text{Long}}\{A_{j,H,W}\} \right\|_{L^p(\mathbb{R}^d)} \\ & \lesssim \left(\sum_j \left\| \sum_H \sum_{W \in \mathcal{W}_{H,l-j}} T_{j,W,\text{Long}}\{A_{j,H,W}\} \right\|_{L^p(\mathbb{R}^d)}^p \right)^{1/p}. \end{aligned} \quad (3.4.12)$$

Thus

$$\left\| \sum_{j,H} \sum_{W \in \mathcal{W}_{H,l-j}} T_{j,W,\text{Long}} A_{j,H,W} \right\|_{L^p(\mathbb{R}^d)} \lesssim 2^{-l\varepsilon} \left(\sum_{j,H} \left\| \sum_{W \in \mathcal{W}_{H,l-j}} A_{j,H,W} \right\|_{L^\infty(\mathbb{R}^d)}^p |W| \right)^{1/p}. \quad (3.4.13)$$

For a fixed l , and each $W \in \mathcal{W}_H$, there exists a unique j such that $W \in \mathcal{W}_{H,l-j}$, which implies that the supports of the functions $\{A_{j,H,W}\}$ in the sums above are almost disjoint, and thus the simple bound $\|A_{j,H,W}\|_{L^\infty(\mathbb{R}^d)} \lesssim H$ gives that

$$\sum_j \left\| \sum_{W \in \mathcal{W}_{H,l-j}} A_{j,H,W} \right\|_{L^\infty(\mathbb{R}^d)}^p |W| \lesssim H^p |\Omega_H|, \quad (3.4.14)$$

and thus that

$$\left\| \sum_{j,H} \sum_{W \in \mathcal{W}_{H,l-j}} T_{j,W,\text{Long}} A_{j,H,W} \right\|_{L^p(\mathbb{R}^d)} \lesssim 2^{-l\varepsilon} (\sum_H H^p |\Omega_H|)^{1/p} \lesssim 2^{-l\varepsilon} \|f\|_{L^p(\mathbb{R}^d)}. \quad (3.4.15)$$

Summing in l trivially using the triangle inequality and the geometric decay in l gives that

$$\left\| \sum_j \sum_H \sum_{W \in \mathcal{W}_H} T_{j,W,\text{Long}} A_{j,H,W} \right\|_{L^p(\mathbb{R}^d)} \lesssim \|f\|_{L^p(\mathbb{R}^d)}, \quad (3.4.16)$$

which controls the long-range interactions in full, completing the proof of Theorem 3.2.

3.5 Quasi-Radial Multipliers and Local Smoothing

Jongchon Kim [25] has extended the bounds of Heo-Nazarov-Seeger to quasi-radial multipliers, i.e. multipliers of the form $(a \circ r)(\xi)$, where $r : \dot{\mathbb{R}}^d \rightarrow [0, \infty)$ is a smooth, homogeneous function of order one, such that the co-sphere $S = \{\xi : r(\xi) = 1\}$ is a hypersurface with non-vanishing Gauss curvature. Such multipliers retain the translation and dilation symmetries of the family of radial Fourier multiplier operators, but are no longer *rotation-invariant*.

Using the same identities as for radial Fourier multiplier operators, one can verify that

$$\|a \circ r\|_{M^p(\mathbb{R}^d)} \gtrsim \sup_R \|k_R\|_{L^q(\mathbb{R}^d)}, \quad (3.5.1)$$

where k_R is the Fourier transform of $\chi(\cdot)(a \circ r)(R\cdot)$, for some fixed $\chi \in C_c^\infty(\mathbb{R})$. In a paper analyzing quasi-radial multipliers of Bochner-Riesz type, Sanghyuk Lee and Andreas Seeger [27] showed

$$\sup_R \|k_R\|_{L^p(\mathbb{R}^d)} \sim \|a\|_{R^{p,s}[0,\infty)}, \quad (3.5.2)$$

Thus

$$\|a \circ r\|_{M^p(\mathbb{R}^d)} \gtrsim \|a\|_{R^{p,s}[0,\infty)}. \quad (3.5.3)$$

Jongchon Kim [25] has showed the converse in the range $1 < p < 2(d-1)/(d+1)$, the same range as considered by Heo, Nazarov and Seeger in the last section.

Theorem 3.11. *Suppose $1 < p < 2(d-1)/(d+1)$, and that $r : \dot{\mathbb{R}}^d \rightarrow [0, \infty)$ is smooth and homogeneous of order one. Then for any regulated function a ,*

$$\|a \circ r\|_{M^{p,q}(\mathbb{R}^d)} \lesssim \|a\|_{R_d^{q,s}[0,\infty)}.$$

The proof is an adaption of the proof of [16]. However, it is more difficult to work

directly with convolution kernels k in this problem, since, unlike for radial multipliers, whose kernels are also radial, the kernels k corresponding to quasi-radial Fourier multiplier operators need not be quasi-radial. It is here that we introduce a technique that has proved essential to an analysis of multiplier operators in settings lacking the full symmetry properties of radial multipliers: a reduction of the study of multipliers to an analysis of wave equations. Using the Fourier inversion formula, we write

$$\begin{aligned}
 (a \circ r)f(x) &= \int a(r(\xi))e^{2\pi i\xi \cdot (x-y)} f(y) dy dx \\
 &= \iint \widehat{a}(t)e^{2\pi i[t r(\xi) + \xi \cdot (x-y)]} f(y) dy dx dt \\
 &= \int \widehat{a}(t)(w_{2\pi t} * f)(x) dt
 \end{aligned} \tag{3.5.4}$$

where $\widehat{w}_t(\xi) = e^{it r(\xi)}$. As t varies, $u = w_t * f$ solves the wave equation $\partial_t u = iPu$, where P is the Fourier multiplier operator whose symbol is the function r . Define

$$\chi_{x,t} = \text{Trans}_x \left(\int_{-\infty}^{\infty} \psi(s) w_{t-s} ds \right), \tag{3.5.5}$$

where the Fourier transform of ψ is non-negative and vanishing to high order at the origin. Then, using oscillatory integral techniques akin to Lemma 3.10, one can obtain inner product estimates on the quantities $\langle \chi_{x_1,t_1}, \chi_{x_2,t_2} \rangle$ that show such terms are negligible unless $|x_1 - x_2| + |t_1 - t_2| \lesssim 1$. We can then adapt the proof of [16], performing a density decomposition using the Euclidean metric on $\mathbb{R}^{d+1}$, and thus obtain single scale bounds for the quasi-radial multipliers. An atomic decomposition analogous to that discussed in Section 3.4 then yields Jongchon Kim's result.

We note that the endpoint analysis of radial multipliers we have been discussing is very closely related to the regularity of solutions to wave equations. In particular, if $\widehat{w}_t = e^{it|\xi|}$, then for any f , the function $u = w_t * f$ solves the half-wave equation $\partial_t u = iPu$ where

$P = \sqrt{-\Delta}$. The Littlewood-Paley pieces $(P_k u)(\cdot, t)$ are Fourier multipliers with symbol $\chi(\cdot/2^k)e^{it|\xi|}$, which are radial, and thus can be written in terms of spherical averages. The analysis of Proposition 3.4 can then be used to show that for $2(d-1)/(d-3) < q < \infty$,

$$\|P_k u\|_{L^q(\mathbb{R}^d \times [1,2])} \lesssim 2^{ks} \|f\|_{L^q(\mathbb{R}^d)} \quad \text{where } s = (d-1)(1/2 - 1/q) - 1/q. \quad (3.5.6)$$

With more work involving the atomic decomposition methods discussed in Section 3.4, Heo, Nazarov, and Seeger are able to combine the frequency scales, and thus prove the local smoothing estimate

$$\|u\|_{L^q(\mathbb{R}^d \times [1,2])} \lesssim \|f\|_{W^{s,q}(\mathbb{R}^d)}. \quad (3.5.7)$$

These are the current sharpest *endpoint* local smoothing bounds for the wave equation in any dimension. Local smoothing is thus closely connected to radial multiplier bounds. In Chapter 5 we reverse this connection, using local smoothing bounds to bound spectral multipliers.

To end our discussion of radial multipliers, notice that the Fourier multiplier operator P above is an elliptic operator on $\mathbb{R}^d$, satisfying the Assumption A we introduced in Chapter 2. For such an operator, the equation $\partial_t u = iPu$ is hyperbolic, and its solutions have wavelike properties. In the compact manifold setting, we will rely on a similar decomposition to reduce the study of spectral multipliers to a hyperbolic equation on the manifold. Together with a study of the geometry of the manifold upon which we study these multipliers, we are able to extend the methods we have discussed in the previous three sections to compact manifolds. We will now return to a further discussion of spectral multipliers on compact manifolds.

Chapter 4

Wave Equations on Compact Manifolds

4.1 An Analogue of the Radial Multiplier Conjecture on Compact Manifolds

We now return to the study of spectral multipliers on compact manifolds. In a heuristic sense, one can think of the Fourier multipliers studied in the last Chapter as a ‘high frequency’ limiting case of multipliers on manifolds, i.e. so that for any compact manifold X , and any operator $P \in \text{COP}_1^+(X)$ with principal symbol $p(x_0, \xi)$, locally around $x_0 \in X$ the operators P/R behave more and more like the Fourier multiplier operator on $\mathbb{R}^d$ with symbol $p(x_0, \cdot)$ as $R \rightarrow \infty$. In particular, the following transplantation result holds.

Theorem 4.1. *Suppose X is a d -dimensional compact manifold, and $P \in \text{COP}_1^+(X)$. For each $x_0 \in X$, if we choose a basis for $T_{x_0}^*X$, which identifies the vector space with $\mathbb{R}^d$, and let $p_{x_0} : \mathbb{R}^d \rightarrow [0, \infty)$ be the principal symbol of p restricted to $T_{x_0}^*X$, then for any regulated function a ,*

$$\|a \circ p_{x_0}\|_{M^p(\mathbb{R}^d)} \lesssim \|a\|_{M_{\text{Dil}}^p(X, P)}.$$

Proof. Mitjagin’s original proof can be found in [31], published in German. See [23] for

a version of the inequality in the English mathematics literature. $\square$

Thus if P satisfies Assumption A, it follows from Theorem 4.1 and (3.5.3) that

$$\|a\|_{M_{\text{Dil}}^p(X,P)} \gtrsim \|a\|_{R^{p,s}(\mathbb{R}^d)}, \quad (4.1.1)$$

with $s = (d-1)(1/p - 1/2)$.

We might conjecture that one can reverse (4.1.1) in the same range as for the radial multiplier conjecture, i.e. proving that $\|a\|_{M_{\text{Dil}}^p(X,P)} \lesssim \|a\|_{R^{p,s}(\mathbb{R}^d)}$ for $1 < p < 2d/(d+1)$. However, this is unlikely to be possible on a general manifold. Such a result would imply the Bochner-Riesz conjecture in the full range, i.e. that if $m^\delta(\lambda) = (1 - \lambda^2)^{\delta/2}$, then $\|m^\delta\|_{M_{\text{Dil}}^p(X,P)} < \infty$ for $\delta > d(1/p - 1/2) - 1/2$. But results of Shaoming Guo, Hong Wang, and Ruixiang Zhang [14] and Song Dai, Liuwei Gong, Shaoming Guo, and Ruixiang Zhang [8], with related work of Minicozzi and Sogge [30], prove that certain oscillatory integral operators associated with Bochner-Riesz operators on 3-dimensional Riemannian manifolds fail to be bounded appropriately when the geometry of X induced on P , which we introduce in the next section, does not have constant sectional curvature. These results do not give a direct contradiction, but provide weak evidence that the Bochner-Riesz conjecture may fail on a general manifold in the full range. On the other hand, it is likely that the conjecture holds in the full range on manifolds with constant sectional curvature; results of [1], which obtain the full range of the conjecture on S^d for *zonal inputs*, provide some evidence for the truth of the conjecture. Nonetheless, in this thesis we do describe positive results for this analogue of the radial multiplier conjecture, and our results can be applied to compact manifolds, such as zonal manifolds and the rank one symmetric spaces, which do not have constant sectional curvature.

4.2 Geometries Induced by Elliptic Operators

We plan to study spectral multipliers of an operator $P \in \text{COP}_1^+(X)$ on a compact manifold X via the use of solutions of the wave equation $\partial_t u = iPu$ on M , which allow us to exploit geometric information about the manifold M and thus extend the results of Heo, Nazarov, and Seeger to the setting of compact manifolds when Assumption A and Assumption B of Chapter 2 hold. But what is the geometry of the manifold we should be using? If the operator $P = \sqrt{-\Delta}$ is induced by a Riemannian metric on X , the geometric structure is clear, since X already has a Riemannian geometry. Given a more general operator P , we will use the principal symbol of P to give X a *Finsler geometry*. In this section we describe the bare essentials of Finsler geometry needed for the arguments which occur later on in the thesis. We mainly refer to a textbook on Finsler geometry written by David Dai-Wai Bao, Shiing-Shen Chern, and Zhongmin Shen [3] for a reference to further details.

Let X be a d -dimensional manifold. We denote an element of the tangent bundle TX by (x, v) , where $x \in X$ and $v \in T_x X$, and an element of the cotangent bundle T^*X by (x, ξ) , with $\xi \in T_x^* X$. A *Finsler metric* on X is a function $F : TX \rightarrow [0, \infty)$, which is homogeneous of order one and smooth on $TX - 0$, and such that for each $x \in X$, the function $F_x : T_x^* X \rightarrow [0, \infty)$ is a *strictly convex* norm, in the sense that for $(x, v) \in TX - 0$, the Hessian of F_x^2 in the v variable is positive-definite, i.e. in coordinates, the matrix with entries

$$g_{ij}(x, v) = \frac{1}{2} \frac{\partial^2 F^2}{\partial v^i \partial v^j}(x, v) \quad (4.2.1)$$

is positive definite. These coefficients give rise to a 2-tensor g on the manifold TX , which approximates the Finsler metric near v to second order.

Lemma 4.2 (The Fundamental Inequality for Finsler Metrics). *For any $v, w \in T_x X$,*

$$\sum g_{ij}(x, v) v^i w^j \leq F(x, v) F(x, w).$$

Proof. By Euler's homogeneous function theorem (see Appendix 7), we can write

$$\sum g_{ij}(x, v)v^i w^j = (1/2) \sum \partial_{v_i} F^2(x, v)w^i = F(x, v) \sum \partial_{v_i} F(x, v)w^i. \quad (4.2.2)$$

To complete the proof, we must thus show that $\sum \partial_{v_i} F(x, v)w^i \leq F(x, w)$. But this follows by the triangle inequality $F(x, v + tw) \leq F(x, v) + tF(x, w)$. $\square$

A Finsler metric gives each of the tangent spaces of a manifold the structure of a norm space. Such a structure naturally gives the dual space a dual norm $F_* : T^*X \rightarrow [0, \infty)$. The strict convexity of F gives rise to a way of a natural way of identifying TX and T^*X . In coordinates, we define a matrix with entries

$$g^{ij}(x, \xi) = \frac{1}{2} \frac{\partial^2 F_*^2}{\partial \xi_i \partial \xi_j}(x, \xi). \quad (4.2.3)$$

These coefficients define a 2-tensor g on T^*X , which we see is naturally dual to the 2-tensor g we defined on TX .

Lemma 4.3. *For a Finsler manifold X , define maps from TX to T^*X and T^*X to TX , defined in coordinates for $v \in T_x X$ and $\xi \in T_x^* X$ by setting*

$$v_i^b = \sum g_{ij}(x, v)v^j \quad \text{and} \quad \xi_\#^i = \sum g^{ij}(x, v)\xi_j.$$

*Then the maps $v \mapsto v^b$ and $\xi \mapsto \xi_\#$ are diffeomorphisms between $TX - 0$ to $T^*X - 0$, and for each $x \in X$ and each $v \in T_x X$, v^b is the unique co-vector ξ such that $\xi(v) = F(x, v)^2$ and $F^*(x, \xi) = F(x, v)$. Conversely for $\xi \in T_x^* X$, $\xi_\#$ is the unique unit vector v such that $\xi(v) = F^*(x, \xi)^2$ and $F(x, v) = F^*(x, \xi)$. If δ_a^b is the Kronecker delta function, then for any a, b, v , and ξ ,*

$$\delta_a^b = \sum g_{ai}(x, v)g^{bj}(x, v^b) = \sum g^{bi}(x, \xi)g_{ia}(x, \xi_\#).$$

Proof. For each $\xi \in T_x^*X$, there is a unique vector $v = f(\xi) \in T_xX$ such that $F(v) = F^*(x, \xi)$ and $\xi(v) = F(x, v)^2$. The existence of ξ follows by definition of the dual norm, and the uniqueness follows by strict convexity of the Finsler norm F . By the method of Lagrangian multipliers and the fundamental inequality, it must be true in coordinates that

$$\xi_i = \sum g_{ij}(x, f(\xi))f(\xi)^j, \quad (4.2.4)$$

i.e. it must be true that $\xi = f(\xi)^b$. Euler's homogeneous function theorem shows that the derivative of the function $(\xi, v) \mapsto \xi - v^b$ in the variable v is the matrix with entries $\{g_{ij}(x, v)\}$, which is invertible, and so it follows from the implicit function theorem that the function f is a diffeomorphism. Differentiating the equation $\xi = f(\xi)^b$ and applying Euler's homogeneous function theorem gives that for each a ,

$$\begin{aligned} \delta_a^b &= \sum \frac{\partial g_{ai}}{\partial v^j}(x, f(\xi)) \frac{\partial f^j}{\partial \xi_b}(\xi) f^i(\xi) + g_{ai}(x, f(\xi)) \frac{\partial f^i}{\partial \xi_b}(\xi) \\ &= \sum g_{ai}(x, f(\xi)) \frac{\partial f^i}{\partial \xi_b}(\xi). \end{aligned} \quad (4.2.5)$$

Differentiating the equation $F^*(x, \xi)^2 = F(x, f(\xi))^2$ in ξ_a and ξ_b , and using (4.2.5) to simplify, we obtain that if δ_{ab} is the Kronecker delta function, then

$$\begin{aligned} 2g^{ab}(x, \xi) &= \sum 2g_{ij}(x, f(\xi))(\partial_{\xi_a} f^i)(\xi)(\partial_{\xi_b} f^j)(\xi) + g_{ij}(x, f(\xi))f(\xi)^j(\partial_{\xi_a} \partial_{\xi_b} f^i)(\xi) \\ &= \sum 2g_{ij}(x, f(\xi))(\partial_{\xi_a} f^i)(\xi)(\partial_{\xi_b} f^j)(\xi) + \xi_i(\partial_{\xi_a \xi_b} f^i)(\xi) \\ &= 2 \frac{\partial f^a}{\partial \xi_b} + 0. \end{aligned} \quad (4.2.6)$$

Since f is homogeneous of order one, this is only possible if $f(\xi)^a = \sum g^{ab}(x, \xi)\xi_b$, i.e. if $f(\xi) = \xi_\#$. Substituting this into (4.2.6) gives that

$$\delta_a^b = \sum g_{ai}(x, \xi_\#)g^{ib}(x, \xi). \quad (4.2.7)$$

The remaining part of the argument, that for each $v \in T_x X$, $v^\flat$ is the unique co-vector ξ such that $\xi(v) = F(x, v)^2$ and $F^*(x, \xi) = F(x, v)$, follows by Euler's homogeneous function theorem and the fundamental inequality. $\square$

The map defined in Lemma 4.3 is a Finsler variant of the musical isomorphism in Riemannian geometry, though the musical isomorphism of Riemannian geometry is linear rather than just homogeneous, because the metric $g(x, v)$ on X depends on the direction v rather than just the point x . The coordinate change is closely related to the *Legendre transform*. Namely, the Legendre transform of $F^2/2$ can be calculated as the function on T^*X given by

$$\sup_{v \in TX} \left\{ \xi(v) - \frac{F(v)^2}{2} \right\} = \frac{F^*(\xi)^2}{2}. \quad (4.2.8)$$

This fact will reappear in the study of geodesics. Since the Legendre transform preserves strict convexity, it follows that F^* is strictly convex.

Now let $P \in \text{COP}_1^+(X)$ satisfy Assumption A. The principal symbol of P is a function $p : T^*M \rightarrow [0, \infty)$, and the following lemma applies, which shows that the principal symbol of any element of $\text{COP}_1^+(X)$ gives X a Finsler geometry.

Lemma 4.4. *Suppose $p : T^*M \rightarrow [0, \infty)$ is homogeneous of order one on $T^*M - 0$, and for each $x \in M$, the co-sphere $S_x^* = \{\xi \in T_x^*M : p(x, \xi) = 1\}$ has non-vanishing Gaussian curvature. Then*

$$F(x, v) = \{\xi(v) : p(x, \xi) = 1\}$$

is a Finsler metric on M , and the dual metric F_ on T^*M is equal to p .*

Proof. For each $x \in U_0$, the co-sphere $S_x^* = \{\xi \in T_x^*M : p(x, \xi) = 1\}$ has non-vanishing Gaussian curvature. We claim that all principal curvatures of S_x^* must actually be *positive*. This follows from a simple modification of an argument found in Chapter 2 of [17]. Indeed, if we fix an arbitrary point $v_0 \in T_x^*M$, and consider the smallest closed ball $B \subset T_x^*M$

centered at v_0 and containing S_x^* , then the sphere ∂B must share the same tangent plane as S_x^* at some point. All principal curvatures of ∂B are positive, and at this point all principal curvatures of S_x^* must be greater than the principal curvatures of ∂B , since S_x^* curves away faster than ∂B in all directions. By continuity, we conclude that the principal curvatures are everywhere positive. Thus for each $x \in M$ and $\xi \in T_x^*M - \{0\}$, the coefficients

$$g^{ij}(x, \xi) = (1/2)(\partial^2 p^2 / \partial \xi_i \partial \xi_j) \quad (4.2.9)$$

form a positive-definite matrix. But inverting the procedure of the previous lemma shows that the dual norm $F(x, v) = \sup_{\xi \in S_x^*} \xi(v)$ is also strictly convex, and thus gives a Finsler metric on X with $F_* = p$. $\square$

A Finsler metric gives a length to each tangent vector on the manifold, and can thus be used to define the lengths of curves $c : I \rightarrow U_0$ by the formula

$$L(c) = \int_I F(c, \dot{c}), \quad (4.2.10)$$

which is invariant under re-parameterization. An analysis of length minimizing curves naturally leads to a theory of geodesics on a Finsler manifold, i.e. to a theory of critical points in the space of paths between two points. The theory of geodesics on Finsler manifolds is similar to the Riemannian case, except for the interesting quirk that a geodesic from a point p to a point q need not necessarily be a geodesic when considered as a curve from q to p . We must therefore consider both *forward* and *backward* geodesics*. We define the *forward distance* $d_+ : X \times X \rightarrow [0, \infty)$ by taking the infima of paths between

*If walking up a hill is more strenuous than walking down a hill, then the optimal path to climb from the bottom of a mountain to its summit need not be the optimal path to descend from the peak to the bottom. Indeed, Makoto Matsumoto obtained a mathematical model for such a situation by constructing a Finsler metric upon the surface of the mountain such that geodesics correspond to optimal paths of ascent and descent [29]. Similar Finsler manifolds can be constructed to model optimal paths involving sailing under the influence of varying wind conditions, the theory being known as Zermelo's navigation problem.

points. This function is a *quasi-metric*, as it satisfies the triangle inequality, but is not necessarily symmetric. We define the *backward distance* $d_- : X \times X \rightarrow [0, \infty)$ by setting $d_-(p, q) = d_+(q, p)$. On a general Finsler manifold one has $d_+ \neq d_-$, and these functions are distinct quasi-metrics on X , though a compactness argument shows both are proportional to one another on compact manifolds. Geodesics from a point p_0 to a point p_1 need not be geodesics from p_1 to p_0 when reversed. A metric can be obtained by setting $d_X = d_X^+ + d_X^-$, and we will use this definition as the canonical metric on a Finsler manifold X in the rest of this thesis.

A standard approach to an analysis of geodesics in Riemannian geometry is to rely on the *fundamental theorem of Riemannian geometry*, which posits the existence of a torsion-free linear connection on the tangent bundle of a Riemannian manifold X compatible with the metric, and using the connection to derive a first variation formula. A similar approach works for Finsler geometry, aside from the fact that (a) the connection is not defined on the tangent bundle and (b) the analogue of the fundamental theorem in general *fails*. Because of (b), there is no canonical connection, used in the Finsler geometry literature, and depending on the application and personal preference one of several can be used, the most popular being the Cartan, Berwald, and Chern connections, the first being metric-compatible, but which can have torsion, and the latter two being torsion free, but which are almost metric-compatible. We use the Chern connection in what follows.

We will need to review the basic notions of vector bundles. Recall that a n -dimensional *vector bundle* B over a manifold X is a manifold equipped with a projection map $\pi : B \rightarrow X$ and, for each $x \in X$, a vector space structure on the fiber $B_x = \pi^{-1}(x)$ which makes B_x an n -dimensional vector space, such that B is *locally trivial*, i.e. there exists a cover of X by sets U and for each U , a diffeomorphism between $\pi^{-1}(U)$ and $\mathbb{R}^n \times U$ such that for each $x \in X$, the diffeomorphism restricts to a map between U_x and $\mathbb{R}^n \times \{x\}$ which is a linear. The archetypal example of a vector bundle on a manifold X is the tangent bundle

TX . Given a smooth map $f : X \rightarrow Y$ between two manifolds, and a vector bundle B over Y , one can define the *pullback bundle* f^*B over X , such that $(f^*B)_x$ is identified with $B_{f(x)}$, put together in a topologically compatible way*. For a vector bundle B over a manifold X , we let $\Gamma(B)$ denote the vector space of all *sections* of B , i.e. smooth maps $s : X \rightarrow B$ such that $s(x) \in B_x$ for each x . If B is a vector bundle over X , and B' is a submanifold of B which is also a vector bundle, then B' is called a *sub-bundle*

We now define the Chern connection. A *linear connection* on B is an association, with each $s \in \Gamma(B)$ and $v \in T_x X$, of a vector $\nabla_v(s) \in B_x$, which is linear in v and s , and such that for each vector field $V \in \Gamma(X)$ and each section $s \in \Gamma(B)$, $\nabla_V(s) \in \Gamma(B)$. The Chern connection is defined on the vector bundle $\pi^*(TX)$, a vector bundle over TX , where $\pi : TX \rightarrow X$ is the projection map to the base of a tangent vector. The bundle $\pi^*(TX)$ can be viewed as a sub-bundle of $T(TX)$, and so $\Gamma(\pi^*(TX))$ can be viewed as a subspace of $\Gamma(T(TX))$. This bundle has a distinguished section $l \in \Gamma(\pi^*(TX))$, given in coordinates by $l(x, v)_i = v_i$. Define the Cartan tensor A , a symmetric 3-tensor on TX given in coordinates by

$$A_{ijk} = \frac{F}{2} \frac{\partial^3 F^2}{\partial v^i \partial v^j \partial v^k}. \quad (4.2.11)$$

Note that if X is a Riemannian manifold, and $F(x, v) = \langle v, v \rangle$ is the Finsler metric induced by the Riemannian metric, then the Cartan tensor A vanishes identically, but on a general Finsler manifold this need not be the case.

Proposition 4.5. *Let X is a Finsler manifold, and consider the projection map $\pi : TX \rightarrow X$. Then there exists a unique linear connection ∇ on the pullback bundle $\pi^*(TM)$, viewed as a bundle over TM , which is torsion-free, in the sense that for any sections $V, w \in$*

*Formally, one can define f^*B as the submanifold $\{(x, v) : f(x) = \pi(v)\}$ of $X \times B$, where $\pi : B \rightarrow Y$ is the base projection map of B .

$\Gamma(\pi^*(TM))$, identified with vector fields on TM ,

$$\nabla_V W - \nabla_W V = [V, W],$$

and almost metric compatible, in the sense that for $V, W, U \in \Gamma(\pi^*(TM))$,

$$V(g(W, U)) = g(\nabla_V W, U) + g(W, \nabla_V U) + A(\nabla_V U, W, U).$$

Proof. See Theorem 2.4.1 of [3]. □

Now we derive a first variation formula for the energy of a curve.

Proposition 4.6. *Consider a variation $c : (-\varepsilon, \varepsilon) \times I \rightarrow X$ on a Finsler manifold, not necessarily fixed at the endpoints, where $I = [a, b]$ is an interval. Define $T = \partial_t c$ be the tangent field, and $U = \partial_s c$ the variation field. Define*

$$E(s) = \frac{1}{2} \int_I F(c, \dot{c})^2 = \frac{1}{2} \int_I g_T(T, T).$$

Then

$$E'(s) = g_T(U, T)|_a^b - \int_I g_T(U, \nabla_T T).$$

Proof. Almost metric compatibility gives that

$$\partial_s \{g_T(T, T)\} = U\{g_T(T, T)\} = 2g_T(\nabla_U T, T), \quad (4.2.12)$$

and since $[U, T] = 0$, torsion free and almost metric compatibility implies that

$$g_T(\nabla_U T, T) = g_T(\nabla_T U, T) = \partial_t \{g_T(U, T)\} - g_T(U, \nabla_T T). \quad (4.2.13)$$

The Cartan tensor does not appear here, since $A_T(U, V, W) = 0$ if $T \in \{U, V, W\}$, by the

Euler homogeneous function theorem. Differentiating under the integral sign, we find that

$$E'(s) = \frac{1}{2} \int_I \partial_s \{g_T(T, T)\} = g_T(U, T)|_a^b - \int_I g_T(U, \nabla_T T), \quad (4.2.14)$$

which finishes the proof. $\square$

A differential formula for geodesics immediately follows.

Corollary 4.7. *A constant speed curve $c : I \rightarrow X$ is a geodesic (i.e. a critical point of any length variation fixing endpoints) if and only if it satisfies the equation $\nabla_T T = 0$, where $T = \dot{c}$. Alternatively, if we assume that $F(c, \dot{c}) = 1$, and we set $x(t) = c(t)$ and $\xi(t) = \dot{c}(t)^b$, then c is a geodesic if and only if it satisfies Hamilton's equations for motion*

$$\dot{x} = -(\partial_\xi F_*)(x, \xi) \quad \text{and} \quad \dot{\xi} = (\partial_x F_*)(x, \xi).$$

Proof. The necessity and sufficiency of $\nabla_T T = 0$ follows from the first variation formula. To obtain the second equation, we apply the theory of Euler-Lagrange equations and Hamiltonian mechanics, as described in Sections 14 and 15 of [2], which we briefly describe. Consider a functional E on curves $c : I \rightarrow X$, of the form

$$E(c) = \int_I L(c, \dot{c}) dt, \quad (4.2.15)$$

for some convex function $L : TX \rightarrow \mathbb{R}$, where $(c, \dot{c})$ is the tangent vector of c . Given c , consider a curve in T^*X defined by $p(t) = (d_v L)(c, \dot{c})$. If we let $H : T^* \rightarrow \mathbb{R}$ denote the Legendre transform of the function L , then c is an extremizer of E if and only if $(c(t), p(t))$ is an integral curve of the Hamiltonian vector field $(-\partial_\xi H, \partial_x H)$ on T^*X , and the quantity H is conserved along this integral curve. The Euler-Lagrange equation for the energy

functional

$$E(c) = \frac{1}{2} \int_I F(c, \dot{c})^2 \quad (4.2.16)$$

has Lagrangian $L(x, v) = F(x, v)^2/2$, and we find that if $p(t) = (d_v F^2)(c(t), \dot{c}(t))$, then by Euler's Homogeneous function theorem, we find that $p = \dot{c}^\flat$, i.e. because

$$p_i = \frac{\partial(F^2/2)}{\partial v^i}(c(t), \dot{c}(t)) = \sum_j g_{ij}(c(t), \dot{c}(t)) \dot{c}(t)^j. \quad (4.2.17)$$

We calculated in (4.2.8) that the Legendre transform of L is equal to $H(x, \xi) = F^*(x, \xi)^2/2$, and the corresponding Hamiltonian vector field is thus

$$(-F^*(x, \xi) \partial_\xi F^*(x, \xi), F^*(x, \xi) \partial_x F^*(x, \xi)). \quad (4.2.18)$$

The curve c is a geodesic if and only if the curve $(x(t), \xi(t))$ is an integral curve to this vector field. We see that $F^*(x, \xi) = F(c, \dot{c}) = 1$, and so the vector field along $(x(t), \xi(t))$ simplifies to

$$(-\partial_\xi F^*(x, \xi), \partial_x F^*(x, \xi)), \quad (4.2.19)$$

from which the Hamiltonian equations for x and ξ immediately follow. $\square$

Remark. The Hamiltonian equations derived in this corollary will reoccur in our study of wave propagation, and will tell us that high frequency wave packet solutions to the equation $\partial_t u = iP$ travel along geodesics of the Finsler metric.

Just as in the Riemannian case, one can also obtain a second variation formula involving a theory of Jacobi fields along geodesics, but this takes us too far afield. Since we will not use the second variation formula in what follows, we simply assume one of its consequences, that geodesics are length minimizing up until the development of cut points on a manifold, which is at least as big as the *injectivity radius* of the manifold,

i.e. the minimum time at which the geodesic flow defined by the Hamiltonian equations above fails to be injective from a point on the manifold*.

4.3 Fourier Integral Operator Techniques

For any (possibly unbounded) self-adjoint operator P on a Hilbert space H , we can use the spectral theorem for such operators to define a bounded operator $a(P)$ on H for each bounded function $a : \sigma(P) \rightarrow \mathbb{C}$. If $a : \mathbb{R} \rightarrow \mathbb{C}$ is in the Wiener algebra $A(\mathbb{R})$, then the Fourier inversion formula

$$a(P) = \int_{-\infty}^{\infty} \widehat{a}(t) e^{2\pi i t P} dt \quad (4.3.1)$$

holds, in a certain sense made precise in the lemma below.

Lemma 4.8. *Suppose the Fourier transform of a bounded, continuous function $a : \mathbb{R} \rightarrow \mathbb{C}$ is integrable. Let H be a Hilbert space, and let P be a self-adjoint operator on H , so we may use spectral theory to define $a(P)$. Then for any $x, y \in H$,*

$$\langle a(P)x, y \rangle = \int_{-\infty}^{\infty} \widehat{a}(t) \langle e^{2\pi i t P} x, y \rangle dt,$$

where the right hand side converges absolutely because $e^{2\pi i t P}$ is a unitary operator.

Proof. We recall that the operators $a(P)$ and $e^{2\pi i t P}$ are defined in terms of a spectral resolution[†] E on $\mathbb{R}$ such that for any $x, y \in H$,

$$\langle Px, y \rangle = \int_{\sigma(P)} \lambda dE_{x,y}(\lambda). \quad (4.3.2)$$

*See Proposition 8.2.1 of [3] for a proof that such geodesics are length minimizing.

[†]See Theorem 13.33 of [34] for a proof of the spectral decomposition for unbounded operators.

We then define a bounded operator $a(P)$ such that for $x, y \in H$,

$$\langle a(P)x, y \rangle = \int_{\sigma(P)} a(\lambda) dE_{x,y}(\lambda). \quad (4.3.3)$$

We may apply the Fourier inversion formula to write

$$\int_{\sigma(P)} a(\lambda) dE_{x,y}(\lambda) = \int_{\sigma(P)} \left(\int_{-\infty}^{\infty} \widehat{a}(t) e^{2\pi i t \lambda} dt \right) dE_{x,y}(\lambda). \quad (4.3.4)$$

Provided that $\widehat{a} \in L^1(\mathbb{R})$, the double integral is absolutely integrable, and so we may apply Fubini's theorem to interchange the order of integration, which gives the required result. $\square$

Thus spectral multipliers of a self-adjoint operator P can be written as averages of the semigroup $\{e^{2\pi i t P}\}$ under very weak assumptions on the operator. The catch is that for a general operator P , it is difficult to understand the behavior of the operator e^{itP} given its abstract definition via spectral theory. Nonetheless, in this chapter we will see that if $P \in \text{COP}_1^+(X)$, then e^{itP} is a *Fourier integral operator*, which roughly means that the kernel of the operator is well approximated by oscillatory integral representation *for high frequency inputs*, which is something we can exploit to obtain both geometric and frequential information about the operator. In this section, we briefly describe the relevant components we will need from the theory of Fourier integral operators for the proofs in the latter part of the thesis.

We will require methods of decomposition of functions in space and frequency simultaneously (a decomposition in *phase space*). A *conic subset* of $\mathbb{R}^d$ is a set Γ such that for each $\rho > 0$, $\rho \cdot \Gamma \subset \Gamma$. Let Ω be a bounded, open subset of $\mathbb{R}^d$. Temporarily define a *micro-localization operator* on a *conic subset* $\Gamma \subset \mathbb{R}^d \times \mathbb{R}^d$ to be an operator of the form $\Psi_\Gamma f = \psi \cdot \alpha(D)\{f\}$, where $\psi \in C_c^\infty(\mathbb{R}^d)$ and $\alpha(D)$ is the Fourier multiplier operator whose

symbol $\alpha : \dot{\mathbb{R}}^d \rightarrow \mathbb{C}$ is smooth and homogeneous of order zero, and $\text{supp}(\psi) \times \text{supp}(\alpha) \subset \Gamma$.

We will say such a multiplier is *microlocally supported* on $\text{supp}(\psi) \times \text{supp}(\phi)$.

To motivate the definition of Fourier integral operators, let Ω_X and Ω_Y be open subsets of $\mathbb{R}^n$ and $\mathbb{R}^m$, and consider a Schwartz operator A from Ω_Y to Ω_X , whose kernel is an oscillatory integral defined by an expression

$$K_A(x, y) = \int_{\mathbb{R}^p} s(x, y, \theta) e^{i\Phi(x, y, \theta)} d\theta, \quad (4.3.5)$$

where $s : \Omega_X \times \Omega_Y \times \dot{\mathbb{R}}^p \rightarrow \mathbb{C}$ is a symbol of some fixed order, and $\Phi : \Omega_X \times \Omega_Y \times \dot{\mathbb{R}}^p \rightarrow \mathbb{R}$ is a *phase function* in the sense of Appendix 8. Define the *canonical relation* of A , the set

$$C_\Phi = \{(x, \nabla_x \Phi(x, y, \theta), y, -\nabla_y \Phi(x, y, \theta)) : \nabla_\theta \Phi(x, y, \theta) = 0\}, \quad (4.3.6)$$

Under the assumption that the vectors $\partial_{\theta_1} \{\nabla_{x, y, \theta} \Phi\}, \dots, \partial_{\theta_N} \{\nabla_{x, y, \theta} \Phi\}$ are everywhere linearly independent when $\nabla_\theta \Phi = 0$, C_Φ is a smooth, $n + m$ dimensional manifold*. Such a phase Φ is called *non degenerate*, and we will only deal with such phases in the sequel.

If $(x_0, \xi_0, y_0, \eta_0) \notin C_\Phi$, then, integrating by parts, one can show there exists conic sets $\Gamma_X \subset \Omega \times \dot{\mathbb{R}}^d$ and $\Gamma_Y \subset \Omega \times \dot{\mathbb{R}}^d$ such that for any micro-localization operators Ψ_X and Ψ_Y which micro-localize on to Ψ_X and Ψ_Y respectively, the kernel of the operator $\Gamma_X \circ A \circ \Gamma_Y$ is a smooth function, and so the behavior of the operator on rapidly oscillating ‘high-frequency’ inputs is negligible. The canonical relation of an oscillatory integral operator thus tells us pairs of points in phase space the location where the majority of the support of high-frequency wave packets are mapped to in phase space by the operator A . The main lesson of the theory of Fourier integral operators is that, to a large extent, the canonical

*It is actually a *Lagrangian submanifold* of $T^*\mathbb{R}^n \times T^*\mathbb{R}^m$, i.e. for any $p \in C_\Phi$ and any tangent vectors $v, w \in T_p C_\Phi$, $d\theta(v, w) = 0$, where θ is the *canonical one form* on $T^*(\mathbb{R}^m \times \mathbb{R}^n)$ (see Proposition 0.5.4 of [42], the form also being called the *tautological one form* or *Liouville one form*). Thus the canonical relation of any Fourier integral operator must be a Lagrangian submanifold, though we will only use this fact implicitly.

relation *determines* the behavior of the operator A .

It is natural to view the canonical relation C_Φ associated with a phase Φ as a submanifold of $T^*\Omega_Y \times T^*\Omega_X$, which is the right way to view C_Φ in a ‘coordinate invariant’ way. Indeed, C_Φ is in direct correspondence to all possible points of the *wavefront sets** of the kernels of operators definable by the phase Φ , and the wave-front set of the kernel of an operator from a manifold Y to a manifold X is a subset of $T^*Y \times T^*X$.

A Fourier integral operator is precisely an operator whose kernel is microlocally expressible in oscillatory integrals of the form above. A *Fourier integral operator* of order μ from $\Omega_Y \subset \mathbb{R}^m$ to $\Omega_X \subset \mathbb{R}^n$ associated with a $n + m$ dimensional ‘Lagrangian’ submanifold C of $T^*W_X \times T^*W_Y$ is an operator A for which there exists covers of T^*W_X and T^*W_Y by open conic sets $\{\Gamma_X\}$ and $\{\Gamma_Y\}$, such that for each such pair Γ_X and Γ_Y , and any micro-localization operators Ψ_X and Ψ_Y which micro-localize on to Ψ_X and Ψ_Y , the kernel of the operator $\Psi_X \circ A \circ \Psi_Y$ is defined by an oscillatory integral of the kind defined above, with a symbol of order $\mu + p/2 - (n + m)/4$, and with a phase whose canonical relation is contained in the set C . We denote the set of all Fourier integral operators of this kind by $I^\mu(\Omega_X \times \Omega_Y, C)$.

A Fourier integral operator from a manifold Y to a manifold X is precisely an operator which a Fourier integral operator ‘in coordinates’. To state the theorem, we must define what it means to view a Fourier integral operator ‘in coordinates’. Let A be a Schwartz operator from a manifold X to a manifold Y , and consider open sets $U_X \subset X$ and $U_Y \subset Y$, together with diffeomorphisms $F^X : U_X \rightarrow \Omega_X$ and $F^Y : U_Y \rightarrow \Omega_Y$. We define an operator A^{Ω_X, Ω_Y} from Ω_Y to Ω_X by setting

$$A^{\Omega_X, \Omega_Y} f = A\{f \circ F_Y\} \circ F_X^{-1}. \quad (4.3.7)$$

*See Chapter 8 of [20] for a definition and discussion of the wavefront set of a distribution. In particular, Theorem 8.1.9 of that chapter gives the relation between the canonical relation of a phase function and the wavefront set of the distributions it can define via oscillatory integrals.

If $X = Y$ and $\Omega_X = \Omega_Y = \Omega$, then we will also denote A^{Ω_X, Ω_Y} by A^Ω .

A Fourier integral operator from an m -dimensional manifold Y to an n -dimensional manifold X of order μ , associated with an $n + m$ dimensional Lagrangian submanifold C of $T^*X \times T^*Y$ is precisely an operator for which there exists covers $\{U_X\}$ and $\{U_Y\}$ of X and Y , with U_X diffeomorphic to $\Omega_X \subset \mathbb{R}^n$ and U_Y to $\Omega_Y \subset \mathbb{R}^m$ by diffeomorphisms F^X and F^Y , such that for any such pair Ω_X and Ω_Y , the operator $A^{\Omega_X, \Omega_Y} \in I^\mu(\Omega_X \times \Omega_Y, C^{\Omega_X, \Omega_Y})$, where

$$C^{\Omega_X, \Omega_Y} = \{(x, \xi; y, \eta) : (x, F_*^X(x)^T \xi, y, F_*^Y(y)^T \eta) \in C\}. \quad (4.3.8)$$

We denote the family of Fourier integral operators of order μ associated with a given manifold C by $I^\mu(X \times Y, C)$.

The *equivalence of phase theorem* for Fourier integral operators tells us that the particular phase used to define Fourier integral operators is largely irrelevant to the analysis of the operator, as long as the canonical relation is shared by the phase.

Theorem 4.9. *Suppose $A \in I^\mu(X^n \times Y^m, C)$, and consider $U_X \subset X$ and $U_Y \subset Y$ diffeomorphic to $\Omega_X \subset \mathbb{R}^n$ and $\Omega_Y \subset \mathbb{R}^m$ respectively. Then $A^{\Omega_X, \Omega_Y} \in I^\mu(\Omega_X \times \Omega_Y, C^{\Omega_X, \Omega_Y})$. Moreover, suppose $\Gamma_X \subset T^*\Omega_X - 0$ and $\Gamma_Y \subset T^*\Omega_Y - 0$ are conic sets, and that for some $p \geq 0$ there exists a non-degenerate phase function $\Phi : \Omega_X \times \Omega_Y \times \dot{\mathbb{R}}^p \rightarrow \mathbb{R}$ with $C_\Phi \cap (\Gamma_X \times \Gamma_Y) = C^{\Omega_X, \Omega_Y} \cap (\Gamma_X \times \Gamma_Y)$. Then for any micro-localization operators Ψ_X and Ψ_Y which micro-localize onto Γ_X and Γ_Y respectively, there exists a smooth symbol $s : \mathbb{R}^n \times \mathbb{R}^m \times \dot{\mathbb{R}}^p \rightarrow \mathbb{C}$ of order $\mu + p/2 - (n+m)/4$, and a smooth function $K^\psi \in C^\infty(\mathbb{R}^n \times \mathbb{R}^m)$, such that the kernel K of $\Psi_X \circ A^{\Omega_X, \Omega_Y} \circ \Psi_Y$ is given by*

$$K(x, y) = \int_{\dot{\mathbb{R}}^p} s(x, y, \theta) e^{i\Phi(x, y, \theta)} d\theta + K^\psi(x, y).$$

Proof. See Theorem 25.1.5 of [22]. □

The simplest class of Fourier integral operators are the *pseudo-differential operators* of order μ on a manifold X . They are precisely the elements of $I^\mu(X \times X, \Delta_X)$, whose canonical relation is the diagonal $\Delta_X = \{(x, \xi, x, \xi) : (x, \xi) \in T^*X\}$. In any coordinate system, and for any diffeomorphism $\phi : \mathbb{R}^d \rightarrow \mathbb{R}^d$, $\Phi(x, y, \xi) = \xi \cdot (\phi(x) - \phi(y))$ parameterizes Δ_X , and so since $p = d$, $n = d$, and $m = d$, we conclude from Theorem 4.9 that for any pseudo-differential operator T of order μ on $\mathbb{R}^d$, there exists a symbol s of order μ , and $K_T^\psi \in C^\infty(\mathbb{R}^d \times \mathbb{R}^d)$, so that

$$K_T(x, x') = \int s(x, x', \xi) e^{i\xi \cdot (\phi(x) - \phi(x'))} d\xi + K_T^\psi(x, x'), \quad (4.3.9)$$

where s is a symbol of order μ .

For our purposes, the most important Fourier integral operators are associated with solutions to wave equations on manifolds.

Theorem 4.10. *Fix $P \in COP_1^+(X)$ satisfying Assumption A, and let $p : T^*X \rightarrow \mathbb{R}$ denote its principal symbol. Let $\{\alpha_t\}$ be the co-geodesic flow on T^*X given by the Finsler metric on X induced by p . Define the canonical relation $C \subset T^*(X \times \mathbb{R}) \times T^*X$ by setting*

$$C = \{(x, t, \xi, \tau; x', \xi') : (x, \xi) = \alpha_t(x', \xi') \text{ and } \tau = p(x', \xi')\}.$$

and define

$$C_t = \{(x, \xi; x', \xi') : (x, \xi) = \alpha_t(x', \xi')\}.$$

Then $e^{itP} \in I^0(X \times X, C_t)$, and if we define the solution operator $Af(x, t) = e^{itP}f(x)$ from X to $X \times \mathbb{R}$, which solves the wave equation $\partial_t = iP$, then $A \in I^{1/4}((X \times \mathbb{R}) \times X, C)$.

Proof. It suffices to construct a parametrix for the half-wave equation given by oscillatory integrals that fit into the canonical relations above. One can find such parametrices in many sources, such as Theorem 29.1.1 of [22] or Theorem 4.1.2 of [42]. $\square$

Given this theorem, the equivalence of phase theorem implies multiple useful oscillatory integral representations for the wave propagators e^{itP} .

- Consider an open subset U of X diffeomorphic to $\Omega \subset \mathbb{R}^d$. Let $p : T^*\Omega - 0 \rightarrow (0, \infty)$ be the principal symbol of P in the coordinate system Ω . Consider a smooth hypersurface Σ of $T^*\Omega \times \Omega$, such that for each $(x, \xi) \in T^*\Omega$, the slice $\Sigma_{x, \xi} = \{x' : (x, \xi, x') \in \Sigma\}$ is a smooth hypersurface in X passing through (x, ξ) co-normal to ξ . The theory of Hamilton-Jacobi equations implies the local existence of a function ϕ , homogeneous of order one, satisfying the *Eikonal equation*

$$p(x, \nabla_x \phi(x, x', \xi)) = p(x', \xi) \quad (4.3.10)$$

such that for $x' \in \Sigma_{x, \xi}$, $\nabla_x \phi(x, x', \xi)$ is normal to $\Sigma_{x, \xi}$ and $\phi(x, x', \xi) = 0$. If we define $\Phi(x, t, x', \xi) = \phi(x, x', \xi) + tp(x, \xi)$, then $C_\Phi = C^\Omega$, where C is the canonical relation of the solution operator A , and so there exists $\varepsilon > 0$ and a symbol s of order zero and $K^\psi \in C^\infty(\mathbb{R}^{d+1} \times \mathbb{R}^d)$, such that for $|t| \leq \varepsilon$ the kernel of the operator A^Ω is given by

$$K(x, t, x') = \int s(x, t, x', \xi) e^{i[\phi(x, x', \xi) + tp(x, \xi)]} d\xi + K^\psi(x, t, x'). \quad (4.3.11)$$

See Chapter 4 of [42] for further details about the construction of this parametrix.

- Suppose that $t \in \mathbb{R}$, and $|t|$ is smaller than the injectivity radius on the manifold X with respect to its Finsler metric. Consider an open subset U of X diffeomorphic to $\Omega \subset \mathbb{R}^d$. Then the phase

$$\Phi(x, x', t, \tau) = \tau(|t| - d_+(x', x)) \quad (4.3.12)$$

for $\tau > 0$ satisfies $C_\Phi = C_t^\Omega$ for $t \neq 0$, and so there exists a symbol s of order $\frac{d-1}{2}$,

and $K_\psi \in C^\infty(\mathbb{R}^{d+1} \times \mathbb{R}^d)$ such that the kernel of the operator $(e^{itP})^\Omega$ is given by

$$K(x, t, x') = \int_0^\infty s(x, t, x', \tau) e^{i\tau(|t| - d_+(x', x))} d\tau + K^\psi(x, t, x'). \quad (4.3.13)$$

The problem with this oscillatory integral representation is that it begins to break down as $t \rightarrow 0$, and as t approaches times near the injectivity radius of the manifold.

- Because of the semigroup property $C_t \circ C_s = C_{t+s}$, we can obtain oscillatory integral representations of e^{itP} by composing the oscillatory integral representations we have given above for A . We do not carry out the details, because as $t \rightarrow \infty$ the number of compositions required grows linearly, which means the oscillatory integrals are harder and harder to control in a uniform way as $t \rightarrow \infty$, and we will not use such compositions in our arguments.

We conclude this section with some technical calculations, that result in some useful estimates for certain Fourier integral operators that will arise later in the thesis. Our first calculation concerns ‘Littlewood-Paley projections’ on a manifold with respect to an eigenfunction decomposition of an elliptic operator. Recall that for two sets U and V , $U \Subset V$ means the closure of U is a compact subset of V .

Theorem 4.11. *Let X be a compact manifold, and consider any $P \in COP_1^+(X)$. Consider an open subset V of X diffeomorphic to a bounded subset Ω of $\mathbb{R}^d$, and an open set $U \Subset V$. Fix $q \in C_c^\infty(0, \infty)$ and $R > 0$, and define $Q = q(P/R)$. Then we can find a pseudo-differential operator Q_U such that for any function u with $\text{supp}(u) \subset U$,*

$$\text{supp}(Q_U u) \subset V$$

and

$$\|Qu - Q_U u\|_{C^M(X)} \lesssim_{N,M} R^{-N} \|u\|_{L^1(X)} \quad \text{for all } N, M \geq 0,$$

and such that the coordinatized operator Q_U^Ω is a pseudo-differential operator whose symbol is supported on $\{(x, \xi) : R/4 \leq \xi \leq 4R\}$.

Proof. We proceed with a similar approach to Theorem 4.3.1 of [42]. It will be helpful to introduce a basis $\{e_k\}$ for $L^2(X)$, with $e_k \in C^\infty(X)$ for each k and $Pe_k = \lambda_k e_k$. Using Theorem 4.10, there exists $\varepsilon > 0$ such that for any function u supported on U , for $|t| < \varepsilon$ and u supported on U , we can write

$$e^{2\pi i t P} u = W(t)u + W_\psi(t)u,$$

where the kernel $K_t(x, y)$ of $W_\psi(t)^{\Omega_1, \Omega_2}$ is a smooth function of t , x , and y for any pair of coordinate systems Ω_1 and Ω_2 on X , where $\text{supp}(W(t)u) \subset V$ for any input u , and the kernel K of W^Ω is given by

$$K(t, x, y) = \int s_0(t, x, y, \xi) e^{2\pi i [\phi(x, y, \xi) + t p(y, \xi)]}$$

for a order zero symbol s_0 with $\text{supp}_{x, y}(s_0) \Subset \Omega$. We fix $\rho \in C_c^\infty(\mathbb{R})$ equal to one in a neighborhood of the origin and with $\rho(t) = 0$ for $|t| \geq \varepsilon/2$. Using the Fourier inversion formula, we write

$$\begin{aligned} Qu &= \int R\widehat{q}(Rt) e^{2\pi i t P} u \, dt \\ &= \int R\widehat{q}(Rt) \{ \rho(t)W(t) + \rho(t)R(t) + (1 - \rho(t))e^{2\pi i t P} \} u \, dt \\ &= Q_I u + Q_{II} u + Q_{III} u. \end{aligned} \tag{4.3.14}$$

The rapid decay of $\widehat{q}$ implies that the function $\psi(t) = R\widehat{q}(Rt)(1 - \rho(t))$ satisfies bounds of the form $\|\partial_t^N \psi\|_{L^1(\mathbb{R})} \lesssim_X R^{-M}$, and so

$$|\widehat{\psi}(\lambda)| \lesssim_{N, M} R^{-N} \lambda^{-M}. \tag{4.3.15}$$

Since $Q_{III} = \widehat{\psi}(-P)$, we can write

$$\begin{aligned} P^L Q_{III} u &= \sum_{\lambda} \widehat{\psi}(-\lambda_k) (P^L e_k)(x) \langle e_k, u \rangle \\ &= \sum_{\lambda} \lambda_k^L \widehat{\psi}(-\lambda_k) e_k(x) \langle e_k, u \rangle. \end{aligned} \quad (4.3.16)$$

The rapid decay of $\widehat{\psi}$, and the bound

$$\begin{aligned} |\langle e_k, u \rangle| &\leq \|e_k\|_{L^\infty(X)} \|u\|_{L^1(X)} \\ &\lesssim \|e_k\|_{H^{d/2+1}(X)} \|u\|_{L^1(X)} \\ &\lesssim \|(1 + P)^{d/2+1} e_k\|_{L^2(X)} \|u\|_{L^1(X)} \\ &= (1 + \lambda_k)^{d/2+1} \|u\|_{L^1(X)}, \end{aligned} \quad (4.3.17)$$

imply that

$$\|Q_{III} u\|_{C^M(X)} \lesssim_{N,M} R^{-N} \|u\|_{L^1(X)}. \quad (4.3.18)$$

Since q vanishes near the origin, integration by parts in t yields that for any pair of coordinate systems Ω_1 and Ω_2 , the kernel K_t of $W_\psi(t)^{\Omega_1, \Omega_2}$ satisfies

$$\left| \int R \widehat{q}(Rt) \rho(t) \partial_x^\alpha \partial_y^\beta K_t(x, y) dt \right| \lesssim_N R^{-N}, \quad (4.3.19)$$

and thus a partition of unity argument together with Schur's test for integral operators (see Theorem 7.4) implies that

$$\|Q_{II} u\|_{C^M(X)} \lesssim_{N,M} R^{-N} \|u\|_{L^1(X)}. \quad (4.3.20)$$

Now we expand the definition of Q_I , writing

$$K_{Q_I^\Omega}(t, x, x') = \iint R \widehat{q}(Rt) \rho(t) s_0(t, x, y, \xi) e^{2\pi i[\phi(x, y, \xi) + tp(y, \xi)]} d\xi dt. \quad (4.3.21)$$

We perform a Fourier series expansion, writing

$$c_n(x, y, \xi) = \int_{-\pi}^{\pi} \rho(t) s_0(t, x, y, \xi) e^{-2\pi i n t} dt. \quad (4.3.22)$$

Then the symbol estimates for s_0 , and the compact support of ρ imply that

$$|\partial_{x,y}^\alpha \partial_\xi^\beta c_n(x, y, \xi)| \lesssim_{\alpha,\beta,N} |n|^{-N} \langle \xi \rangle^{-\beta}. \quad (4.3.23)$$

Using Fourier inversion we can write

$$\begin{aligned} K_{Q_I^\Omega}(x, y) &= \iint \sum_n R \widehat{q}(Rt) c_n(x, y, \xi) e^{2\pi i [\phi(x,y,\xi) + t[n+p(y,\xi)]]} d\xi dt \\ &= \int \sum_n q((n + p(y, \xi))/R) c_n(x, y, \xi) e^{2\pi i \phi(x,y,\xi)} d\xi \\ &= \int \tilde{\sigma}(x, y, \xi) e^{2\pi i \phi(x,y,\xi)} d\xi, \end{aligned} \quad (4.3.24)$$

where

$$\tilde{\sigma}(x, y, \xi) = \sum_{n \in \mathbb{Z}} q\left(\frac{n + p(y, \xi)}{R}\right) c_n(x, y, \xi). \quad (4.3.25)$$

The n th term of this sum is supported on $R/4 - n \leq p(y, \xi) \leq 4R - n$, so in particular, if $n > 4R$ then the term vanishes. For $-\infty < n \leq 4R$, we have estimates of the form

$$\left| \partial_{x,y}^\alpha \partial_\xi^\beta \left\{ q\left(\frac{n + p(y, \xi)}{R}\right) c_n(x, y, \xi) \right\} \right| \lesssim_{\alpha,\beta,N} |n|^{-N} \langle \xi \rangle^{-\beta}. \quad (4.3.26)$$

For $n \geq R/8$ and $n \leq -4R$, $|\xi| \sim n$ on the support of $q((n + p(y, \xi))/R)$, and so the rapid decay of n implies that

$$\left| \partial_{x,y}^\alpha \partial_\xi^\beta \left\{ q\left(\frac{n + p(y, \xi)}{R}\right) c_n(x, y, \xi) \right\} \right| \lesssim_{\alpha,\beta,N,M} |n|^{-N} \langle \xi \rangle^{-M}. \quad (4.3.27)$$

This means that if we define

$$\sigma(x, y, \xi) = \sum_{-4R \leq n \leq R/8} q\left(\frac{n + p(y, \xi)}{R}\right) c_n(x, y, \xi). \quad (4.3.28)$$

and define a Schwartz operator Q_U to be the unique operator with support compactly contained in $V \times V$ such that

$$K_{Q_U^\Omega}(x, y) = \int \sigma(x, y, \xi) e^{2\pi i \phi(x, y, \xi)} d\xi, \quad (4.3.29)$$

then

$$\left| \partial_{x,y}^\alpha \partial_\xi^\beta \{\tilde{\sigma} - \sigma\}(x, y, \xi) \right| \lesssim_{\alpha, \beta, N, M} R^{-N} \langle \xi \rangle^{-M}, \quad (4.3.30)$$

and so

$$\|(Q_I - Q_U)u\|_{C^M(X)} \lesssim_{N, M} R^{-N} \|u\|_{L^1(X)}. \quad (4.3.31)$$

Combining (4.3.18), (4.3.20), and (4.3.31), we conclude that

$$\|(Q - Q_U)u\|_{C^M(X)} \lesssim_{N, M} R^{-N} \|u\|_{L^1(X)}. \quad (4.3.32)$$

We have thus verified the required properties of Q . □

We can use the Littlewood-Paley projections to localize the phase support of oscillatory integrals for general Fourier integral operators.

Lemma 4.12. *For each $R \geq 1$, consider a family of pseudo-differential operators Q_R , with kernels*

$$K_{Q_R}(x, y) = R^n \int_{\mathbb{R}^n} q_R(x, y, \xi) e^{iR\xi \cdot (x-y)} d\xi,$$

on $\mathbb{R}^n$, where the functions $\{q_R\}$ are all supported on $\mathbb{R}^d \times \mathbb{R}^d \times \{\xi : 1/10 \leq |\xi| \leq 10\}$, and have uniformly bounded derivatives of all orders. Consider an oscillatory integral

operator A with kernel

$$K_A(x, y) = \int s(x, y, \theta) e^{i\Phi(x, y, \theta)} d\theta$$

from $\mathbb{R}^n$ to $\mathbb{R}^m$, where the (x, y) support of s is compact, and suppose that whenever $\nabla_\theta \Phi = 0$ on this support, $\nabla_y \Phi \neq 0$. If $L > 0$ is sufficiently large, and we fix $\chi \in C_c^\infty(\mathbb{R}^p)$ equal to one for $1/L \leq |\xi| \leq L$, and define an operator $\tilde{A}$ with kernel

$$K_{\tilde{A}}(x, y) = \int \chi(\theta/R) s(x, y, \theta) e^{i\Phi(x, y, \theta)} d\theta,$$

then for all $N \geq 0$,

$$\|A \circ Q_R - \tilde{A} \circ Q_R\|_{L^1(\mathbb{R}^n) \rightarrow C^M(\mathbb{R}^m)} \lesssim_{N, M} R^{-N}$$

Since the kernels of A and $\tilde{A}$ are compactly supported, for any $p, q \in [1, \infty]$ and $s \geq 0$ we have that for all $N \geq 0$,

$$\|A \circ Q_R - \tilde{A} \circ Q_R\|_{L^p(\mathbb{R}^n) \rightarrow W^{q, s}(\mathbb{R}^m)} \lesssim_{p, q, s, N} R^{-N}.$$

Proof. Choose a small quantity $\delta > 0$, and write $\text{supp}(s) = \Gamma_1 \cup \Gamma_2$, where Γ_1 and Γ_2 are two conical sets such that $|(\nabla_\theta \Phi)(x, y, \theta)| \geq \delta$ for $(x, y, \theta) \in \Gamma_1$, and $\delta|\theta| \leq |(\nabla_y \Phi)(x, y, \theta)| \leq \delta^{-1}|\theta|$ for $(x, y, \theta) \in \Gamma_2$. Consider a partition of unity $\{\psi_1, \psi_2\}$ of $\text{supp}(s)$ subordinate to Γ_1 and Γ_2 , where ψ_1 and ψ_2 are homogeneous of order zero in θ . For $j \in \{0, 1\}$ define T_j to be the operator with kernel

$$K_j(x, y) = \int (1 - \chi(\theta/R)) \psi_j(x, y, \theta) s(x, y, \theta) e^{i\Phi(x, y, \theta)}. \quad (4.3.33)$$

Integration by parts in θ implies that

$$|\partial_{x, y}^\alpha K_1(x, y)| \lesssim_{\alpha, N} R^{-N} \quad \text{for all } \alpha \text{ and } N \geq 0, \quad (4.3.34)$$

and so integration by parts in y and Schur's Test implies that

$$\|T_1 \circ Q_R\|_{L^1(\mathbb{R}^d) \rightarrow C^M(\mathbb{R}^d)} \lesssim_{N,M} R^{-N} \quad \text{for all } N, M \geq 0. \quad (4.3.35)$$

If s is a symbol of order μ , we write

$$\begin{aligned} K_{T_2 \circ Q_R}(x, y) &= R^{p+d+\mu} \int (1 - \chi(\theta)) \psi_j(x, z, \theta) [R^{-\mu} s(x, z, R\theta)] \\ &\quad q_R(z, y, \xi) e^{iR[\Phi(x, y, \theta) + \xi \cdot (x-y)]} d\theta d\xi dz. \end{aligned} \quad (4.3.36)$$

If $L \geq 20/\delta$, and $|\theta| \geq L$, then for $(x, z, \theta) \in \Gamma_2$

$$|\nabla_z \{\Phi(x, z, \theta) + \xi \cdot (z - y)\}| \geq \delta|\theta| - |\xi| \geq (\delta/2)|\theta|. \quad (4.3.37)$$

Integrating by parts in z sufficiently many times thus gives rapid decay in θ for $|\theta| \geq L$, in addition to arbitrary negative powers of R . Conversely, if $|\theta| \leq 1/L$, then

$$|\nabla_z \{\Phi(x, z, \theta) + \xi \cdot (z - y)\}| \geq |\xi| - \delta^{-1}|\theta| \geq 1/2, \quad (4.3.38)$$

so integration by parts in z introduces arbitrary negative powers of R into the equation in this range. Thus integrating by parts in z arbitrarily many times, and then integrating trivially in the θ and ξ variables, gives that

$$|\partial_{x,y}^\alpha K_{T_2 \circ Q_R}(x, y)| \lesssim_{\alpha,N} R^{-N} \quad \text{for all } \alpha \text{ and } N \geq 0. \quad (4.3.39)$$

Thus Schur's Test gives that

$$\|T_2 \circ Q_R\|_{L^1(\mathbb{R}^d) \rightarrow C^M(\mathbb{R}^d)} \lesssim_{N,M} R^{-N} \quad \text{for all } N, M \geq 0. \quad (4.3.40)$$

But together, (4.3.35) and (4.3.40), and the fact that $A = T_1 + T_2 + \tilde{A}$ imply that

$$\|A \circ Q_R - \tilde{A} \circ Q_R\|_{L^1(\mathbb{R}^n) \rightarrow C^M(\mathbb{R}^m)} \lesssim_{N,M} R^{-N} \quad \text{for all } N, M \geq 0, \quad (4.3.41)$$

which completes the proof. $\square$

4.4 Periodic Geodesics and Assumption B

Using the Fourier integral operator methods we have described, we can now discuss the relation of Assumption B to the geometry of the manifold X induced by the operator P .

Theorem 4.13. *Suppose that all the eigenvalues of P occur in an arithmetic progression of the form $\{a + nb\}$. Then all geodesics on X with respect to the Finsler geometry induced by P are closed, and have length $1/bn$ for some $n \geq 1$.*

Proof. Because the eigenvalues of P occur in the arithmetic progression $\{a + nb\}$, we know that $e^{2\pi i P/b} = e^{2\pi i a/b} I$. In particular, it follows that the canonical relation of the operator $e^{2\pi i P/b}$ must be equal to the canonical relation of the identity, i.e. the diagonal

$$\Delta_X = \{(x, \xi; x, \xi) : (x, \xi) \in T^*X\}. \quad (4.4.1)$$

But Theorem 4.10 says that the canonical relation of $e^{2\pi i P/b}$ is equal to $\{(x, \xi; x', \xi') : (x, \xi) = \alpha_{1/b}(x', \xi')\}$, where α is the geodesic flow on X with respect to the Finsler geometry of P . So it follows that $\alpha_{1/b}$ is the identity map, which implies the required result. $\square$

The converse is *almost* true.

Theorem 4.14. *Let X be a compact manifold, let $P \in \text{COP}_1^+(X)$, and suppose that all geodesics are closed and have length $1/bn$. Let $a \in \mathbb{Z}$ denote the Maslov index* of the*

*See [9] for a basic discussion of Maslov indices and their connection to Morse indices.

geodesic flow on X , or equivalently, the Morse index of a generic closed geodesic on X . Then there exists $\tilde{P} \in COP_1^+(X)$ commuting with P , such that $P - \tilde{P}$ is order zero and $e^{2\pi i \tilde{P}/b} = I$. If the sub-principal symbol of P is equal to $\pi a/2b$, then $P - \tilde{P}$ is order -1 .

Proof. See Section 29.2 of [22]. □

If P and $\tilde{P}$ are as in Theorem 4.14, then the eigenvalues of $\tilde{P}$ must be integer multiples of b . In the second case of that theorem, since $P - \tilde{P}$ is an operator of order -1 , it follows that for any L^2 normalized eigenfunction e of P , with $Pe = \lambda e$, and with $\tilde{P}e = (n/b)e$ (e must be an eigenfunction of $\tilde{P}$ since P and $\tilde{P}$ commute),

$$|\lambda - n/b| = \|(P - \tilde{P})e\|_{L^2(X)} \lesssim \|e\|_{H^{-1}(X)} \lesssim \|(1 + \tilde{P})^{-1}e\|_{L^2(X)} \lesssim n^{-1}. \quad (4.4.2)$$

It follows from this inequality that if C is suitably large, then the eigenvalues of P must lie in eigenbands of the form $[n/b - C/n, n/b + C/n]$, and thus ‘almost’ lie in the arithmetic progression $\mathbb{Z}/b$. In the case where the sub-principal symbol is a constant not equal to $\pi a/2b$, then the eigenvalues of P will lie in a shifted arithmetic progression; this is the case, for instance, when $P = \sqrt{-\Delta}$ on a Riemannian manifold; for instance, on the sphere, the eigenvalues of P are equal to

$$\sqrt{k(k+d)} = k + \frac{d-1}{2} + O(k^{-1}). \quad (4.4.3)$$

The constant term $(d-1)/2$ arises precisely because $d-1$ is the Maslov index* of the geodesic flow on S^d .

In what follows, we need the fact that the wave equation is *completely periodic* in order to control the wave equation for large times, rather than just contained in eigen-

*This is connected to the fact that on a closed geodesic on S^d (a great circle) each point has a single conjugate point (it’s antipode), and each pair of conjugate points can be joined by a $d-1$ parameter family of geodesics.

bands like above, which is why we cannot replace Assumption B with a more geometrical assumption on the periodicity of eigenvalues.

Chapter 5

The Boundedness of Spectral Multipliers with Compact Support

In this chapter, we prove Theorem 2.4, which we restate below:

Theorem 2.4. *Suppose X is a compact d -dimensional manifold, and $P \in \text{COP}_1^+(X)$ satisfies Assumptions A and B. Then for $1 < p < 2(d-1)/(d+1)$, if $s = (d-1)(1/p - 1/2)$, then for any regulated a with $\text{supp}(a) \subset [1/2, 2]$,*

$$\|a\|_{M_{\text{Dil}}^p(X,P)} \sim \|a\|_{R^{s,p}[0,\infty)} \sim \left(\int_0^\infty |\widehat{a}(t)|^p \langle t \rangle^s dt \right)^{1/p}.$$

In Section 4.1 we saw that for $P \in \text{COP}_1^+(X)$ satisfying Assumption A,

$$\|a\|_{M_{\text{Dil}}^p(X,P)} \gtrsim \|a\|_{R^{s,p}[0,\infty)}, \quad (5.0.1)$$

and so it suffices to prove that $\|a\|_{M_{\text{Dil}}^p(X,P)} \lesssim \|a\|_{R^{s,p}[0,\infty)}$. Define $a_R(\lambda) = a(\lambda/R)$. In Section 5.1, we will see from elementary functional analysis that

$$\sup_{R \leq 1} \|a_R\|_{M^p(X,P)} \lesssim \|a\|_{l^\infty[0,\infty)} \lesssim \|a\|_{R^{s,p}[0,\infty)}. \quad (5.0.2)$$

On the other hand, the upper bound

$$\sup_{R \geq 1} \|a_R\|_{M^p(X, P)} \lesssim \|a\|_{R^{s, P}[0, \infty)}. \quad (5.0.3)$$

requires a more in depth analysis than (5.0.2). It is here we apply the wave equation transform, writing

$$a_R(P) = \int_{-\infty}^{\infty} R\widehat{a}(Rt)e^{2\pi i t P} dt, \quad (5.0.4)$$

which reduces $a_R(P)$ to studying averages associated with solutions to the half-wave equation $\partial_t = 2\pi i P$ on X . We obtain (5.0.3) by studying averages to the wave equation using several methods, including:

- (A) Quasi-orthogonality estimates for averages of solutions to the half-wave equation on X , discussed in Section 5.2, which arise from a study of the geometry of the Finsler metric on X .
- (B) Variants of the density-decomposition arguments, which we introduced in Section 3.3, to control the ‘small time behavior’ of solutions to the half-wave equation. We carry out these arguments in section 5.3
- (C) A new strategy to reduce the ‘large time behavior’ of the half-wave equation to an endpoint local smoothing inequality for the half-wave equation on X , described in Section 5.4.

Equations (5.0.2) and (5.0.3) thus imply Theorem 2.4.

5.1 Preliminary Setup

Without loss of generality, by translating and dilating our multiplier, we can assume that the eigenvalues of the operator P are all integers. Fix a constant $\varepsilon_X > 0$, strictly smaller

than the injectivity radius of the geodesic flow on the manifold X induced by the geometry of P . Our goal is to prove inequalities (5.0.2) and (5.0.3). Proving (5.0.2) is simple because the operators $a_R(P)$ are smoothing operators, uniformly for $0 < R < 1$.

Lemma 5.1. *Let X be a compact d -dimensional manifold, and suppose $P \in COP_1^+(X)$. If $1/2d < 1/p - 1/2 \leq 1/2$, and if a is a regulated function, then*

$$\sup_{R \leq 1} \|a_R\|_{M_{Diff}^p(X,P)} \lesssim \|a\|_{l^\infty(\Lambda)} \lesssim \|a\|_{R^{p,s}[0,\infty)}.$$

Proof. Let $T_R = m_R(P)$. Recall that Λ is the set of eigenvalues of P . The set $\Lambda \cap [0, 2]$ is finite. For each $\lambda \in \Lambda$, choose a finite orthonormal basis $\mathcal{E}_\lambda$. Then we can write

$$T_R = \sum_{\lambda \in \Lambda} \sum_{e \in \mathcal{E}_\lambda} \langle f, e \rangle e. \quad (5.1.1)$$

Since $\mathcal{V}_\lambda \subset C^\infty(X)$, Hölder's inequality implies

$$\|\langle f, e \rangle e\|_{L^p(X)} \leq \|f\|_{L^p(X)} \|e\|_{L^{p'}(X)} \|e\|_{L^p(X)} \lesssim_\lambda \|f\|_{L^p(X)}. \quad (5.1.2)$$

But this means that

$$\|T_R f\|_{L^p(X)} \leq \sum_{\lambda \in \Lambda_P \cap [0,2]} \sum_{e \in \mathcal{E}_\lambda} |m(\lambda/R)| \|\langle f, e \rangle e\|_{L^p(X)} \lesssim \|m\|_{l^\infty(\Lambda)} \|f\|_{L^p(X)}. \quad (5.1.3)$$

For $1/p - 1/2 > 1/2d$, the Sobolev embedding theorem and (2.1.9) imply that

$$\|a\|_{l^\infty(\Lambda)} \lesssim \|a\|_{W^{s,p'}[0,\infty)} \lesssim \|a\|_{R^{s,p}[0,\infty)}, \quad (5.1.4)$$

which completes the proof. $\square$

We now begin to reduce the analysis of $a_R(P)$ for $R \geq 1$ to the study of the wave

equation. Fix a bump function $q \in C_c^\infty(\mathbb{R})$ with $\text{supp}(q) \subset [1/4, 4]$ and $q(\lambda) = 1$ for $\lambda \in [1/2, 2]$, and define $Q_R = q(P/R)$. We write

$$T_R = T_R \circ Q_R = \int_{\mathbb{R}} R \widehat{m}(Rt) (e^{2\pi i t P} \circ Q_R) dt, \quad (5.1.5)$$

and view the operators $(e^{2\pi i t P} \circ Q_R)$ as ‘frequency localized’ wave propagators.

Because all eigenvalues of P are integers, it follows that $e^{2\pi i(t+n)P} = e^{2\pi i t P}$ for any $t \in \mathbb{R}$ and $n \in \mathbb{Z}$. Let I_0 denote the interval $[-1/2, 1/2]$. We may then write

$$T_R = \int_{I_0} b_R(t) (e^{2\pi i t P} \circ Q_R) dt, \quad (5.1.6)$$

where $b_R : I_0 \rightarrow \mathbb{C}$ is the periodic function

$$b_R(t) = \sum_{n \in \mathbb{Z}} R \widehat{m}(R(t+n)). \quad (5.1.7)$$

We split our analysis of T_R into two regimes: regime I and regime II. In regime I, we analyze the behavior of the wave equation over times $0 \leq |t| \leq \varepsilon_X$ by decomposing this time interval into length $1/R$ pieces, and analyzing the interactions of the wave equations between the different intervals. In regime II, we analyze the behavior of the wave equation over times $\varepsilon_X \leq |t| \leq 1$. Here we need not perform such a decomposition, since the $R^{s,p}$ norm gives better control on the function b_R over these times.

Lemma 5.2. *Fix $\varepsilon > 0$. Let $\mathcal{T}_R = \mathbb{Z}/R \cap [-\varepsilon, \varepsilon]$ and define $I_t = [t - 1/R, t + 1/R]$. For any function $a : [0, \infty) \rightarrow \mathbb{C}$, define a periodic function $b : I_0 \rightarrow \mathbb{C}$ by setting*

$$b(t) = \sum_{n \in \mathbb{Z}} R \widehat{m}(R(t+n)).$$

Then we can write $b = \left(\sum_{t_0 \in \mathcal{T}_R} b_{t_0}^I \right) + b^I$, where

$$\text{supp}(b_{t_0}^I) \subset I_{t_0} \quad \text{and} \quad \text{supp}(b_R^I) \subset I_0 \setminus [-\varepsilon, \varepsilon].$$

Moreover, we have

$$\left(\sum_{t_0 \in \mathcal{T}_R} \left[\|b_{t_0}^I\|_{L^p(I_0)} \langle R t_0 \rangle^s \right]^p \right)^{1/p} \lesssim R^{1/p'} \|a\|_{R^{s,p}[0,\infty)}$$

and

$$\|b^I\|_{L^p(I_0)} \lesssim R^{1/p' - (d-1)(1/p-1/2)} \|a\|_{R^{s,p}[0,\infty)}.$$

Proof of Lemma 5.2. The intervals $\{I_{t_0} : t_0 \in \mathcal{T}_R\}$ cover $[-\varepsilon, \varepsilon]$, and so we may consider an associated partition $\mathbb{I}_{[-\varepsilon, \varepsilon]} = \sum_{t_0} \chi_{t_0}$ where $\text{supp}(\chi_{t_0}) \subset I_{t_0}$ and $|\chi_{t_0}| \leq 1$. Define $b_{t_0}^I = \chi_{t_0} b_j$ and $b^I = (1 - \mathbb{I}_{[-\varepsilon, \varepsilon]})b$. Then $b = \sum_{t_0} b_{t_0}^I + b^I$, and the support assumptions are satisfied. It remains to prove the required norm bounds for these choices. For each $n \in \mathbb{Z}$, define a function $b_n : I_0 \rightarrow \mathbb{C}$ by setting $b_n(t) = R\widehat{m}(R(t+n))$. Then $b = \sum_n b_n$. Moreover,

$$\begin{aligned} & \left(\sum_{n \neq 0} [\langle Rn \rangle^s \|b_n\|_{L^p(I_0)}]^p \right)^{1/p} \\ & \sim \left(\int_{|t| \geq 1/2} [\langle Rt \rangle^s |R\widehat{m}(Rt)|]^p \right)^{1/p} \\ & = R^{1/p'} \left(\int_{|t| \geq R/2} [|t|^s \widehat{m}(t)]^p \right)^{1/p} \leq R^{1/p'} C_p(m). \end{aligned} \tag{5.1.8}$$

Write $b_{t_0}^I = \sum_n b_{t_0,n}^I$ and $b^I = \sum_n b_n^I$, where $b_{t_0,n}^I = \chi_{t_0} b_n$ and $b_n^I = \mathbb{I}_{I_0 \setminus [-\varepsilon, \varepsilon]} b_n$. Then

$$\begin{aligned} \|b_0^I\|_{L^p(I_0)} &= \left(\int_{\varepsilon \leq |t| \leq 1/2} |R\widehat{m}(Rt)|^p \right)^{1/p} \\ &= R^{1/p'} \left(\int_{R\varepsilon \leq |t| \leq R/2} |\widehat{m}(t)|^p \right)^{1/p} \lesssim R^{1/p' - s} C_p(m). \end{aligned} \tag{5.1.9}$$

Using (5.1.8), (5.1.9), and Hölder's inequality, we conclude that

$$\begin{aligned}
\|b^{II}\|_{L^p(I_0)} &\leq \sum_n \|b_n^{II}\|_{L^p(I_0)} \\
&\leq \|b_0^{II}\|_{L^p(I_0)} + \sum_{n \neq 0} \left[|Rn|^s \|b_n^{II}\|_{L^p(I_0)} \right] \frac{1}{|Rn|^s} \\
&\leq \|b_0^{II}\|_{L^p(I_0)} + R^{-s} \left(\sum_{n \neq 0} [|Rn|^s \|b_n\|_{L^p(I_0)}]^p \right)^{1/p} \\
&\lesssim R^{1/p' - s} C_p(m).
\end{aligned} \tag{5.1.10}$$

A similar calculation shows that

$$\begin{aligned}
\|b_{t_0}^I\|_{L^p(I_0)} &\leq \sum_n \|b_{t_0,n}^I\|_{L^p(I_0)} \\
&= \|b_{t_0,0}^I\|_{L^p(I_0)} + \sum_{n \neq 0} \|b_{t_0,n}^I\|_{L^p(I_0)} \\
&\lesssim \|b_{t_0,0}^I\|_{L^p(I_0)} + R^{-s} \left(\sum_{n \neq 0} |Rn|^s \|b_{t_0,n}^I\|_{L^p(I_0)}^p \right)^{1/p} \\
&\lesssim \|b_{t_0,0}^I\|_{L^p(I_0)} + \left(\sum_{n \neq 0} |Rn|^s \|b_{t_0,n}^I\|_{L^p(I_0)}^p \right)^{1/p}.
\end{aligned} \tag{5.1.11}$$

Using (5.1.11), we calculate that

$$\begin{aligned}
&\left(\sum_{t_0 \in \mathcal{T}_R} \left[\|b_{t_0}^I\|_{L^p(I_0)} \langle R t_0 \rangle^s \right]^p \right)^{1/p} \\
&\lesssim \left(\sum_{t_0 \in \mathcal{T}_R} \left[\|b_{t_0,0}^I\|_{L^p(I_0)} \langle R t_0 \rangle^s \right]^p + \sum_{n \neq 0} \left[|Rn|^s \|b_{t_0,n}^I\|_{L^p(I_0)} \right]^p \right)^{1/p} \\
&\lesssim \left(\int_{\mathbb{R}} [\langle R t \rangle^s R \widehat{m}(R t)]^p dt \right)^{1/p} \\
&\lesssim R^{1/p'} C_p(m).
\end{aligned} \tag{5.1.12}$$

Since each function $b_{t_0}^I$ is supported on a length $1/R$ interval, we have

$$\|b_{t_0}^I\|_{L^1(I_0)} \lesssim R^{-1/p'} \|b_{t_0}^I\|_{L^p(I_0)}, \tag{5.1.13}$$

and substituting this inequality into (5.1.12) completes the proof. $\square$

The following proposition, a kind of L^p square root cancellation bound, implies (5.0.3) once we take Lemma 5.2 into account. Since we already proved (5.0.2), this proposition completes the proof of Theorem 2.4, and will take the remainder of the chapter.

Proposition 5.3. *Let X be a d -dimensional compact manifold, and consider $P \in \text{COP}_1^+(X)$ satisfying Assumptions A and B. Fix $R > 0$ and a constant ε_X smaller than the injectivity radius of X with respect to the geodesic flow on X with respect to the Finsler metric induced by P , and suppose $1/(d-1) < 1/p - 1/2 < 1/2$. Consider any function $b : I_0 \rightarrow \mathbb{C}$, and write $b = \sum_{t_0 \in \mathcal{T}_R} b_{t_0}^I + b^I$, where $\text{supp}(b_{t_0}^I) \subset I_{t_0}$ and $\text{supp}(b^I) \subset I_0 \setminus [-\varepsilon_X, \varepsilon_X]$. Define operators $T^I = \sum_{t_0 \in \mathcal{T}_R} T_{t_0}^I$ and T^I , where*

$$T_{t_0}^I = \int b_{t_0}^I(t)(e^{2\pi i t P} \circ Q_R) dt \quad \text{and} \quad T^I = \int b^I(t)(e^{2\pi i t P} \circ Q_R) dt.$$

Then if $s = (d-1)(1/p - 1/2)$, then

$$\|T^I\|_{L^p \rightarrow L^p} \lesssim R^{-1/p'} \left(\sum_{t_0 \in \mathcal{T}_R} \left[\|b_{t_0}^I\|_{L^p(I_0)} \langle R t_0 \rangle^s \right]^p \right)^{1/p} \quad (5.1.14)$$

and

$$\|T^I\|_{L^p \rightarrow L^p} \lesssim R^{s-1/p'} \|b^I\|_{L^p(I_0)}. \quad (5.1.15)$$

Remark. One should view this proposition as a discretization of Theorem 2.4, and thus compared to Lemma 3.5 in the argument of Heo, Nazarov, and Seeger.

Proposition 5.3 splits the main bound of the paper into two regimes: regime I and regime II. Noting that we require weaker bounds in (5.1.15) than in (5.1.14), the operator T^I will not require as refined an analysis as for the operator T^I , and we obtain bounds on T^I by a reduction to an endpoint local smoothing inequality in Section 5.4. On the other hand, to obtain more refined estimates for the L^p norms of quantities of the form $f = T^I u$, we consider a decompositions of the form $u = \sum_{x_0} u_{x_0}$, where $u_{x_0} : X \rightarrow \mathbb{C}$ is

supported on a ball $B(x, 1/R)$ of radius $1/R$ centered at x_0 . We then have $f = \sum_{(x_0, t_0)} f_{x_0, t_0}$, where $f_{x_0, t_0} = T_{t_0}^I u_{x_0}$. To control f we must establish an L^p square root cancellation bound for the functions $\{f_{x_0, t_0}\}$. In the next section, we study the L^2 quasi-orthogonality of these functions, which we use as a starting point to obtain the required square root cancellation.

5.2 Quasi-Orthogonality Estimates For Wave-Packets

The discussion at the end of Section 5.1 motivates us to consider estimates for functions obtained by taking averages of the wave equation over a small time interval, with initial conditions localized to a particular part of space. In this section, we study the L^2 orthogonality of such quantities. One should compare the results of this section to the results of Lemma 3.10. We do not exploit periodicity of the Hamiltonian flow in this section since we are only dealing with estimates for the half wave-equation for *small times*. Our results here thus hold for any manifold X , and any operator P whose principal symbol has co-spheres with non-vanishing Gaussian curvature. In order to prove the required quasi-orthogonality estimates of the functions $\{f_{x_0, t_0}\}$, we find a new connection between the Finsler geometry we introduced in section 4.2 and the behavior of the operator P . In particular, recall the quasi-metrics d_+ and d_- introduced in that section.

Proposition 5.4. *Let X be a compact manifold of dimension d , and suppose $P \in COP_1^+(X)$ satisfies the curvature assumptions of Theorem 2.4. Suppose $\varepsilon_X > 0$ is chosen smaller than the injectivity radius of X . Then for all $R \geq 0$, the following estimates hold:*

- (Pointwise Estimates) Fix $|t_0| \leq \varepsilon_X$ and $x_0 \in X$. Consider any two L^1 normalized functions $c : \mathbb{R} \rightarrow \mathbb{C}$ and $u : X \rightarrow \mathbb{C}$ with $\text{supp}(c) \subset I_{t_0}$ and $\text{supp}(u) \subset B(x_0, 1/R)$. Define $S_{x_0, t_0} : X \rightarrow \mathbb{C}$ by setting

$$S_{x_0, t_0} = \int c(t)(e^{2\pi i t P} \circ Q_R)\{u\} dt.$$

Then for any $K \geq 0$, and any $x \in X$,

$$|S_{x_0, t_0}(x)| \lesssim_K \frac{R^d}{\langle R d_X(x_0, x) \rangle^{\frac{d-1}{2}}} \max_{\pm} \left\langle R |t_0 \pm d_X^{\pm}(x, x_0)| \right\rangle^{-K}.$$

- (*Quasi-Orthogonality Estimates*) Fix $|t_0 - t_1| \leq \varepsilon_X$, and $x_0, x_1 \in X$. Consider L^1 normalized functions $c_0, c_1 : \mathbb{R} \rightarrow \mathbb{C}$ and $u_0, u_1 : X \rightarrow \mathbb{C}$ such that, for each $v \in \{0, 1\}$, $\text{supp}(c_v) \subset I_{t_v}$ and $\text{supp}(u_v) \subset B(x_v, 1/R)$. Define $S_{x_v, t_v} : X \rightarrow \mathbb{C}$ by setting

$$S_{x_v, t_v} = \int c_v(t) (e^{2\pi i t P} \circ Q_R) \{u_v\} dt.$$

Then for any $K \geq 0$,

$$|\langle S_{x_0, t_0}, S_{x_1, t_1} \rangle| \lesssim_K \frac{R^d}{\langle R d_X(x_0, x_1) \rangle^{\frac{d-1}{2}}} \max_{\pm} \left\langle R |(t_0 - t_1) \pm d^{\pm}(x_0, x_1)| \right\rangle^{-K}.$$

Remark. Just as in Lemma 3.10, the pointwise estimate of Lemma 5.4 tells us that the function S is concentrated on a geodesic annulus of radius $|t_0|$ centered at x_0 and thickness $O(1/R)$. The quasi-orthogonality estimate tells us that the two functions S_{x_0, t_0} and S_{x_1, t_1} are only significantly correlated with one another if the two annuli on which the majority of the support lies are internally or externally tangent to one another, depending on whether t_0 and t_1 have the same or opposite sign respectively.

Proof of Proposition 5.4. To simplify notation, in the following proof we will suppress the use of R as an index, for instance, writing Q for Q_R . For both the pointwise and quasi-orthogonality estimates, we want to consider the operators in coordinates, so we can use Theorem 4.9 to understand the wave propagators in terms of various oscillatory integrals. Consider a small quantity $\varepsilon'_X < \varepsilon_X$, to be fixed later.

We begin with a proof of the pointwise bounds. Write $S = S_{x_0, t_0}$. Start by covering

X by a finite family of suitably small open sets $\{V_\alpha\}$, such that for each α , there is a coordinate chart U_α compactly containing V_α and with $N(V_\alpha, 1.1\varepsilon_X) \subset U_\alpha$. Let $\{\eta_\alpha\}$ be a partition of unity subordinate to $\{V_\alpha\}$. Given $u : M \rightarrow \mathbb{C}$, write $u = \sum_\alpha u_\alpha$, where $u_\alpha = \eta_\alpha u$.

Let us first address the case where $\varepsilon'_X \leq |t_0| \leq \varepsilon_X$. Lemmas 4.11 and 4.12 imply that if we define

$$S_\alpha = \int c(t)(Q_\alpha \circ W_\alpha(t) \circ Q_\alpha)\{u_\alpha\} dt, \quad (5.2.1)$$

where Q_α is a pseudo-differential operator whose symbol in the coordinate system U_α is supported on $\{R/4 \leq |\xi| \leq 4R\}$, and

$$K_{W_\alpha}(x, y) = R^{\frac{d+1}{2}} \int_{-\infty}^{\infty} s(t, x, y, R\tau) e^{iR\tau\Phi(x, t, x', \tau)} d\xi, \quad (5.2.2)$$

where s is smooth, and compactly supported on $L^{-1} \leq |\tau| \leq L$, and Φ is defined in (4.3.12), then for all $N \geq 0$,

$$\|S - \sum_\alpha S_\alpha\|_{L^\infty(X)} \lesssim_N R^{-N}. \quad (5.2.3)$$

This error is negligible to our analysis if N is chosen appropriately large. Integrating by parts if $|t| - d^+(y, x)$ and $|t| - d^-(y, x)$ is suitably large, and then taking in absolute values gives the required bounds for each of the terms S_α , completing the proof of the required bounds in this range.

Now we address the case $|t_0| \leq \varepsilon_X$. Lemmas 4.11 and 4.12 imply that if we define

$$S_\alpha = \int c(t)(Q_\alpha \circ W_\alpha(t) \circ Q_\alpha)\{u_\alpha\} dt, \quad (5.2.4)$$

where Q_α is as above, and

$$K_{W_\alpha}(x, y) = R^d \int s(t, x, y, \theta) e^{i[\phi(x, y, \xi) + tp(y, \xi)]} d\xi,$$

where s is smooth and supported on $L^{-1} \leq |\theta| \leq L$, and ϕ is a solution to the Eikonal equation (4.3.10), then for all $N \geq 0$,

$$\|S - \sum_{\alpha} S_{\alpha}\|_{L^{\infty}(X)} \lesssim_N R^{-N}. \quad (5.2.5)$$

The error in (5.2.5) is again negligible. We bound each of the functions $\{S_{\alpha}\}$ separately, combining the estimates using the triangle inequality. We continue by expanding out the implicit integrals in the definition of S_{α} . In the coordinate system U_{α} , we can write

$$\begin{aligned} S_{\alpha}(x) = & \int c(t) \sigma(x, \eta) e^{2\pi i \eta \cdot (x-y)} \\ & s(t, y, z, \xi) e^{2\pi i [\phi(y, z, \xi) + t p(z, \xi)]} \\ & \sigma(z, \theta) e^{2\pi i \theta \cdot (z-w)} (\eta_{\alpha} u)(w) \\ & dt dy dz dw d\theta d\xi d\eta. \end{aligned} \quad (5.2.6)$$

The integral in (5.2.6) looks highly complicated, but can be simplified considerably by noticing that most variables are quite highly localized. In particular, oscillation in the η variable implies that the amplitude is negligible unless $|x - y| \lesssim 1/R$, oscillation in the θ variable implies that the amplitude is negligible unless $|z - w| \lesssim 1/R$, and the support of u implies that $|w - x_0| \lesssim 1/R$. Define

$$k_1(t, x, z, \xi) = \int \sigma(x, \eta) s(t, y, z, \xi) e^{2\pi i [\eta \cdot (x-y) + \phi(y, z, \xi) - \phi(x, z, \xi)]} dy d\eta, \quad (5.2.7)$$

and

$$\begin{aligned} k_2(t, \xi) = & \int k_1(t, x, z, \xi) \sigma(z, \theta) (\eta_{\alpha} u)(w) \\ & e^{2\pi i [\theta \cdot (z-w) + \phi(x, z, \xi) - \phi(x, x_0, \xi) + t p(z, \xi) - t p(x_0, \xi)]} d\theta dw, \end{aligned} \quad (5.2.8)$$

and then set

$$a(x, \xi) = \int c(t) k_2(t, R\xi) e^{2\pi i[(t-t_0)p(x_0, R\xi)]} dt dz, \quad (5.2.9)$$

so that $\text{supp}_\xi(a) \subset \{\xi : 1/8 \leq |\xi| \leq 8\}$, and

$$S_\alpha(x) = R^d \int a(x, \xi) e^{2\pi i R[\phi(x, x_0, \xi) + t_0 p(x_0, \xi)]} d\xi. \quad (5.2.10)$$

Integrating by parts in η and θ in (5.2.7) and (5.2.8) gives that for all multi-indices α ,

$$|\partial_\xi^\alpha k_1(t, x, z, \xi)| \lesssim_\alpha R^{-|\alpha|} \quad \text{and} \quad |\partial_\xi^\alpha k_2(z, \xi)| \lesssim_\alpha R^{-|\alpha|}. \quad (5.2.11)$$

Using the bounds in (5.2.11) with the fact that $\text{supp}(c)$ is contained in a $O(1/R)$ neighborhood of t_0 in (5.2.9) then implies $|\partial_\xi^\alpha a(x, \xi)| \lesssim_\alpha 1$ for all α .

We now account for angular oscillation of the integral by working in a kind of 'polar coordinate' system. First we find $\lambda : V_\alpha \times S^{d-1} \rightarrow (0, \infty)$ such that for all $|\xi| = 1$,

$$p(x_0, \lambda(x_0, \xi)\xi) = 1. \quad (5.2.12)$$

If $\tilde{a}(x, \rho, \eta) = a(x, \rho \lambda(x_0, \xi) \xi) \det[\lambda(x_0, \xi)I + \xi(\nabla_\xi \lambda)(x_0, \xi)^T]$, then

$$S_\alpha(x) = R^d \int_0^\infty \rho^{d-1} \int_{|\xi|=1} \tilde{a}(x, \rho, \xi) e^{2\pi i R \rho [t_0 + \phi(x, x_0, \lambda(x_0, \xi)\xi)]} d\xi d\rho. \quad (5.2.13)$$

Define $\Phi : S^{d-1} \rightarrow \mathbb{R}$ by setting $\Phi(\xi) = \phi(x, x_0, \lambda(x_0, \xi)\xi)$. We claim that, if ε'_X is chosen appropriately small, then in the ξ variable, Φ has exactly two critical points $|\xi^+|^{-1}\xi^+$ and $|\xi^-|^{-1}\xi^-$, where $\xi^+ \in S_{x_0}^*$ is the co-vector corresponding to the forward geodesic from x_0 to x , and $\xi^- \in S_{x_0}^*$ is the co-vector corresponding to the backward geodesic from x_0 to x .

Moreover,

$$\Phi(|\xi^+|^{-1}\xi^+) = d_X^+(x_0, x) \quad \text{and} \quad \Phi(|\xi^-|^{-1}\xi^-) = -d_X^-(x_0, x), \quad (5.2.14)$$

and the Hessian at each of these points is non-degenerate, with each eigenvalue of the Hessian having magnitude exceeding a constant multiple of $d_X^\pm(x_0, x)$. We prove that these properties hold for Φ in Proposition 5.5 of the following section, via a series of geometric arguments. It then follows from the principle of stationary phase that

$$S_\alpha(x) = \sum_{\pm} \frac{R^d}{\langle R d_X^\pm(x_0, x) \rangle^{\frac{d-1}{2}}} \int_0^\infty \rho^{\frac{d-1}{2}} a_\pm(x, \rho) e^{2\pi i R \rho [t_0 \pm d_X^\pm(x_0, x)]} d\rho, \quad (5.2.15)$$

where $a_\pm$ is supported on $|\rho| \sim 1$, and for all α , $|\partial_\rho^\alpha a_\pm| \lesssim_\alpha 1$. Integrating by parts in the ρ variable if $t_0 \pm d_X^\pm(x_0, x)$ is large, we conclude that

$$|S_\alpha(x)| \lesssim \frac{R^d}{\langle R d_X(x_0, x) \rangle^{\frac{d-1}{2}}} \sum_{\pm} \langle R |t_0 \pm d_X(x_0, x)| \rangle^{-K}. \quad (5.2.16)$$

Combining (5.2.5) and (5.2.16) completes the proof of the pointwise bounds.

The quasi-orthogonality arguments are obtained by a largely analogous method, and so we only sketch the proof. Write S_ν for S_{x_ν, t_ν} . One major difference is that we can use the self-adjointness of the operators Q , and the unitary group structure of $\{e^{2\pi i t P}\}$, to write

$$\begin{aligned} \langle S_0, S_1 \rangle &= \int c_0(t) c_1(s) \langle (e^{2\pi i t P} \circ Q)\{u_0\}, (e^{2\pi i s P} \circ Q)\{u_1\} \rangle \\ &= \int c_0(t) c_1(s) \langle (e^{2\pi i (t-s) P} \circ Q^2)\{u_0\}, u_1 \rangle \\ &= \int c(t) \langle (e^{2\pi i t P} \circ Q^2)\{u_0\}, u_1 \rangle, \end{aligned} \quad (5.2.17)$$

where $c(t) = \int c_0(u) c_1(u - t) du$, by Young's inequality, satisfies

$$\|c\|_{L^1(\mathbb{R})} \lesssim \|c_0\|_{L^1(\mathbb{R})} \|c_1\|_{L^1(\mathbb{R})} \leq 1 \quad (5.2.18)$$

and $\text{supp}(c) \subset [(t_0 - t_1) - 4/R, (t_0 - t_1) + 4/R]$. After this, one proceeds exactly as in the proof of the pointwise estimate. We write the inner product as

$$\sum_{\alpha} \int c(t) \langle (e^{2\pi i t P} \circ Q^2) \{ \eta_{\alpha} u_0 \}, u_1 \rangle. \quad (5.2.19)$$

Then we use Lemmas 4.11 and 4.12 to replace $e^{2\pi i t P} \circ Q^2$ with $Q_{\alpha}^2 \circ W_{\alpha}(t) \circ Q_{\alpha}^2$, modulo a negligible error. The integral

$$\sum_{\alpha} \int c(t) \langle (Q_{\alpha}^2 \circ W_{\alpha}(t) \circ Q_{\alpha}^2) \{ \eta_{\alpha} u_0 \}, u_1 \rangle \quad (5.2.20)$$

is then only non-zero if both the supports of u_0 and u_1 are compactly contained in U_{α} . Thus we can switch to the coordinate system of U_{α} , in which we can express the inner product by oscillatory integrals of the exact same kind as those occurring in the pointwise estimate. Integrating away the highly localized variables as in the pointwise case, and then applying stationary phase in polar coordinates proves the required estimates. $\square$

It remains to classify the critical points that arose in Proposition 5.4.

Proposition 5.5. *Fix a bounded open set $U_0 \subset \mathbb{R}^d$. Consider a Finsler metric $F : U_0 \times \mathbb{R}^d \rightarrow [0, \infty)$ on U_0 , and its dual metric $F_* : U_0 \times \mathbb{R}^d \rightarrow [0, \infty)$, which extends to a Finsler metric on an open set containing the closure of U_0 . Fix a suitably small constant $r > 0$. Let U be an open subset of U_0 with diameter at most r which is geodesically convex (any two points are joined by a minimizing geodesic). Let $\phi : U \times U \times \mathbb{R}^d \rightarrow \mathbb{R}$ solve the eikonal equation*

$$F_*(x, \nabla_x \phi(x, y, \xi)) = F_*(y, \xi),$$

such that $\phi(x, y, \xi) = 0$ for $x \in H(y, \xi)$, where $H(y, \xi) = \{x \in U : \xi \cdot (x - y) = 0\}$. For each $x_0, x_1 \in U$, let $S_{x_0}^ = \{\xi \in \mathbb{R}^d : F_*(x_0, \xi) = 1\}$ be the co-sphere at x_0 , and define*

$\Psi : S_{x_0}^* \rightarrow \mathbb{R}$ by setting

$$\Psi(\xi) = \phi(x_1, x_0, \xi).$$

Then the function Ψ has exactly two critical points, at ξ^+ and ξ^- , where the Legendre transform of ξ^+ is the tangent vector of the forward geodesic from x_0 to x_1 , and the Legendre transform of ξ^- is the tangent vector of the backward geodesic from x_1 to x_0 . Moreover,

$$\Psi(\xi^+) = d_X^+(x_0, x_1) \quad \text{and} \quad \Psi(\xi^-) = -d_X^-(x_0, x_1),$$

and the Hessians H_+ and H_- of Ψ at these critical points, viewed as quadratic maps from $T_{\xi_{\pm}} S_{x_0}^* \rightarrow \mathbb{R}$ satisfy

$$H_+(\zeta) \geq C d_X^+(x_0, x_1) |\zeta| \quad \text{and} \quad H_-(\zeta) \leq -C d_X^-(x_0, x_1) |\zeta|,$$

where the implicit constant is uniform in x_0 and x_1 .

If Ψ is as above, then the function $\Phi : S^{d-1} \rightarrow \mathbb{R}$ obtained by setting $\Phi(\xi) = \phi(x, x_0, F_*(x, \xi)^{-1} \xi)$ is precisely the kind of function that arose as a phase in Proposition 5.4, where F_* was the principal symbol p of the pseudo-differential operator we were considering. Since critical points and the Hessians of maps at critical points are stable under diffeomorphisms, and the map $\xi \mapsto F_*(x, \xi)^{-1} \xi$ is a diffeomorphism from $S_{x_0}^*$ to S^{d-1} , classifying the critical points of the map Ψ implies the required properties of the map Φ used in Proposition 5.4.

In order to prove 5.5, we rely on a geometric interpretation of Ψ following from Hamilton-Jacobi theory.

Lemma 5.6. *Consider the setup to Proposition 5.5. For any $\xi \in S_{x_0}^*$,*

$$|\Psi(\xi)| = \begin{cases} \text{the length of the shortest curve from } H(x_0, \xi) \text{ to } x_1 & : \text{ if } \Psi(\xi) > 0, \\ \text{the length of the shortest curve from } x_1 \text{ to } H(x_0, \xi) & : \text{ if } \Psi(\xi) < 0. \end{cases}$$

Proof. We rely on a construction of ϕ from Proposition 3.7 of [51], which we briefly describe. Fix x_0 and ξ . Then there is a unique co-vector field $\omega : H(x_0, \xi) \rightarrow \mathbb{R}^d$ which is everywhere perpendicular to $H(x_0, \xi)$, with $\omega(x_0) = \xi$ and with $F_*(x, \omega(x)) = F_*(x_0, \xi)$ for all $x \in H(x_0, \xi)$. There exists a unique point $x(\xi) \in H(x_0, \xi)$ and a unique $t(\xi) \in \mathbb{R}$, such that the unit speed geodesic γ on X with $\gamma(0) = x(\xi)$ and $\gamma'(0) = \omega(x(\xi))_\#$ satisfies $\gamma(t(\xi)) = x_1$. We then have $\Psi(\xi) = t(\xi)$. If $t(\xi)$ is negative, then $\gamma|_{[t(\xi), 0]}$ is a geodesic from x_1 to x_0 , and if $t(\xi)$ is positive, $\gamma|_{[0, t(\xi)]}$ is a geodesic from x_0 to x_1 . Because γ is a geodesic, the geometric interpretation then follows if U is a suitably small neighborhood such that geodesics are length minimizing. $\square$

It follows immediately from Lemma 5.6 that

$$\Psi(\xi_+) = d_X^+(x_0, x_1) \quad \text{and} \quad \Psi(\xi_-) = -d_X^-(x_0, x_1). \quad (5.2.21)$$

A simple geometric argument also shows $\Psi(\xi^+)$ is the maximum value of Ψ on $S_{x_0}^*$, and $\Psi(\xi^-)$ is the minimum value on $S_{x_0}^*$, so that these two points are both critical. Indeed, the point x_0 lies in $H(x_0, \xi)$ for all $\xi \in \mathbb{R}^d - \{0\}$. Thus the shortest curve from x_0 to x_1 is always longer than the shortest curve from $H(x_0, \xi)$ to x_1 . Similarly, the shortest curve from x_1 to x_0 is always longer than the shortest curve from x_1 to $H(x_0, \xi)$. Thus

$$-d_X^-(x_0, x_1) \leq \Psi(\xi) \leq d_X^+(x_0, x_1), \quad (5.2.22)$$

and so $\Psi(\xi^-) \leq \Psi(\xi) \leq \Psi(\xi^+)$ for all $\xi \in S_{x_0}^*$. All that remains is to prove that ξ^+ and ξ^- are the

only critical points of Ψ , and that these critical points are appropriately non-degenerate.

In order to simplify proofs, we employ a structural symmetry to reduce the number of cases we need to analyze. Namely, if one defines the reverse Finsler metric $F_\rho(x, v) = F(x, -v)$, then $F_\rho^*(x, \xi) = F_*(x, -\xi)$, and so the associated function Ψ^ρ which is the analogue of Ψ for F^ρ satisfies $\Psi^\rho(\xi) = -\Psi(-\xi)$. The critical points of Ψ and Ψ^ρ are thus directly related to one another, which allows us without loss of generality to study only points with $\Psi \leq 0$ (and thus only study geodesics beginning at x_1).

Proof that ξ^+ and ξ^- are the only critical points. Fix $\xi^* \in T_{x_0}X - \{\xi^\pm\}$. Using the notation defined above, let $x_* = x(\xi^*)$ and $t_* = t(\xi^*)$. Using the symmetry above, we may assume without loss of generality that $\Psi(\xi^*) \leq 0$. Since ξ^- is not perpendicular to $H(x_0, \xi^*)$ at x_0 , we have $x_* \neq x_0$. If $\Psi(\xi^*) < 0$, let γ be the unique unit speed forward geodesic with $\gamma(0) = x_1$ and $\gamma(t_*) = x_*$. If $\Psi(\xi^*) = 0$, let γ be the unique unit speed forward geodesic with $\gamma'(0)$ equal to the Legendre transform of ξ^* . Pick η such that $\eta \cdot (x_* - x_0) \neq 0$, and then, for t suitably close to t_* , define a smooth map

$$\xi(t) = \frac{\xi^* + a(t)\eta}{F_*(x_0, \xi^* + a(t)\eta)}, \quad (5.2.23)$$

into $S_{x_0}^*$, where

$$a(t) = -\frac{\xi^* \cdot (\gamma(t) - x_0)}{\eta \cdot (\gamma(t) - x_0)} \quad (5.2.24)$$

is defined so that $\gamma(t) \in H(x_0, \xi(t))$. Then $\Psi(\xi(t)) = -t$, and differentiation at $t = t_*$ gives $D\Psi(\xi^*)(\xi'(t_*)) = -1$. In particular, $D\Psi(\xi^*) \neq 0$, so ξ^* is not a critical point. $\square$

We now analyze the non-degeneracy of the critical points ξ^+ and ξ^- .

Proof that ξ^+ and ξ^- are non-degenerate. Using symmetry, it suffices without loss of generality to analyze the critical point ξ^- rather than ξ^+ . Let $H_- : T_{\xi^-}S_{x_0}^* \rightarrow \mathbb{R}$ be the Hessian of Ψ at ξ^- . Let $v_0 = \xi_\#^-$, and using the notation of the last argument, let $l = t(\xi^-)$. Consider

a curve $\xi(a)$ valued in $S_{x_0}^*$ with $\xi(0) = \xi^-$ and $\zeta = \xi'(0)$ for some $\zeta \in T_{\xi^-}^*S_{x_0}$. Then the second derivative of $\Psi(\xi(a))$ at $a = 0$ is $H_-(\zeta)$, and so our proof would be complete if we could show that the function $L(a) = -\Psi(\xi(a))$, which is the length of the shortest geodesic from x_1 to $H(x_0, \xi(a))$, satisfies $L''(0) \geq C\|\zeta\|$, for some $C > 0$ uniform in x_0, x_1 , and ζ .

Consider the partial function $\text{Exp}_{x_1} : \mathbb{R}^d \rightarrow U_0$ obtained from the exponential map at x_1 . If r is chosen suitably small, there exists a neighborhood V of the origin in $\mathbb{R}^d$ such that $E = \text{Exp}_{x_1}|_V$ is a diffeomorphism between V and U . Unlike in Riemannian manifolds, in general E is only C^1 at the origin, though smooth on $V - \{0\}$. However, for $|v| = 1$ and $t > 0$ we can write $E(tv) = (\pi \circ \varphi)(x_1, v^\flat, t)$, where the partial function $\varphi : (T^*U_0 - 0) \times \mathbb{R} \rightarrow (T^*U_0 - 0)$ is the flow induced by the the Hamiltonian vector field $(\partial_\xi F_*, -\partial_x F_*)$ on $T^*U_0 - 0$, and π is the projection map from $T^*U_0 - 0$ to U_0 . Where defined, φ is a smooth function since the Hamiltonian vector field is smooth, and so it follows by homogeneity and the pre-compactness of U that for each α , the partial derivatives $(\partial^\alpha E_x)(v)$ are uniformly bounded for $v \neq 0$ and $x \in U$. It thus follows from the inverse function theorem that there exists a constant $A > 0$ such that for all x_1 and x in U with $x \neq x_1$,

$$|\partial_j G_{x_1}(x)| \leq A \quad \text{and} \quad |(\partial_j \partial_k G_{x_1})(x)| \leq A. \quad (5.2.25)$$

We can also pick A to be large enough that for all $x \in U$, and all $v, w \in \mathbb{R}^d - \{0\}$,

$$A^{-1}|w|^2 \leq \sum_{ij} g_{ij}(x, v) w^i w^j \leq A|w|^2. \quad (5.2.26)$$

and such that for all $x \in U$ and $v \in \mathbb{R}^d - \{0\}$,

$$|\partial_{x_k} g_{ij}(x, v)| \leq A. \quad (5.2.27)$$

Since $\zeta \in T_{\xi^-}^*S_{x_0}$, and $S_{x_0}^* = \{\xi : F_*(x_0, \xi)^2 = 1\}$, by Euler's homogeneous function

theorem we have

$$\sum_{i,j} g^{ij}(x_0, \xi^-) \xi_j'(0) \xi_j^- = \frac{1}{2} \sum_{i,j} \frac{\partial^2 F_*^2}{\partial \xi_i \partial \xi_j}(x_0, \xi^-) \xi_i \xi_j^- = \frac{1}{2} \sum \frac{\partial F_*^2}{\partial \xi_j}(x_0, \xi^-) \xi_j = 0. \quad (5.2.28)$$

Differentiating $F_*(x_0, \xi(a))^2 = 1$ twice with respect to a at $a = 0$ yields that

$$\sum_{i,j} g_{ij}(x_0, \xi^-) \xi_i''(0) \xi_j^- = - \sum_{i,j} g_{ij}(x_0, \xi^-) \xi_i^i \xi_j^j \quad (5.2.29)$$

Define vectors $n(a)$ by setting $n^i(a) = \sum g^{ij}(x_0, \xi^-) \xi_j(a)$. Then $n(a)$ is the normal vector to $H(x_0, \xi(a))$ with respect to the inner product with coefficients $g_{ij}(x_0, v_0)$. Let $u(a)$ be the orthogonal projection of v_0 onto the hyperplane $\{v : \xi(a) \cdot v = 0\}$ with respect to the inner product $g_{ij}(x_0, v_0)$. If we define

$$c(a) = \sum g_{ij}(x_0, v_0) v_0^i n^j(a) = \sum g^{ij}(x_0, \xi^-) \xi_i^- \xi_j(a), \quad (5.2.30)$$

then $u(a) = v_0 - c(a)n(a)$. Note that (5.2.28) and (5.2.29) imply $c(0) = 1$, $c'(0) = 0$, and $c''(0) \leq -|\zeta|^2/A$. Also $n(0) = v_0$ and $|n'(0)| \leq A|\zeta|$.

Let $x(s) = x_0 + su(a)$ and let $R(s)$ be the length of the geodesic from x_1 to $x(s)$. To control $R(s)$, define $y(s) = G(x(s))$, and consider the variation $A(s, t) = E(ty(s))$, defined so that $t \mapsto A(s, t)$ is the geodesic from x_1 to $x(s)$. The Gauss Lemma for Finsler manifolds (see Lemma 6.1.1 of [3]) implies that

$$A^{-1}R(s) \leq |y(s)| \leq AR(s). \quad (5.2.31)$$

Define $T(s) = (\partial_t A)(s, 1)$ and $V(s) = (\partial_s A)(s, 1)$. Then the first variation formula for geodesics implies

$$R(s)R'(s) = \sum_{i,j} g_{ij}(x(s), T(s)) V^i(s) T^j(s). \quad (5.2.32)$$

Again, the Gauss Lemma implies

$$A^{-1}R(s) \leq |T(s)| \leq AR(s). \quad (5.2.33)$$

We can write

$$T^i(s) = \sum_{i,j} (\partial_j E^i)(y(s)) y^j(s) \quad (5.2.34)$$

and

$$V^i(s) = \sum_{i,j,k} (\partial_j E^i)(y(s)) (\partial_k G^j)(x(s)) u^k(a) = u^i(a). \quad (5.2.35)$$

In particular, $T(0) = v_0$, so $R'(0) = \sum g_{ij}(x_0, v_0) u^i(a) v_0^j = 1 - c(a)^2$. The Cauchy-Schwarz inequality applied to (5.2.32) also tells us that

$$|R'(s)| \lesssim_A |u(a)|. \quad (5.2.36)$$

Note that $u(0) = 0$, so if a is small enough, then we have $|u(a)| \leq l/2$. Taylor's theorem applied to (5.2.36), noting $R(0) = l$ then gives that for $|s| \leq 1$,

$$l/2 \leq R(s) \leq 2l. \quad (5.2.37)$$

Differentiating (5.2.32) tells us that

$$\begin{aligned} R'(s)^2 + R''(s)R(s) &= \sum_{i,j,k} \left[(\partial_{x_k} g_{ij})(x(s), T(s)) u^i(a) T^j(s) u^k(a) \right] \\ &\quad + \left[(\partial_{v_k} g_{ij})(x(s), T(s)) u^i(a) T^j(s) (\partial_s T^k)(s) \right] \\ &\quad + \left[g_{ij}(x(s), T(s)) u^i(a) (\partial_s T^j)(s) \right]. \end{aligned} \quad (5.2.38)$$

Write the right hand side as I + II + III. Using (5.2.27), (5.2.33), (5.2.37), and the triangle

inequality gives

$$|I| \lesssim R(s)|u(a)|^2 \lesssim l|u(a)|^2. \quad (5.2.39)$$

Since $\partial_{v_k} g_{ij} = (1/2)(\partial^3 F^2 / \partial v_i \partial v_j \partial v_k)$, applying Euler's homogeneous function theorem when summing over j implies that

$$II = 0. \quad (5.2.40)$$

Applying the Cauchy-Schwarz inequality and (5.2.25), we find

$$|III| \lesssim |u(a)||T'(s)| \lesssim |u(a)|^2[|y(s)| + 1] \lesssim (l + 1)|u(a)|^2. \quad (5.2.41)$$

But now combining (5.2.39), (5.2.40), (5.2.41), and rearranging (5.2.38) shows that

$$|R''(s)| \lesssim l^{-1}|u(a)|^2 + (1 + l^{-1})|u(a)|^2 \lesssim l^{-1}|u(a)|^2. \quad (5.2.42)$$

Taylor's theorem implies there exists $B > 0$ depending only on A and d such that

$$|R(s) - (R(0) + sR'(0))| \leq Bl^{-1}|u(a)|^2 s^2. \quad (5.2.43)$$

Since $R(0) = 1$, $R'(0) = 1 - c(a)^2$, $c(0) = 1$, $u(0) = 0$, $c'(0) = 0$, $|u'(0)| \leq A|\zeta|$, and $c''(0) \leq -|\zeta|^2/A$, we conclude from (5.2.43) that as $a \rightarrow 0$, if $s > 0$ then

$$\begin{aligned} R(-s) &\leq l - sR'(0) + Bl^{-1}|u(a)|^2 s^2 \\ &\leq l - s(A^{-1}|\zeta|^2 a^2) + A^2 Bl^{-1}|\zeta|^2 s^2 a^2 + O(a^3(s + l^{-1}s^2)). \end{aligned} \quad (5.2.44)$$

For all s , $L(a) \leq R(s)$. Optimizing by picking $s = l/2A^3B$ gives

$$L(a) \leq R(-s) \leq l - l|\zeta|^2 a^2 / 4A^4 B + O(a^3). \quad (5.2.45)$$

Taking $a \rightarrow 0$ and using that $L(0) = l$ gives that $L''(0) \leq -l|\zeta|^2/4A^4B$, so setting $C = 1/4A^4B$, we find we have proved what was required. $\square$

5.3 Analysis of Regime I via Density Methods

We now begin obtaining bounds for the operator T^I specified in Proposition 5.3 by using the quasi-orthogonality estimates of Proposition 5.4. Given an input $u : X \rightarrow \mathbb{C}$, we consider a maximal $1/R$ separated subset $\mathcal{X}_R$ of X , and then consider a decomposition $u = \sum_{x_0 \in \mathcal{X}_R} u_{x_0}$ with respect to some partition of unity, where $\text{supp}(u_{x_0}) \subset B(x_0, 1/R)$. The balls $\{B(x_0, 1/R) : x_0 \in \mathcal{X}_R\}$ have finite overlap, and so

$$\|u\|_{L^p(X)} \sim \left(\sum_{x_0 \in \mathcal{X}_R} \|u_{x_0}\|_{L^p(X)}^p \right)^{1/p}. \quad (5.3.1)$$

If we set $S_{x_0, t_0} = T_{t_0}^I \{u_{x_0}\}$, then

$$\|T^I u\|_{L^p(X)} = \left\| \sum_{(x_0, t_0) \in \mathcal{X}_R \times \mathcal{T}_R} S_{x_0, t_0} \right\|_{L^p(X)}. \quad (5.3.2)$$

In this subsection, we use the quasi-orthogonality estimates of the last section to obtain L^2 estimates on partial sums of the quantities on the right hand side of (5.3.2), where the partial sum is taken over a ‘sparse’ subset of indices. We say a set $\mathcal{E} \subset \mathcal{X}_R \times \mathcal{T}_R$ has *density type* (A_0, A_1) if for any set $B \subset \mathcal{X}_R \times \mathcal{T}_R$ with $1/R \leq \text{diam}(B) \leq A_1/R$,

$$\#(\mathcal{E} \cap B) \leq RA_0 \text{diam}(B).^* \quad (5.3.3)$$

*This definition of density is chosen because it is ‘scale-invariant’ as we change the parameter R , matching the definition given in Section 3.3 when $R = 1$. Indeed, if $X = \mathbb{R}^d$, $\mathcal{X}_R = (\mathbb{Z}/R)^d$, and $\mathcal{T}_R = \mathbb{Z}/R$, then a set $\mathcal{E} \subset (\mathbb{Z}/R)^d \times (\mathbb{Z}/R)$ has density type (A_0, A_1) if and only if $R \mathcal{E} \subset \mathbb{Z}^d \times \mathbb{Z}$ has density type (A_0, A_1) .

To obtain L^p bounds from these L^2 bounds, in the next section we will perform a *density decomposition* to break up $\mathcal{X}_R \times \mathcal{T}_R$ into families of indices with controlled density, and then apply Proposition 5.7 on each subfamily to control (5.3.2) via an interpolation.

Proposition 5.7. *Fix $A \geq 1$. Consider a set $\mathcal{E} \subset \mathcal{X}_R \times \mathcal{T}_R$. For each $(x_0, t_0) \in \mathcal{E}$, consider $S_{x_0, t_0} : X \rightarrow \mathbb{C}$ such that the family $\{S_{x_0, t_0}\}$ satisfies the pointwise and quasi-orthogonality estimates of Proposition 5.4. Write $\mathcal{E} = \bigcup_{k=0}^{\infty} \mathcal{E}_k$, where*

$$\mathcal{E}_0 = \{(x, t) \in \mathcal{E} : 0 \leq |t| \leq 1/R\}$$

and for $k > 0$, define

$$\mathcal{E}_k = \{(x, t) \in \mathcal{E} : 2^{k-1}/R < |t| \leq 2^k/R\}.$$

Suppose that for each k , the set $\mathcal{E}_k$ has density type $(D, 2^k)$. Then

$$\left\| \sum_k \sum_{(x_0, t_0) \in \mathcal{E}_k} 2^{k \frac{d-1}{2}} S_{x_0, t_0} \right\|_{L^2(X)}^2 \lesssim R^d \log(D) D^{\frac{2}{d-1}} \sum_k 2^{k(d-1)} \#\mathcal{E}_k.$$

Proof. Write $F = \sum_k F_k$, where

$$F_k = 2^{k \frac{d-1}{2}} \sum_{(x_0, t_0) \in \mathcal{E}_k} S_{x_0, t_0}. \quad (5.3.4)$$

Our goal is to bound $\|F\|_{L^2(M)}$. Applying the Cauchy-Schwarz inequality, we have

$$\|F\|_{L^2(X)}^2 \leq \log(D) \left(\sum_{k \leq \log(D)} \|F_k\|_{L^2(X)}^2 + \left\| \sum_{k \geq \log(D)} F_k \right\|_{L^2(X)}^2 \right). \quad (5.3.5)$$

Without loss of generality, increasing the implicit constant in the final result by applying the triangle inequality, we can assume that $\{k : \mathcal{E}_k \neq \emptyset\}$ is 10-separated, and that all values of t with $(x, t) \in \mathcal{E}$ are positive. Thus if F_k and $F_{k'}$ are both nonzero functions, then $k = k'$ or $|k - k'| \geq 10$.

Let us estimate $\langle F_k, F_{k'} \rangle$ for $k \geq k' + 10$. We write

$$\langle F_k, F_{k'} \rangle = \sum_{(x_0, t_0) \in \mathcal{E}_k} \sum_{(x_1, t_1) \in \mathcal{E}_{k'}} 2^{k \frac{d-1}{2}} 2^{k' \frac{d-1}{2}} \langle S_{x_0, t_0}, S_{x_1, t_1} \rangle. \quad (5.3.6)$$

For each $(x_0, t_0) \in \mathcal{E}_k$, and each $k' \leq k - 10$, consider the set

$$\mathcal{G}_0(x_0, t_0, k') = \{(x_1, t_1) \in \mathcal{E}_{k'} : |(t_0 - t_1) - d_X^-(x_0, x_1)| \leq 2^{k'+5}/R\}, \quad (5.3.7)$$

Also consider the sets of indices

$$\mathcal{G}_l^+(x_0, t_0, k') = \{(x_1, t_1) \in \mathcal{E}_{k'} : 2^l/R < |(t_0 - t_1) + d_X^+(x_0, x_1)| \leq 2^{l+1}/R\}. \quad (5.3.8)$$

and

$$\mathcal{G}_l^-(x_0, t_0, k') = \{(x_1, t_1) \in \mathcal{E}_{k'} : 2^l/R < |(t_0 - t_1) - d_X^-(x_0, x_1)| \leq 2^{l+1}/R\}. \quad (5.3.9)$$

If we set

$$\mathcal{G}_0(x_0, t_0, k') = (\mathcal{G}_l^+(x_0, t_0, k') \cup \mathcal{G}_l^-(x_0, t_0, k')) \quad (5.3.10)$$

and

$$\begin{aligned} \mathcal{G}_l(x_0, t_0, k') &= \left(\mathcal{G}_l^+(x_0, t_0, k') \cup \mathcal{G}_l^-(x_0, t_0, k') \right) \\ &\quad - \bigcup_{r < l} \left(\mathcal{G}_r^+(x_0, t_0, k') \cup \mathcal{G}_r^-(x_0, t_0, k') \right). \end{aligned} \quad (5.3.11)$$

Then $\mathcal{E}_{k'}$ is covered by $\mathcal{G}_0(x_0, t_0, k')$ and $\mathcal{G}_l(x_0, t_0, k')$ for $k' + 5 \leq l \leq 10 \log R$. Define

$$B_0(x_0, t_0, k') = \sum_{(x_1, t_1) \in \mathcal{G}_0(x_0, t_0, k')} 2^{k \frac{d-1}{2}} 2^{k' \frac{d-1}{2}} |\langle S_{x_0, t_0}, S_{x_1, t_1} \rangle|, \quad (5.3.12)$$

and

$$B_l(x_0, t_0, k') = \sum_{(x_1, t_1) \in \mathcal{G}_l(x_0, t_0, k')} 2^{k \frac{d-1}{2}} 2^{k' \frac{d-1}{2}} |\langle S_{x_0, t_0}, S_{x_1, t_1} \rangle|. \quad (5.3.13)$$

We thus have

$$\langle F_k, F_{k'} \rangle \leq \sum_{(x_0, t_0) \in \mathcal{E}_k} B_0(x_0, t_0, k') + \sum_{(x_0, t_0) \in \mathcal{E}_k} \sum_{k'+5 \leq l \leq 10 \log R} B_l(x_0, t_0, k'). \quad (5.3.14)$$

Using the density properties of $\mathcal{E}$, we can control the size of the index sets $\mathcal{G}_\bullet(x_0, t_0, k')$, and thus control the quantities $B_\bullet(x_0, t_0, k')$. The rapid decay of Proposition 5.4 means that only $\mathcal{G}_0(x_0, t_0, k')$ needs to be estimated rather efficiently:

- Start by bounding the quantities $B_0(x_0, t_0, k')$. If $(x_1, t_1) \in \mathcal{G}_0(x_0, t_0, k')$, then

$$|d_X^-(x_0, x_1) - (t_0 - t_1)| \leq 2^{k'+5}/R, \quad (5.3.15)$$

Thus if we consider $\text{Ann}(x_0, t_0, k') = \{x_1 : |d_X^-(x_0, x_1) - t_0| \leq 2^{k'+8}/R\}$, which is a geodesic annulus of radius $\sim 2^k/R$ and thickness $O(2^{k'}/R)$, then

$$\mathcal{G}_0(x_0, t_0, k') \subset \text{Ann}(x_0, t_0, k') \times [2^{k'}/R, 2^{k'+1}/R]. \quad (5.3.16)$$

The latter set is covered by $O(2^{(k-k')(d-1)})$ balls of radius $2^{k'}/R$, and so the density properties of $\mathcal{E}_{k'}$ implies that

$$\#(\mathcal{G}_0(x_0, t_0, k')) \lesssim D 2^{(k-k')(d-1)} 2^{k'}. \quad (5.3.17)$$

Since $k \geq k' + 10$, for $(x_1, t_1) \in \mathcal{G}_0(x_0, t_0, k')$ we have $d_X(x_0, x_1) \gtrsim 2^k/R$ and so

$$\langle S_{x_0, t_0}, S_{x_1, t_1} \rangle \lesssim R^d 2^{-k(\frac{d-1}{2})}. \quad (5.3.18)$$

But putting together (5.3.17) and (5.3.18) gives that

$$\begin{aligned} B_0(x_0, t_0, k') &\leq (2^{k\frac{d-1}{2}} 2^{k'\frac{d-1}{2}})(D2^{(k-k')(d-1)} 2^{k'})(R^d 2^{-k(\frac{d-1}{2})}) \\ &= DR^d 2^{k(d-1)} 2^{-k'\frac{d-3}{2}}. \end{aligned}$$

Thus for each k , since $d \geq 4$,

$$\sum_{(x_0, t_0) \in \mathcal{E}_k} \sum_{k' \in [\log(D), k-10]} B_0(x_0, t_0, k') \lesssim R^d 2^{k(d-1)} \#\mathcal{E}_k. \quad (5.3.19)$$

- Next we bound $B_l(x_0, t_0, k')$ for $k' + 5 \leq l \leq k - 5$. The set $\mathcal{G}_l^+(x_0, t_0, k')$ is empty in this case. Thus

$$\mathcal{G}_l(x_0, t_0, k') \subset (\text{Ann} \cup \text{Ann}') \times [t_0 - 2^{k'}/R, t_0 + 2^{k'}/R], \quad (5.3.20)$$

where

$$\text{Ann} = \{x \in X : |d_X^-(x_0, x) - (t_0 - 2^{k'})/R| \leq 100 \cdot 2^l/R\} \quad (5.3.21)$$

and

$$\text{Ann}' = \{x \in X : |d_X^-(x_0, x) - (t_0 + 2^{k'})/R| \leq 100 \cdot 2^l/R\}, \quad (5.3.22)$$

These are geodesic annuli of thickness $O(2^l/R)$ and radius $\sim 2^k$. Thus $\mathcal{G}_l(x_0, t_0, k')$ is covered by $O(2^{(l-k')} 2^{(k-k')(d-1)})$ balls of radius $2^{k'}/R$, and the density of $\mathcal{E}_{k'}$ implies that

$$\#(\mathcal{G}_l(x_0, t_0, k')) \lesssim RD 2^{(l-k')} 2^{(k-k')(d-1)} 2^{k'}/R = D2^l 2^{(k-k')(d-1)}. \quad (5.3.23)$$

For $(x_1, t_1) \in \mathcal{G}_l(x_0, t_0, k')$, $d_X(x_0, x_1) \sim 2^k/R$, and thus Proposition 5.4 implies

$$|\langle S_{x_0, t_0}, S_{x_1, t_1} \rangle| \lesssim R^d 2^{-k \frac{d-1}{2}} 2^{-lK}. \quad (5.3.24)$$

Thus for any $K \geq 0$,

$$\begin{aligned} B_l(x_0, t_0, k') &\lesssim_K \left(D 2^l 2^{(k-k')(d-1)} \right) R^d 2^{k \frac{d-1}{2}} 2^{k' \frac{d-1}{2}} \left(2^{-k \frac{d-1}{2}} 2^{-lK} \right) \\ &\lesssim D R^d 2^l 2^{k(d-1)} 2^{-k' \frac{d-1}{2}} 2^{-lK}. \end{aligned} \quad (5.3.25)$$

Picking $K > 1$, we conclude that

$$\sum_{(x_0, t_0) \in \mathcal{E}_k} \sum_{k' \in [\log(D), k-10]} \sum_{l \in [k'+10, k-5]} B_l(x_0, t_0, k') \lesssim R^d 2^{k(d-1)} \# \mathcal{E}_k. \quad (5.3.26)$$

- Finally, let's bound $B_l(x_0, t_0, k')$ for $k-5 \leq l \leq 10 \log R$. If either $(x_1, t_1) \in \mathcal{G}_l^-(x_0, t_0, k')$ or $(x_1, t_1) \in \mathcal{G}_l^+(x_0, t_0, k')$, then $d_X(x_0, x_1) \lesssim 2^l/R$. So $\mathcal{G}_l(x_0, t_0, k')$ is covered by $O(2^{(l-k')d})$ balls of radius $2^{k'}/R$, and thus

$$\#(\mathcal{G}_l(x_0, t_0, k')) \lesssim R D 2^{(l-k')d} (2^{k'}/R) = D 2^{(l-k')d} 2^{k'}. \quad (5.3.27)$$

For $(x_1, t_1) \in \mathcal{G}_l(x_0, t_0, k')$, we have no good control over $d_X(x_0, t_1)$ aside from the trivial estimate $d_X(x_0, x_1) \lesssim 1$. Thus Proposition 5.4 yields a bound of the form

$$|\langle S_{x_0, t_0}, S_{x_1, t_1} \rangle| \lesssim R^d 2^{-lK}. \quad (5.3.28)$$

Thus we conclude that

$$\begin{aligned} B_l(x_0, t_0, k') &\lesssim_N R^d 2^{k \frac{d-1}{2}} 2^{k' \frac{d-1}{2}} \left(D 2^{(l-k')d} 2^{k'} \right) (2^{-lN}) \\ &= D R^d 2^{k \frac{d-1}{2}} 2^{-k' \frac{d-1}{2}} 2^{-lN} \end{aligned} \quad (5.3.29)$$

Picking $K > d$, we conclude that

$$\sum_{(x_0, t_0) \in \mathcal{E}_k} \sum_{k' \in [\log(D), k-10]} \sum_{l \in [k+10, \log R]} B_l(x_0, t_0, k') \lesssim R^d. \quad (5.3.30)$$

The three bounds (5.3.19), (5.3.26) and (5.3.30) imply that

$$\sum_k \sum_{k' \in [\log(D), k]} |\langle F_k, F_{k'} \rangle| \lesssim R^d \sum_k 2^{k(d-1)} \# \mathcal{E}_k. \quad (5.3.31)$$

In particular, combining (5.3.31) with (5.3.5), we have

$$\|F\|_{L^2(X)}^2 \lesssim \log(D) \left(\sum_k \|F_k\|_{L^2(X)}^2 + R^d \sum_k 2^{k(d-1)} \# \mathcal{E}_k \right). \quad (5.3.32)$$

Next, consider some parameter a to be determined later, and decompose the interval $[2^k/R, 2^{k+1}/R]$ into the disjoint union of length D^a/R intervals of the form

$$I_{k,\mu} = [2^k/R + (\mu - 1)D^a/R, 2^k/R + \mu D^a/R] \quad \text{for } 1 \leq \mu \leq 2^k/D^a. \quad (5.3.33)$$

We thus consider a further decomposition $\mathcal{E}_k = \bigcup \mathcal{E}_{k,\mu}$, where $F_k = \sum F_{k,\mu}$. As before, increasing the implicit constant in the Proposition, we may assume without loss of generality that the set $\{\mu : \mathcal{E}_{k,\mu} \neq \emptyset\}$ is 10-separated. We now estimate

$$\sum_{\mu \geq \mu' + 10} |\langle F_{k,\mu}, F_{k,\mu'} \rangle|. \quad (5.3.34)$$

For $(x_0, t_0) \in \mathcal{E}_{k,\mu}$ and $l \geq 1$, define

$$\mathcal{H}_l(x_0, t_0, \mu') = \left\{ (x_1, t_1) \in \mathcal{E}_{k,\mu'} : \frac{2^l D^a}{2R} \leq \max(d_X(x_0, x_1), t_0 - t_1) \leq \frac{2^l D^a}{R} \right\}. \quad (5.3.35)$$

Then $\bigcup_{l \geq 1} \mathcal{H}_l(x_0, t_0, \mu')$ covers $\bigcup_{\mu \geq \mu' + 10} \mathcal{E}_{k, \mu'}$. Set

$$B'_l(x_0, t_0, \mu') = \sum_{(x_1, t_1) \in \mathcal{H}_l(x_0, t_0, \mu')} 2^{k(d-1)} |\langle S_{x_0, t_0}, S_{x_1, t_1} \rangle|. \quad (5.3.36)$$

Then

$$\langle F_{k, \mu}, F_{k, \mu'} \rangle \leq \sum_{(x_0, t_0) \in \mathcal{E}_{k, \mu}} \sum_l B'_l(x_0, t_0, \mu'). \quad (5.3.37)$$

We now bound the constants B'_l . Pick a constant r such that $d_X \leq 2^r d_X^+$ and $d_X \leq 2^r d_X^-$. As in the estimates of the quantities B_l , the quantities where l is large have negligible magnitude:

- For $l \leq k - a \log_2 D + 10r$, we have $2^l D^a / R \lesssim 2^k / R$. The set $\mathcal{H}_l(x_0, t_0, \mu')$ is covered by $O(1)$ balls of radius $2^l D^a / R$, and density properties imply

$$\#\mathcal{H}_l(x_0, t_0, \mu') \lesssim (RD)(2^l D^a / R) = D^{a+1} 2^l \quad (5.3.38)$$

For $(x_1, t_1) \in \mathcal{H}_l(x_0, t_0, \mu')$, we claim that

$$2^{k(d-1)} |\langle S_{x_0, t_0}, S_{x_1, t_1} \rangle| \lesssim R^d 2^{k(d-1)} (2^l D^a)^{-\frac{d-1}{2}}. \quad (5.3.39)$$

Indeed, for such tuples we have

$$d_X(x_0, x_1) \gtrsim 2^l D^a / R \quad \text{or} \quad \min_{\pm} |d_X^{\pm}(x_0, x_1) - (t_0 - t_1)| \gtrsim 2^l D^a / R, \quad (5.3.40)$$

and the estimate follows from Proposition 5.4 in either case. Since $d \geq 4$, we

conclude that

$$\begin{aligned}
& \sum_{l \in [1, k - a \log_2 D + 10]} B'_l(x_0, t_0, \mu') \\
& \lesssim \sum_{l \in [1, k - a \log_2 D + 10]} R^d (2^{k(d-1)}) (2^l D^a)^{-\frac{d-1}{2}} (D^{a+1} 2^l) \\
& \lesssim \sum_{l \in [1, k - a \log_2 D + 10]} R^d 2^{k(d-1)} 2^{-l \frac{d-3}{2}} D^{1-a(\frac{d-3}{2})} \\
& \lesssim R^d 2^{k(d-1)} D^{1-a(\frac{d-3}{2})}.
\end{aligned} \tag{5.3.41}$$

- For $l > k - a \log_2 D + 10r$, a tuple $(x_1, t_1) \in \mathcal{E}_k$ lies in $\mathcal{H}_l(x_0, t_0, \mu')$ if and only if $2^l D^a / 2R \leq d_X(x_0, x_1) \leq 2^l D^a / R$, since we always have

$$t_0 - t_1 \leq 2^{k+1} / R < 2^{l+r} D^a / 8R. \tag{5.3.42}$$

and so $d_X(x_0, x_1) \geq 2^l D^a / 2R$. And so

$$|(t_0 - t_1) - d_X^-(x_0, x_1)| \geq 2^{-r} 2^l D^a / 2R - 2^{k+1} / R \geq 2^{-r} 2^l D^a / 4R.$$

Also $|(t_0 - t_1) + d_X^+(x_0, x_1)| \geq |d_X^+(x_0, x_1)| \geq 2^{-r} 2^l D^a / 4R$. Thus we conclude from Proposition 5.4 that

$$2^{k(d-1)} |\langle S_{x_0, t_0}, S_{x_1, t_1} \rangle| \lesssim_K R^d 2^{k(d-1)} (2^l D^a)^{-K}. \tag{5.3.43}$$

Now $\mathcal{H}_l(x_0, t_0, \mu')$ is covered by $O((2^{l-k} D^a)^d)$ balls of radius $2^k / R$, and the density properties of $\mathcal{E}_k$ thus imply that

$$\#(\mathcal{H}_l(x_0, t_0, \mu')) \lesssim (RD)(2^{l-k} D^a)^d (2^k / R) \lesssim D^{1+ad} 2^{ld} 2^{-k(d-1)}. \tag{5.3.44}$$

Thus, picking $K > \max(d, 1 + ad)$, we conclude that

$$\begin{aligned} & \sum_{l \geq k - a \log_2 D + 10} B'_l(x_0, t_0, \mu') \\ & \lesssim R^d \sum_{l \geq k - a \log_2 D + 10} (2^{k(d-1)})(2^l D^a)^{-M} D^{1+ad} 2^{ld} 2^{-k(d-1)} \lesssim R^d. \end{aligned} \quad (5.3.45)$$

Combining (5.3.41) and (5.3.45), and then summing over the tuples $(x_0, t_0) \in \mathcal{E}_{k,\mu}$, we conclude that

$$\sum_{\mu \geq \mu' + 10} |\langle F_{k,\mu}, F_{k,\mu'} \rangle| \lesssim R^d \left(1 + 2^{k(d-1)} D^{1-a(\frac{d-3}{2})}\right) \#\mathcal{E}_{k,\mu}. \quad (5.3.46)$$

Now summing in μ , (5.3.46) implies that

$$\|F_k\|_{L^2(X)}^2 \lesssim \sum_{\mu} \|F_{k,\mu}\|_{L^2(X)}^2 + R^d \left(1 + 2^{k(d-1)} D^{1-a(\frac{d-3}{2})}\right) \#\mathcal{E}_k. \quad (5.3.47)$$

The functions in the sum defining $F_{k,\mu}$ are highly coupled, and it is difficult to use anything except the Cauchy-Schwarz inequality to break them apart. Since $\#(\mathcal{T}_R \cap I_{k,\mu}) \sim D^a$, if we set $F_{k,\mu} = \sum_{t \in \mathcal{T}_R \cap I_{k,\mu}} F_{k,\mu,t}$, where

$$F_{k,\mu,t} = \sum_{(x_0,t) \in \mathcal{E}_{k,\mu}} 2^{k\frac{d-1}{2}} S_{x_0,t}. \quad (5.3.48)$$

Then the Cauchy-Schwarz inequality implies that

$$\|F_{k,\mu}\|_{L^2(X)}^2 \lesssim D^a \sum_{t \in \mathcal{T}_R \cap I_{k,\mu}} \|F_{k,\mu,t}\|_{L^2(X)}^2. \quad (5.3.49)$$

Since the elements of $\mathcal{X}_R$ are $1/R$ separated, the functions in the sum defining $F_{k,\mu,t}$ are quite orthogonal to one another; Proposition 5.4 implies that for $x_0 \neq x_1$,

$$|\langle S_{x_0,t}, S_{x_1,t} \rangle| \lesssim R^d (R d_X(x_0, x_1))^{-K} \quad \text{for all } K \geq 0. \quad (5.3.50)$$

Thus

$$\|F_{k,\mu,t}\|_{L^2(X)}^2 \lesssim R^d 2^{k(d-1)} \#(\mathcal{E}_k \cap (X \times \{t\})). \quad (5.3.51)$$

But this means that

$$D^a \sum_{t \in \mathcal{T}_R \cap I_{k,\mu}} \|F_{k,\mu,t}\|_{L^2(X)}^2 \lesssim R^d 2^{k(d-1)} D^a \# \mathcal{E}_{k,\mu}. \quad (5.3.52)$$

Thus (5.3.47), (5.3.49), and (5.3.52) imply that

$$\begin{aligned} \|F_k\|_{L^2(X)}^2 &\lesssim \sum_{\mu} \|F_{k,\mu}\|_{L^2(X)}^2 + R^d \left(1 + 2^{k(d-1)} D^{1-a(\frac{d-3}{2})}\right) \# \mathcal{E}_k \\ &\lesssim R^d \left(2^{k(d-1)} D^a + (1 + 2^{k(d-1)} D^{1-a(\frac{d-3}{2})})\right) \# \mathcal{E}_k. \end{aligned} \quad (5.3.53)$$

Optimizing by picking $a = 2/(d-1)$ gives that

$$\|F_k\|_{L^2(X)}^2 \lesssim R^d 2^{k(d-1)} D^{\frac{2}{d-1}} \# \mathcal{E}_k. \quad (5.3.54)$$

The proof is completed by combining (5.3.32) with (5.3.54). $\square$

Combining the L^2 analysis of Section 5.3 with a density decomposition argument, we can now prove the following Lemma, which completes the analysis of the operator T^I in Proposition 5.3.

Lemma 5.8. *Using the notation of Proposition 5.3, let $T^I = \sum_{t_0 \in \mathcal{T}_R} T_{t_0}^I$, where*

$$T_{t_0}^I = b_{t_0}^I(t) (e^{2\pi i t P} \circ Q_R) dt.$$

Then for $1 \leq p < 2(d-1)/(d+1)$,

$$\|T^I u\|_{L^p(X)} \lesssim R^{-1/p'} \left(\sum_{t_0 \in \mathcal{T}_R} \left[\|b_{t_0}^I\|_{L^p(I_0)} \langle R t_0 \rangle^s \right]^p \right)^{1/p} \|u\|_{L^p(X)}.$$

We prove Lemma 5.8 via a *density decomposition* argument, adapted from the methods of [16]. Given a function $u : X \rightarrow \mathbb{C}$, we use a partition of unity to write

$$u = \sum_{x_0 \in \mathcal{X}_R} u_{x_0}, \quad (5.3.55)$$

where u_{x_0} is supported on $B(x_0, 1/R)$, and

$$\left(\sum_{x_0 \in \mathcal{X}_R} \|u_{x_0}\|_{L^1(X)}^p \right)^{1/p} \lesssim R^{-d/p'} \left(\sum_{x_0 \in \mathcal{X}_R} \|u_{x_0}\|_{L^p(X)}^p \right)^{1/p} \lesssim R^{-d/p'} \|u\|_{L^p(X)}. \quad (5.3.56)$$

Lemma 5.8 follows from the following result.

Lemma 5.9. *Fix $u \in L^p(X)$, consider 2^{1-k} separated subsets $\mathcal{X}_k$ and $\mathcal{T}_k$, and a family of functions $\{S_{x_0, t_0} : X \rightarrow \mathbb{C}\}$ satisfying the pointwise and quasi-orthogonality estimates of Proposition 5.4. Then for any function $c : \mathcal{X}_R \times \mathcal{T}_R \rightarrow \mathbb{C}$, and $1 < p < 2(d-1)/(d+1)$,*

$$\left\| \sum_{(x_0, t_0) \in \mathcal{X}_k \times \mathcal{T}_k} c(x_0, t_0) S_{x_0, t_0} \right\|_{L^p(X)} \lesssim R^{d/p'} \left(\sum_{(x_0, t_0) \in \mathcal{X}_k \times \mathcal{T}_k} \left[|c(x_0, t_0)| \langle R t_0 \rangle^s \right]^p \right)^{1/p},$$

where $s = (d-1)(1/p - 1/2)$.

To see how Lemma 5.9 implies Lemma 5.8, given the setup of Lemma 5.9, set

$$c(x_0, t_0) = \|b_{t_0}\|_{L^1(\mathbb{R})} \|u_{x_0}\|_{L^p(X)} \quad (5.3.57)$$

for $x_0 \in \mathcal{X}_a$ and $t_0 \in \mathcal{T}_b$, and define

$$S_{x_0, t_0} = \frac{1}{\|b_{t_0}\|_{L^1(\mathbb{R})} \|u_{x_0}\|_{L^1(X)}} \int b_{t_0}(t) (e^{2\pi i t P} \circ Q_R) \{u_{x_0}\} dt. \quad (5.3.58)$$

Then Lemma 5.9 and Hölder's inequality implies that

$$\begin{aligned}
\|T^I u\|_{L^p(X)} &\lesssim R^{d/p'} \left(\sum_{(x_0, t_0)} \left[\|b_{t_0}^I\|_{L^1(\mathbb{R})} \|u_{x_0}\|_{L^1(X)} \langle R t_0 \rangle^s \right]^p \right)^{1/p} \\
&\lesssim R^{-1/p'} \left(\sum_{t_0} \left[\|b_{t_0}^I\|_{L^p(I_0)} \langle R t_0 \rangle^s \right]^p \right)^{1/p} \left(\sum_{x_0} \|u_{x_0}\|_{L^p(X)}^p \right)^{1/p} \\
&\lesssim R^{-1/p'} \left(\sum_{t_0} \left[\|b_{t_0}^I\|_{L^p(I_0)} \langle R t_0 \rangle^s \right]^p \right)^{1/p} \|u\|_{L^p(X)}.
\end{aligned} \tag{5.3.59}$$

Thus we have proved Lemma 5.8. We take the remainder of this section to prove Lemma 5.9 using a density decomposition argument.

Proof of Lemma 5.9. For $p = 1$, this inequality follows simply by applying the triangle inequality and applying the pointwise estimates of Proposition 5.4. By the Marcinkiewicz interpolation theorem, to prove the result for $p > 1$, we thus only need only prove a restricted strong type version of this inequality. In other words, we can restrict c to be the indicator function of a set $\mathcal{E} \subset \mathcal{X}_R \times \mathcal{T}_R$. Write $\mathcal{E} = \bigcup_{k \geq 0} \mathcal{E}_k$, where

$$\mathcal{E}_0 = \{(x, t) \in \mathcal{E} : |t| \leq 1/R\} \tag{5.3.60}$$

and for $k > 0$, let

$$\mathcal{E}_k = \{(x, t) \in \mathcal{E} : 2^{k-1}/R < |t| \leq 2^k/R\}. \tag{5.3.61}$$

Write

$$F_k = \sum_{(x_0, t_0) \in \mathcal{E}_k} 2^{k(\frac{d-1}{2})} S_{x_0, t_0}. \tag{5.3.62}$$

Our proof will be completed if we can show that

$$\left\| \sum_k F_k \right\|_{L^p(X)} \lesssim R^{d(1-1/p)} \left(\sum_k 2^{k(d-1)} \# \mathcal{E}_k \right)^{1/p}. \tag{5.3.63}$$

To prove (5.3.63), we perform a density decomposition on the sets $\{\mathcal{E}_k\}$. Applying Lemma

3.7, rescaling our metric so $\mathcal{E}$ is 1-separated, we can obtain a disjoint union

$$\mathcal{E}_k = \bigcup_{u \geq 0} \mathcal{E}_k(u), \quad (5.3.64)$$

where $\mathcal{E}_k(u)$ has density type $(R2^u, 2^k/100R)$, and is covered by $B_{k,u,1}^*, \dots, B_{k,u,N_{k,u}}^*$, where the family $\{B_{k,u,n}\}$ are a family of disjoint balls, each of radius at most $2^k/100R$ such that

$$\sum_n \text{rad}(B_{k,u,n}) \leq 2^{-u}/R \# \mathcal{E}_k. \quad (5.3.65)$$

and such that $\mathcal{E}_k(u)$ is covered by the balls $\{B_{k,u,n}^*\}$, where, for a ball B , B^* denotes the ball with the same center as B , but 5 times the radius. Now write

$$F_{k,u} = \sum_{(x_0, t_0) \in \mathcal{E}_k(u)} 2^{k(\frac{d-1}{2})} S_{x_0, t_0}. \quad (5.3.66)$$

We can apply Lemma 5.7 of the last section to $\{\mathcal{E}_k(u)\}$, which implies that

$$\left\| \sum_k F_{k,u} \right\|_{L^2(X)} \lesssim R^{d/2} \left(u^{1/2} 2^{u(\frac{1}{d-1})} \right) \left(\sum_k 2^{k(d-1)} \# \mathcal{E}_k \right)^{1/2}. \quad (5.3.67)$$

Let $(y_{k,u,n}, t_{k,u,n})$ denote the center of $B_{k,u,n}$. Then the functions

$$\sum_{(x_0, t_0) \in [B_{k,u,n} \cap \mathcal{E}_k(u)]} S_{x_0, t_0} \quad (5.3.68)$$

have mass concentrated on the geodesic annulus $\text{Ann}_{k,u,n} \subset X$ with center $y_{k,u,n}$, with radius $t_{k,u,n} \sim 2^k/R$, and with thickness $5 \text{ rad}(B_{k,u,n})$. Thus

$$\sum_n |\text{Ann}_{k,u,n}| \lesssim \sum_n (2^k/R)^{d-1} \text{rad}(B_{k,u,n}) \leq (2^k/R)^{d-1} R^{-1} 2^{-u} \# \mathcal{E}_k. \quad (5.3.69)$$

If we set $\Lambda_u = \bigcup_k \bigcup_n \text{Ann}_{k,u,n}$, then

$$|\Lambda_u| \lesssim R^{-d} 2^{-u} \sum_k 2^{k(d-1)} \# \mathcal{E}_k \quad (5.3.70)$$

Since $1/p - 1/2 > 1/(d-1)$, so that $s > 1$, Hölder's inequality implies that

$$\begin{aligned} \left\| \sum_k F_{k,u} \right\|_{L^p(\Lambda_u)} &\lesssim |\Lambda_u|^{1/p-1/2} \left\| \sum_k F_{k,u} \right\|_{L^2(\Lambda_{k,u})} \\ &\lesssim \left(R^{-d} 2^{-u} \sum_k 2^{k(d-1)} \# \mathcal{E}_k \right)^{1/p-1/2} \\ &\quad \left(R^d \left(u 2^{u(\frac{2}{d-1})} \right) \sum_k 2^{k(d-1)} \# \mathcal{E}_k \right)^{1/2} \\ &= R^{d(1-1/p)} \left(u^{1/2} 2^{-u(\frac{s-1}{d-1})} \right) \left(\sum_k 2^{k(d-1)} \# \mathcal{E}_k \right)^{1/p} \\ &\lesssim R^{d/p'} 2^{-u\varepsilon} \left(\sum_k 2^{k(d-1)} \# \mathcal{E}_k \right)^{1/p} \end{aligned} \quad (5.3.71)$$

for some suitable small $\varepsilon > 0$. For each $(x_0, t_0) \in \mathcal{E}_{k,a,b}(u) \cap B_{k,u,n}$, using the pointwise bounds of Proposition 5.4 we calculate that

$$\begin{aligned} \|S_{x_0,t_0}\|_{L^1(\Lambda(u)^c)} &= R^d \int_{\text{Ann}_R^c} \langle R d_X(x, x_0) \rangle^{-(\frac{d-1}{2})} \langle R |t_0 - d_X(x, x_0)| \rangle^{-M} dx \\ &\lesssim R^{(\frac{d+1}{2}-M)} \int_{5 \text{ rad}(B_{k,u,n})}^{O(1)} (t_{k,u,n} + s)^{(\frac{d-1}{2})} s^{-M} ds \\ &\lesssim R^{(\frac{d+1}{2}-M)} \text{rad}(B_{k,u,n})^{1-M} t_{k,u,n}^{(\frac{d-1}{2})} \\ &\lesssim 2^{k(\frac{d-1}{2})} (R \text{rad}(B_{k,u,n}))^{1-M}. \end{aligned} \quad (5.3.72)$$

Thus

$$\|S_{x_0,t_0}\|_{L^1(\text{Ann}_n^c)} \lesssim 2^{k(\frac{d-1}{2})} (R \text{rad}(B_{k,u,n}))^{1-M} \quad (5.3.73)$$

Because the set of points in $\mathcal{E}_k$ is $1/R$ separated, there are at most $O((R \text{rad}(B_{k,u,n}))^{d+1})$ points in $\mathcal{E}_k(u) \cap B_{k,u,n}$, and so the triangle inequality implies that

$$\left\| \sum_{(x_0,t_0) \in \mathcal{E}_k(u) \cap B_{k,u,n}} S_{x_0,t_0} \right\|_{L^1(\Lambda(u)^c)} \lesssim 2^{k(\frac{d-1}{2})} (R \text{rad}(B_{k,u,n}))^{d+2-M}. \quad (5.3.74)$$

Since $\#\mathcal{E}_k \cap B_{k,u,n} \geq R2^u \text{rad}(B_{k,u,n})$, and $\mathcal{E}_k$ is $1/R$ discretized, we must have

$$\text{rad}(B_{k,u,n}) \geq (2^u/2^d)^{\frac{1}{d-1}} 1/R. \quad (5.3.75)$$

Thus

$$\begin{aligned} \left\| \sum_k F_{k,u} \right\|_{L^1(\Lambda(u)^c)} &\lesssim_X \sum_k \sum_n 2^{k(d-1)} (R \text{rad}(B_{k,u,n}))^{d+2-M} \\ &\lesssim \sum_k 2^{k(d-1)} \left(R \min_n \text{rad}(B_{k,u,n}) \right)^{d+1-M} \left(\sum_n R \text{rad}(B_{k,u,n}) \right) \\ &\lesssim \sum_k 2^{k(d-1)} 2^{u(\frac{d+1-M}{d-1})} (2^{-u} \#\mathcal{E}_k) \\ &\lesssim 2^{u(\frac{2-M}{d-1})} \sum_k 2^{k(d-1)} \#\mathcal{E}_k \end{aligned} \quad (5.3.76)$$

Picking $M > 2 + (1/p')(1/p - 1/2)^{-1}$, and interpolating with the bounds on $\|\sum_k F_{k,u}\|_{L^2(X)}$ yields that

$$\begin{aligned} \left\| \sum_k F_{k,u} \right\|_{L^p(\Lambda(u)^c)} &\lesssim \left(2^{u(\frac{2-M}{d-1})} \right)^{2/p-1} \left(R^{d/2} \left(u^{1/2} 2^{u(\frac{1}{d-1})} \right) \right)^{2/p'} \left(\sum_k 2^{k(d-1)} \#\mathcal{E}_k \right)^{1/p} \\ &\lesssim 2^{-u\varepsilon} R^{d/p'} \sum_k \left(\sum 2^{k(d-1)} \#\mathcal{E}_k \right)^{1/p}. \end{aligned} \quad (5.3.77)$$

So now we know

$$\left\| \sum_k F_{k,u} \right\|_{L^p(X)} \lesssim R^{d(1-1/p)} 2^{-u\varepsilon} \left(\sum_k 2^{k(d-1)} \#\mathcal{E}_k \right)^{1/p}. \quad (5.3.78)$$

The exponential decay in u allows us to sum in u to obtain that

$$\left\| \sum_u \sum_k F_{k,u} \right\|_{L^p(X)} \lesssim R^{d/p'} \left(\sum_k 2^{k(d-1)} \#\mathcal{E}_k \right)^{1/p}. \quad (5.3.79)$$

This is precisely the bound we were required to prove. $\square$

5.4 Analysis of Regime II via Local Smoothing

In this section, we bound the operators $\{T^I\}$, by a reduction to an endpoint local smoothing inequality, namely, the inequality that

$$\|e^{2\pi i t P} f\|_{L^{p'}(X) L_t^{p'}(I_0)} \lesssim \|f\|_{L_{s-1/p'}^{p'}}. \quad (5.4.1)$$

This inequality is proved in Corollary 1.2 of [26] for $1 < p < 2(d-1)/(d+1)$ for $P \in \text{COP}_1^+(X)$ satisfying the co-sphere assumption of Theorem 2.4. The range of p here is also precisely the range of p in Theorem 2.4. Alternatively, Lemma 5.9 can be used to prove (5.4.1) independently of [26] in the same range by a generalization of the method of Section 10 of [16].

Lemma 5.10. *Using the notation of Proposition 5.3, let*

$$T^I = \int b^I(t)(e^{2\pi i t P} \circ Q_R) dt.$$

For $1 < p < 2(d-1)/(d+1)$, we then have

$$\|T^I u\|_{L^p(X)} \lesssim R^{s-1/p'} \|b^I\|_{L^p(I_0)} \|u\|_{L^p(X)}.$$

Proof. For each R , the class of operators of the form $\{T^I\}$ formed from a given function b^I is closed under taking adjoints. Indeed, if T^I is obtained from b^I , then $(T^I)^*$ is obtained from the multiplier $\overline{b^I}$. Because of this self-adjointness, if we can prove that

$$\|T^I u\|_{L^{p'}(X)} \lesssim R^{s-1/p'} \|b^I\|_{L^p(I_0)} \|u\|_{L^{p'}(X)}, \quad (5.4.2)$$

then we obtain the required result by duality. We apply this duality because it is easier to exploit local smoothing inequalities in $L^{p'}(X)$ since now $p' > 2$.

Applying Hölder and Minkowski's inequalities, we find that

$$\|T^H u\|_{L^{p'}(X)} \leq \|b^H\|_{L^p(\mathbb{R})} \left\| \left(\int_{I_0} |e^{2\pi i t P}(Q_R u)|^{p'} \right)^{1/p'} \right\|_{L^{p'}(X)}. \quad (5.4.3)$$

Applying the endpoint local smoothing inequality (5.4.1), we conclude that

$$\begin{aligned} \|T^H u\|_{L^{p'}(X)} &\lesssim \|b^H\|_{L^p(\mathbb{R})} \|e^{2\pi i P}(Q_R u)\|_{L_t^{p'} L_x^{p'}} \\ &\lesssim \|b^H\|_{L^p(\mathbb{R})} \|Q_R u\|_{L_{s-1/p'}^q(X)}, \end{aligned} \quad (5.4.4)$$

The operator $L_R = R^{1/p'-s} \langle P \rangle^{s-1/p'} Q_R$ can be written as $l_R(P)$, where $l_R(\lambda) = R^{1/p'-s} \langle \lambda \rangle^{s-1/p'} q(\lambda/R)$. Since the functions l_R are *uniformly* symbols of order zero, i.e. for $R \geq 1$ we have estimates $(\partial^\alpha l_R)(\lambda) \lesssim_\alpha \langle \lambda \rangle^{-\alpha}$ which are uniform in R , Corollary 4.3.2 of [42] implies that the operators $\{L_R\}$ are uniformly bounded on $L^p(X)$. But this means that

$$\|Q_R u\|_{L_{s-1/p'}^q(X)} = R^{s-1/p'} \|L_R u\|_{L^q(X)} \lesssim R^{s-1/p'} \|u\|_{L^p(X)}. \quad (5.4.5)$$

Thus we conclude that

$$\|T^H u\|_{L^{p'}(X)} \lesssim R^{s-1/p'} \|b^H\|_{L^p(I_0)} \|u\|_{L^{p'}(X)}, \quad (5.4.6)$$

which completes the proof. $\square$

Combining Lemma 5.8 and Lemma 5.10 completes the proof of Proposition 5.3, and thus of inequality (5.0.3). Since (5.0.2) was already proven as a consequence of Lemma 5.1, this completes the proof of Theorem 2.4.

Chapter 6

Combining Scales Via Atomic Decompositions

In this chapter, we discuss a prospective method of combining frequency scales for spectral multipliers, proving Theorem 2.5. We take an operator P satisfying Assumptions A, B, and C, and show we can combine estimates between frequency scales under an assumption which is *slightly stronger* than controlling each part of the operator separately. We do this by adapting the techniques of Heo, Nazarov, and Seeger from [15] to the case of compact manifolds, which we previously discussed in Section 3.4. We now restate the setup to Theorem 2.5.

Fix $k \geq 0$, and consider a pair of maximal 2^{-k} separated subsets $\mathcal{X}_k$ and $\mathcal{T}_k$ of X and of $[0, \Pi]$, where $1/\Pi$ is the spacing between points in the arithmetic progression that the eigenvalues of the operator P are contained in. Fix a family of L^1 -normalized functions $\mathbf{b} = \{b_{t_0}\}$ and $\mathbf{u} = \{u_{x_0}\}$, with u_{x_0} supported on a 2^{1-k} neighborhood of x_0 , and b_{t_0} on a 2^{1-k} interval centered at t_0 . Also fix a bump function $q \in C_c^\infty(\mathbb{R})$ with $\text{supp}(q) \subset [1/4, 4]$ and $q(\lambda) = 1$ for $\lambda \in [1/2, 2]$, and define $Q_k = q(P/2^k)$, which we view as ‘frequency localizations’ at a scale 2^k . For each $(x_0, t_0) \in \mathcal{X}_k \times \mathcal{T}_k$, define $S_{x_0, t_0} = \int b_{t_0}(t)(\cos(2\pi itP) \circ$

$Q_k)\{u_{x_0}\}$, a time average of a frequency localized solution to the wave equation $\partial_t^2 u = -P^2 u$. Finally, define an operator A_k from functions on $X_k \times \mathcal{T}_k$ to functions on X by $A_k\{c\} = \sum_{(x_0, t_0) \in X_k \times \mathcal{T}_k} c(x_0, t_0) S_{x_0, t_0}$. We assume uniform bounds on such operators.

Assumption Wave-Bound $_q(P)$: There exists a constant $C_0 > 0$ such that

$$\|A_k\{c\}\|_{L^q(X)} \leq C_0 2^{kd/q'} \left(\sum_{(x_0, t_0) \in X_k \times \mathcal{T}_k} [|c(x_0, t_0)| \langle 2^k t_0 \rangle^s]^q \right)^{1/q}.$$

uniformly in k , b , and u , with $s = (d-1)(1/q - 1/2)$.

As we have seen in the previous chapter, such a bound naturally arises in the study of averages of the wave equation. In particular, Wave-Bound $_q(P)$ implies that for any function a , if $a_k = \chi(t)a(2^k t)$ for some $\chi \in C_c^\infty(\mathbb{R})$ with $1 = \sum \chi(\lambda/2^k)$ for $\lambda \neq 0$, then for any $1 < p \leq q$,

$$\|a_k(P/2^k)\|_{L^p(X) \rightarrow L^p(X)} \lesssim \|a\|_{R^{s,p}[0,\infty)} \quad \text{for } s = (d-1)(1/p - 1/2), \quad (6.0.1)$$

uniformly in k . The main result of this section is that the interactions between these operators is not significant enough for control on the operator $a(P) = \sum a_k(P/2^k)$.

Theorem 2.5. *Let X be a compact manifold, and let $P \in COP_1^+(X)$ satisfy Assumptions A, B, and C. Fix $1 < p < q < 2$ with $q < 2d/(d+1)$, and suppose that Wave-Bound $_q(P)$ holds. Then for any regulated function a , if $s = (d-1)(1/p - 1/2)$,*

$$\|a\|_{M_{dil}^p(X,P)} \sim \|a\|_{R^{s,p}[0,\infty)}.$$

Since the right hand side of the inequality in Theorem 2.5 is invariant under dilations of a , it suffices to prove a bound of the form $\|a(P)\|_{L^p(X) \rightarrow L^p(X)} \lesssim \|a\|_{R^{s,q}[0,\infty)}$. To prove Theorem 2.5, we use an analogue of the technique of atomic decompositions introduced in

Section 3.4. The following lemma describes the properties of this atomic decomposition useful to us, and is proved in an appendix.

Lemma 6.1. *Consider coordinate charts $\{U_\alpha\}$ covering X . Then, for any measurable function $u : X \rightarrow \mathbb{C}$, if we consider the dyadic decomposition $u = \sum u_k$, where $u_k = Q_k u$ is a quasi-mode with eigenvalue 2^k , then we have a further decomposition*

$$u_k = \sum_\alpha \sum_H \sum_{W \in \mathcal{W}_{\alpha,H}} A_{\alpha,k,H,W},$$

where H ranges over powers of 2. For each α and H , $\mathcal{W}_{\alpha,H}$ is a family of almost disjoint dyadic cubes in the coordinate system U_α whose union is a set $\Omega_{\alpha,H}$, such that the following properties hold:

- The 10-fold dilates $\{W^* : W \in \mathcal{W}_{\alpha,H}\}$ have the bounded overlap property.
- Let $l(W)$ denote the side-length of the cube W . If $l(W) = 2^l$, then $A_{\alpha,k,H,W} = 0$ for $k < -l$.
- For each W , $\text{supp}(A_{\alpha,k,H,W}) \subset W$, but as H varies, the functions $\{A_{\alpha,k,H,W}\}$ have almost disjoint support.
- For each H ,

$$\left(\sum_k \sum_\alpha \sum_{W \in \mathcal{W}_{\alpha,H}} \|A_{\alpha,k,H,W}\|_{L^2(X)}^2 \right)^{1/2} \lesssim H |\Omega_{\alpha,H}|,$$

- For any choice of indices $k(\alpha, W)$ for each α and W , we have

$$\left(\sum_\alpha \sum_{W \in \mathcal{W}_{\alpha,H}} |W| \|A_{\alpha,k(\alpha,W),H,W}\|_{L^\infty(X)}^p \right)^{1/p} \lesssim H |\Omega_{\alpha,H}|.$$

- For each α ,

$$\left(\sum_H H^p |\Omega_{\alpha,H}| \right)^{1/p} \lesssim \|u\|_{L^p(X)}.$$

Proof. We mostly adapt the approach of [15] to the compact manifold setting, using a decomposition phrased in terms of a square function of Peetre [32]. Define

$$\mathfrak{S}u(x) = \left(\sum_k \sup_{d(x,x') \leq 100d2^{-k}} |u(x')|^2 \right)^{1/2}. \quad (6.0.2)$$

Then, adapting a Euclidean result of Peetre [32], Seeger [36] proved that $\mathfrak{S}$ is bounded on $L^p(X)$ for all $1 < p < \infty$.

Consider a partition of unity $\{\eta_\alpha\}$, with η_α supported on a pre-compact subset V_α of U_α . For each dyadic H , define $\Omega_{\alpha,H} = \{x \in V_\alpha : \mathfrak{S}u(x) > H\}$. In the coordinate system U_α , define the set $\mathcal{Q}_{\alpha,H,k}$ to be the set of all dyadic cubes Q of side-length 2^{-k} such that $|Q \cap \Omega_{\alpha,H}| \geq |Q|/2$ but $|Q \cap \Omega_{\alpha,2H}| < |Q|/2$. Define

$$\Omega_{\alpha,H}^* = \{x \in V_\alpha : M\mathbb{I}_{\Omega_{\alpha,H}}(x) > 100^{-d}\}. \quad (6.0.3)$$

Then $\Omega_{\alpha,H}^*$ is open, and by the weak L^1 boundedness of the Hardy-Littlewood maximal function, $|\Omega_{\alpha,H}^*| \lesssim |\Omega_{\alpha,H}|$. We perform a Whitney decomposition of the set $\Omega_{\alpha,H}^*$. Let $\mathcal{W}_{\alpha,H}$ be the set of all dyadic cubes W which are maximal among cubes whose 50-fold dilate of W is contained in $\Omega_{\alpha,H}^*$. If we let W^* denote the 10-fold dilate of W , then the sets $\{W^* : W \in \mathcal{W}_{\alpha,H}\}$ have the bounded overlap property.

For each $W \in \mathcal{W}_{\alpha,H}$, define

$$A_{\alpha,k,H,W} = \eta_\alpha u_k \sum_{\substack{Q \in \mathcal{Q}_{\alpha,H,k} \\ Q \subset W}} \mathbb{I}_Q. \quad (6.0.4)$$

Each dyadic cube $Q \in \mathcal{Q}_{\alpha,H,k}$ is contained in some W , because the 50-fold dilate of Q

is contained in $\Omega_{\alpha,H}^*$. Since all dyadic side-length 2^{-k} cubes are contained in $Q_{\alpha,H,k}$ for a unique H , modulo sets of measure zero, we have

$$\eta_\alpha u_k = \sum_H \sum_{W \in \mathcal{W}_{\alpha,H}} A_{\alpha,k,H,W}. \quad (6.0.5)$$

We now verify each of the properties of the decomposition one by one, in the order stated:

- The sets $\{W^*\}$ have the bounded overlap property.
- This property follows immediately from the fact that $A_{\alpha,k,H,W}$ is a sum over side-length 2^{-k} cubes contained in W .
- The sets $Q_{\alpha,H,k}$ are disjoint as H varies, which implies the supports of the atoms $A_{\alpha,k,H,W}$ are disjoint as H varies.
- We have a pointwise bound

$$\sum_\alpha \sum_k \sum_W |A_{\alpha,k,H,W}|^2 \lesssim H^2. \quad (6.0.6)$$

Indeed, if $x \in Q$ for some $Q \in Q_{\alpha,H,k_0}$, then there exists a point $x' \in \Omega_{\alpha,H}$ with $d(x, x') \leq 100d2^{-k_0}$, and thus $\sum_{k \leq k_0} |u_k(x)|^2 \leq \mathfrak{S}f(x') \leq H$. But if $k_0 \leq \infty$ is the suprema of such k_0 for which we can find such a cube Q , then because x is supported on $O(1)$ of the functions $A_{\alpha,k,H,W}$ for each fixed k , we obtain that

$$\sum_\alpha \sum_k \sum_W |A_{\alpha,k,H,W}|^2 \lesssim \sum_{k \leq k_0} |u_k(x)|^2 \leq H^2. \quad (6.0.7)$$

Integrating this pointwise bound over $\Omega_{\alpha,H}^*$ gives the relevant L^2 estimates.

- The pointwise estimates of the last section imply that $|A_{\alpha,k,H,W}| \lesssim H$, and since the supports of the cubes in $\mathcal{W}_{\alpha,H}$ are disjoint for each H and have $\Omega_{\alpha,H}$ as their union, the required estimate follows immediately.

- The last property automatically follows from the L^p boundedness of $\mathfrak{S}$.

We have verified all properties stated by the lemma, which completes the proof. $\square$

To exploit this atomic decomposition, we write

$$a(P)u = \sum_k m_k(P/2^k)\{u_k\} = \sum_\alpha \sum_k \sum_H \sum_{W \in \mathcal{W}_{\alpha,H}} m_k(P/2^k)\{A_{\alpha,k,H,W}\}. \quad (6.0.8)$$

We regroup this sum as

$$\sum_\alpha \sum_k \sum_H \sum_{l \geq 0} \sum_{\substack{W \in \mathcal{W}_{\alpha,H} \\ l(W)=l-k}} a_k(P/2^k)\{A_{\alpha,k,H,W}\}. \quad (6.0.9)$$

For each k and l , we write $a_k(P/2^k) = T_{k,l,\text{Short}} + T_{k,l,\text{Long}}$, where

$$T_{k,l,\text{Short}} = \int_0^\infty \chi(2^{k-l}t) 2^k \widehat{a_k}(2^k t) \cos(2\pi i t P) \quad (6.0.10)$$

and

$$T_{k,l,\text{Long}} = \int_0^\infty (1 - \chi(2^{k-l}t)) 2^k \widehat{a_k}(2^k t) \cos(2\pi i t P). \quad (6.0.11)$$

For $f = Tu$, we thus write $f = f_{\text{Short}} + f_{\text{Long}}$, where

$$f_{\text{Short}} = \sum_\alpha \sum_k \sum_H \sum_{l \geq 0} \sum_{\substack{W \in \mathcal{W}_{\alpha,H} \\ l(W)=l-k}} T_{k,l,\text{Short}}\{A_{\alpha,k,H,W}\} = \sum f_{\alpha,k,H,W,\text{Short}}. \quad (6.0.12)$$

and

$$f_{\text{Long}} = \sum_\alpha \sum_k \sum_H \sum_{l \geq 0} \sum_{\substack{W \in \mathcal{W}_{\alpha,H} \\ l(W)=l-k}} T_{k,l,\text{Long}}\{A_{\alpha,k,H,W}\} = \sum f_{\alpha,k,H,W,\text{Long}}, \quad (6.0.13)$$

and analyze each part using separate techniques.

6.1 Short Range Bounds

To obtain short range bounds, we exploit the propagation speed of the operators $\cos(2\pi itP)$, and the bounded overlap of the sets $W_{\alpha,H}$ for a fixed H . To obtain an L^q bound for f_{Short} , we interpolate between an L^2 bound and an L^1 bound, at a fixed quantity H . So write $f_{\text{Short}} = \sum_{\alpha} \sum_k \sum_H f_{\alpha,k,H}$. In L^2 , different frequencies are orthogonal, so that

$$\|f_{H,\text{Short}}\|_{L^2(X)} \lesssim \left(\sum_k \left\| \sum_{W \in \mathcal{W}_{\alpha,H}} f_{\alpha,k,H,W,\text{Short}} \right\|_{L^2(X)}^2 \right)^{1/2}. \quad (6.1.1)$$

We note that by the finite propagation speed of $\cos(2\pi tP)$, $\langle f_{\alpha,k,H,W,\text{Short}}, f_{\alpha,k,H,W',\text{Short}} \rangle = 0$ if $W^* \cap (W')^* = \emptyset$. By the bounded overlap property of the sets $\{W^*\}$, these functions are almost orthogonal, and so we conclude that

$$\|f_{H,\text{Short}}\|_{L^2(X)} \lesssim \left(\sum_k \sum_{W \in \mathcal{W}_{\alpha,H}} \|f_{\alpha,k,H,W,\text{Short}}\|_{L^2(X)}^2 \right)^{1/2}. \quad (6.1.2)$$

We can write $T_{k,l,\text{Short}} = a_{k,l}(P/2^k)$, where

$$a_{k,l}(t) = [2^l \widehat{\chi}(2^l \cdot) * a_k]. \quad (6.1.3)$$

Now

$$\|a_{k,l}(P/2^k)\|_{L^2(X) \rightarrow L^2(X)} = \|a_{k,l}\|_{L^\infty(\mathbb{R})} \lesssim \|a_k\|_{L^\infty(\mathbb{R})} \lesssim \|a\|_{R^{s,p}[0,\infty)}, \quad (6.1.4)$$

and so

$$\|f_{\alpha,k,H,W}\|_{L^2(X)} = \|T_{k,l,\text{Short}}\{A_{\alpha,k,H,W}\}\|_{L^2(X)} \lesssim \|a\|_{R^{s,p}[0,\infty)} \|A_{\alpha,k,H,W}\|_{L^2(X)}. \quad (6.1.5)$$

So

$$\|f_{H,\text{Short}}\|_{L^2(X)} \lesssim \|a\|_{R^{s,p}[0,\infty)} H |\Omega_H|^{1/2}. \quad (6.1.6)$$

By the finite propagation speed of the wave equation, $f_{H,\text{Short}}$ is supported on the set $\{x : (M\chi_{\Omega_H})(x) \geq 1/10^d\}$, where M is the Hardy-Littlewood maximal function; by the weak L^1 boundedness of M , this set has measure $O(|\Omega_H|)$. Thus we can use Hölder's inequality to conclude that for any $r \in [1, 2]$,

$$\|f_{H,\text{Short}}\|_{L^1(X)} \lesssim \|a\|_{R^{s,p}[0,\infty)} |\Omega_H|^{1/2} H |\Omega_H|^{1/2} = \|a\|_{R^{s,p}[0,\infty)} H |\Omega_H|. \quad (6.1.7)$$

If we interpolate between $r = 1$ and $r = 2$ using Lemma 7.6, which allows us to sum over the dyadic height scales H , we conclude that

$$\|f_{\text{Short}}\|_{L^p(X)} \lesssim \|a\|_{R^{s,p}[0,\infty)} \left(\sum H^p |\Omega_H| \right)^{1/p} \lesssim \|a\|_{R^{s,p}[0,\infty)} \|u\|_{L^p(X)}. \quad (6.1.8)$$

This completes the analysis of the short range interactions.

6.2 Long Range Bounds

The long range interactions require slightly more work. Define an operator

$$S_{l,k} C = \sum_{z_0} \sum_{t_0 \geq 2^{l-k}} C(z_0, t_0) \sum_{x_0 \in Q(z_0)} F_{x_0, t_0}, \quad (6.2.1)$$

where $U_{z_0} = \sum_{x_0 \in Q(z_0)} u_{x_0}$ and $F_{x_0, t_0} = \int b_{t_0}(t) \cos(2\pi t P) \{U_{z_0}\} dt$ for L^1 normalized functions $\{b_{t_0}\}$. The bound $\text{Wave-Bound}_q(P)$ implies an exponential decay in l on the L^p operator norm of $S_{l,k}$.

Lemma 6.2. *Suppose $1 \leq p \leq q$, and that $\text{Wave-Bound}_q(P)$ is true. Then*

$$\|S_{l,k}\{C\}\|_{L^p(X)} \lesssim 2^{-l\varepsilon} \left(\sum_{z_0} \sum_{t_0 \geq 2^{l-k}} \left[\langle 2^k t_0 \rangle^s |C(z_0, t_0)| 2^{(l-k)d/p} \|U_{z_0, t_0}\|_{L^\infty(X)} \right]^p \right)^{1/p},$$

where $s = (d - 1)(1/p - 1/2)$, and

$$\varepsilon = -\left(\frac{d-1}{2}\right)\left(\frac{1/p - 1/q}{1 - 1/q}\right),$$

which, in particular, is positive for $p < q$.

Proof. By applying the triangle inequality, we may assume that the support of C is 10-separated in the z_0 variable. Thus any point x_0 is contained in a unique side-length cube $Q(z(x_0))$ with $z(x_0)$ in $\text{supp}_z(C)$, or is not contained in any such cube. If we define $c(x_0, t_0) = C(z(x_0), t_0)\|u_{x_0}\|_{L^1(X)}$, then we can apply $\text{Wave-Bound}_q(P)$ and Hölder's inequality to conclude that

$$\|S_{l,k}C\|_{L^q(X)} \lesssim \left(\sum_{x_0} \sum_{t_0 \geq 2^{l-k}} \left[|C(z(x_0), t_0)|\|u_{x_0}\|_{L^q(X)} \langle 2^k t_0 \rangle^{s(q)}\right]^q\right)^{1/q}, \quad (6.2.2)$$

where $s(q) = (d - 1)(1/q - 1/2)$. Since the supports of the functions $\{u_{x_0}\}$ are almost disjoint, and since U_{z_0} is supported on a set of measure $2^{(l-k)d}$,

$$\sum_{z(x_0)=z_0} \|u_{x_0}\|_{L^q(X)}^q \lesssim \|U_{z_0}\|_{L^q(X)}^q \lesssim 2^{(l-k)d} \|U_{z_0}\|_{L^\infty(X)}^q. \quad (6.2.3)$$

Substituting this bound into the prior bound yields that

$$\|S_{l,k}C\|_{L^q(X)} \lesssim \left(\sum_{z_0} \sum_{t_0 \geq 2^{l-k}} \left[|C(z_0, t_0)|[2^{(l-k)d/q} \|U_{z_0}\|_{L^\infty(X)}] \langle 2^k t_0 \rangle^{s(q)}\right]^q\right)^{1/q}. \quad (6.2.4)$$

We will interpolate this bound with an L^1 bound with exponential decay, which will yield the result. We should expect F_{t_0, z_0} to be concentrated on a set of measure $O(2^{l-k} t_0^{d-1})$, namely, the annulus Ann_{t_0, z_0} of width $O(2^{l-k})$ upon a sphere of radius t_0 centered at z_0 . By

Hölder's inequality, we find that

$$\begin{aligned}
\|F_{t_0, z_0}\|_{L^1(\text{Ann}_{t_0, z_0})} &\lesssim \left(2^{l-k} t_0^{d-1}\right)^{1/2} \|F_{t_0, z_0}\|_{L^2(X)} \\
&\lesssim 2^{\frac{l-k}{2}} t_0^{\frac{d-1}{2}} \|U_{z_0}\|_{L^2(X)} \\
&\lesssim 2^{(l-k)(\frac{d+1}{2})} t_0^{\frac{d-1}{2}} \|U_{z_0}\|_{L^\infty(X)}.
\end{aligned} \tag{6.2.5}$$

Here we used the fact that $F_{t_0, z_0} = \int b_{t_0}(t)(\cos(2\pi t P) \circ Q_k)\{U_{z_0}\}$, that $\|b_{t_0}\|_{L^1(\mathbb{R})} \leq 1$, and that $\|\cos(2\pi t P) \circ Q_k\|_{L^2(X) \rightarrow L^2(X)} \lesssim 1$. so that we can apply the triangle inequality to conclude that $\|T_{t_0}^l\|_{L^2(X) \rightarrow L^2(X)} \lesssim \|b_{t_0}\|_{L^1(X)} \leq 1$. On the other hand, we can use Lemmas 4.11 and 4.12 to prove that $(\cos(2\pi t P) \circ Q_k)(x, y) \lesssim_N [2^k |d(x, y) - t|]^{-N}$, uniformly for $|t| \lesssim 1$ and k , and thus that

$$\|F_{t_0, z_0}\|_{L^1(\text{Ann}_{t_0, z_0}^c)} \lesssim_N 2^{-lN} 2^{-k} t_0^{d-1} \|U_{z_0}\|_{L^1(X)} \lesssim 2^{-lN} 2^{-k(d+1)} t_0^{d-1} \|U_{z_0}\|_{L^\infty(X)}. \tag{6.2.6}$$

Since $t_0 \geq 2^{l-k}$, we have

$$\|F_{t_0, z_0}\|_{L^1(\text{Ann}_{t_0, z_0}^c)} \lesssim_N 2^{-lN} 2^{-k(\frac{d+3}{2})} t_0^{\frac{d-1}{2}} \|U_{z_0}\|_{L^\infty(X)}, \tag{6.2.7}$$

which is smaller than the L^1 norm bound on Ann_{t_0, z_0} . Summing in t_0 and z_0 using the triangle inequality gives that

$$\begin{aligned}
\|S_{l,k} C\|_{L^1(X)} &\lesssim 2^{(l-k)(\frac{d+1}{2})} \sum_{z_0} \sum_{t_0 \geq 2^{l-k}} t_0^{\frac{d-1}{2}} |C(z_0, t_0)| \|U_{z_0}\|_{L^\infty(X)} \\
&\lesssim 2^{-l(\frac{d-1}{2})} \sum_{z_0} \sum_{t_0 \geq 2^{l-k}} |C(z_0, t_0)| \left[2^{(l-k)d} \|U_{z_0}\|_{L^\infty(X)} \right] \langle 2^k t_0 \rangle^{\frac{d-1}{2}}.
\end{aligned} \tag{6.2.8}$$

The argument is concluded by interpolation. $\square$

We now use this lemma to control the function f_{Long} . Because of the exponential decay

in l given by the lemma above, we may sum in l trivially using the triangle inequality for $1 \leq q < p$. Using L^2 orthogonality, if $f_{\text{Long}} = \sum_k f_{\text{Long},k}$, then we find that

$$\left\| \sum_k f_{\text{Long},k} \right\|_{L^p(X)} = \left(\sum_k \|f_{\text{Long},k}\|_{L^p(X)}^p \right)^{1/p}. \quad (6.2.9)$$

Now write $f_{\text{Long},k} = \sum f_{\text{Long},\alpha,l,k}$ with

$$f_{\text{Long},\alpha,l,k} = \sum_H \sum_{\substack{W \in \mathcal{W}_{\alpha,H} \\ l(W)=l-k}} T_{k,l,\text{Long}} \{A_{\alpha,k,H,W}\}. \quad (6.2.10)$$

Applying Lemma 6.2 with l and k as above, $C(z_0, t_0) = \|b_{t_0}\|_{L^1(X)}$, and with $U_{z_0} = \sum_H A_{\alpha,k,H,Q(z_0)}$, we conclude that

$$\begin{aligned} \|f_{\text{Long},\alpha,l,k}\|_{L^p(X)} &\lesssim 2^{-l\varepsilon} \left(\sum_{z_0} \sum_{t_0 \geq 2^{l-k}} [\langle 2^k t_0 \rangle^s \|b_{t_0}\|_{L^1(X)}]^p [2^{(l-k)d} \|U_{z_0}\|_{L^\infty(X)}^p] \right)^{1/p} \\ &\lesssim \|a\|_{R^{s,p}[0,\infty)} 2^{-l\varepsilon} \left(\sum_H \sum_{\substack{W \in \mathcal{W}_{\alpha,H} \\ l(W)=l-k}} |W| \|A_{\alpha,k,H,W}\|_{L^\infty(X)}^p \right)^{1/p}. \end{aligned} \quad (6.2.11)$$

Summing in k using L^p orthogonality (Lemma 7.5), and summing over α trivially, we find that

$$\|f_{\text{Long},l}\|_{L^p(X)} \lesssim 2^{-l\varepsilon} \|a\|_{R^{s,p}[0,\infty)} \left(\sum_H H^p |\Omega_H| \right) \lesssim 2^{-l\varepsilon} \|a\|_{R^{s,p}[0,\infty)} \|u\|_{L^p(X)}. \quad (6.2.12)$$

Summing in l trivially gives $\|f_{\text{Long}}\|_{L^p(X)} \lesssim \|a\|_{R^{s,p}[0,\infty)} \|u\|_{L^p(X)}$, completing the proof of the long range estimates, and thus the proof.

6.3 Wave Bounds on the Sphere

We end this chapter with a proof that $\text{Wave-Bound}_q(P)$ holds on S^d for $1 < q < 2(d-1)/(d+1)$, which completes the proof of Theorem 2.2.

Lemma 6.3. *The operator P_{SH} satisfies $\text{Wave-Bound}_q(P)$ for $1 < q < 2(d-1)/(d+1)$.*

Proof. Here $\Pi = 1$. If $\varepsilon > 0$, and a given function c satisfies $c(x_0, t_0) = 0$ for $t_0 \in [1/2 - \varepsilon, 1/2 + \varepsilon]$, then the bound has already been proven in this paper; indeed, it follows from Lemma 5.9, where $R = 2^k$, since for $t_0 \leq 1/2$ we can write $2 \cos(2\pi t P) = e^{2\pi i t P} + e^{-2\pi i t P}$, and for $t_0 \geq 1/2$ we can write $2 \cos(2\pi t P) = e^{2\pi i(t-1)P} + e^{-2\pi i(t-1)P}$. Applying Lemma 5.9 to each piece of the exponential, we have that, if f_{x_0, t_0} is as in the definition of the assumption $\text{Wave-Bound}_q(P)$,

$$\left\| \sum_{\substack{(x_0, t_0) \in X_k \times \mathcal{T}_k \\ |t_0| \leq 1/2 - \varepsilon}} c(x_0, t_0) f_{x_0, t_0} \right\|_{L^q(S^d)} \lesssim 2^{kd/q'} \left(\sum_{(x_0, t_0) \in X_k \times \mathcal{T}_k} [|c(x_0, t_0)| \langle R t_0 \rangle^s]^q \right)^{1/q} \quad (6.3.1)$$

and

$$\begin{aligned} & \left\| \sum_{\substack{(x_0, t_0) \in X_k \times \mathcal{T}_k \\ |t_0| \geq 1/2 + \varepsilon}} c(x_0, t_0) f_{x_0, t_0} \right\|_{L^q(S^d)} \\ & \lesssim 2^{kd/q'} \left(\sum_{\substack{(x_0, t_0) \in X_k \times \mathcal{T}_k \\ |t_0| \geq 1/2 + \varepsilon}} [|c(x_0, t_0)| \langle R(t_0 - 1) \rangle^s]^q \right)^{1/q} \\ & \lesssim 2^{kd/q'} \left(\sum_{\substack{(x_0, t_0) \in X_k \times \mathcal{T}_k \\ |t_0| \geq 1/2 + \varepsilon}} [|c(x_0, t_0)| \langle R t_0 \rangle^s]^q \right)^{1/q}. \end{aligned} \quad (6.3.2)$$

It remains to prove the result for c supported on $[1/2 - \varepsilon, 1/2 + \varepsilon]$. To obtain this result, if $Ux = -x$ is the reflection operator on S^d , then for $t \in [1/2 - \varepsilon, 1/2 + \varepsilon]$, for small t the operators $U \circ e^{2\pi i(t+1/2)P}$ have the same conical relation as $e^{2\pi i t P}$, and so writing out these operators using oscillatory integrals following the proof of Proposition 5.4, one can prove that the operators $S_{x_0, t_0} = U \circ \int b_{1/2+t_0} e^{2\pi i t P} \{u_{x_0}\}$ satisfy the pointwise and orthogonality

estimates of Proposition 5.4. Thus we can apply Lemma 5.9 to obtain bounds of the form

$$\begin{aligned}
& \left\| \sum_{\substack{(x_0, t_0) \in \mathcal{X}_k \times \mathcal{T}_k \\ t_0 \in [1/2 - \varepsilon, 1/2 + \varepsilon]}} \langle 2^k t_0 \rangle^{\frac{d-1}{2}} c(x_0, t_0) f_{x_0, t_0} \right\|_{L^q(S^d)} \\
& \lesssim 2^{kd/q'} \left(\sum_{\substack{(x_0, t_0) \in \mathcal{X}_k \times \mathcal{T}_k \\ |t_0| \geq 1/2 + \varepsilon}} \left[|c(x_0, t_0)| \langle R(t_0 - 1/2) \rangle^s \right]^q \right)^{1/q} \\
& \lesssim 2^{kd/q'} \left(\sum_{\substack{(x_0, t_0) \in \mathcal{X}_k \times \mathcal{T}_k \\ |t_0| \geq 1/2 + \varepsilon}} \left[|c(x_0, t_0)| \langle R t_0 \rangle^s \right]^q \right)^{1/q}.
\end{aligned} \tag{6.3.3}$$

Combining these three bounds proves that $\text{Wave-Bound}_q(P)$ holds. $\square$

Remark. We note that this analysis requires an understanding of the global geometric of the geodesic flow on S^d , in particular, that points flow into a conjugate point exactly as they exit a starting point. A similar analysis may yield $\text{Wave-Bound}_q(P)$ for operators on the rank one symmetric spaces, but is unlikely to be easy to obtain on an arbitrary Zoll manifold with periodic geodesic flow, due to the absence of information about the geometric properties of conjugate points on such a manifold.

Proof of Theorem 2.2. Together with Theorem 2.5, Lemma 6.3 shows that for $1 < p < 2(d-1)/(d+1)$, and any regulated function a ,

$$\|a\|_{M_{\text{Dil}}^p(S^d)} = \|a\|_{M_{\text{Dil}}^p(S^d, P_{\text{SH}})} \sim \|a\|_{R^{s,p}[0, \infty]}.$$

All that remains is to finish the proof of Theorem 2.2 is to prove that $\|a\|_{M^p(S^d)} \lesssim \|a\|_{R^{s,p}[0, \infty]}$.

Choose a function $\beta \in C_c^\infty[0, \infty)$ with $\beta(t) = 1$ for $|t| \leq 10d$, and write $a = a_0 + a_1$, where $a_0 = \beta a$ and $a_1 = (1 - \beta)a$. Then a simple argument as in Lemma 5.1 shows that

$$\|a_0\|_{M^p(S^d)} \lesssim \|a_0\|_{L^\infty[0, \infty)} \lesssim \|a\|_{L^\infty[0, \infty)} \lesssim \|a\|_{R^{s,p}[0, \infty)}. \tag{6.3.4}$$

We can write $T_{a_1} = \tilde{a}_1(P_{\text{SH}})$, where $\tilde{a}_1 = \text{Trans}_{-c}\{a_1\}$ with $c = \frac{d-1}{2}$. Thus Theorem 6.1

implies that

$$\|a_1\|_{M^p(S^d)} = \|\tilde{a}_1\|_{M^p(S^d, P_{\text{SH}})} \leq \|\tilde{a}_1\|_{M^p_{\text{Dil}}(S^d, P_{\text{SH}})} \lesssim \|\tilde{a}_1\|_{R^{s,p}[0,\infty)} \lesssim \|a\|_{R^{s,p}[0,\infty)}. \quad (6.3.5)$$

Together, (6.3.4) and (6.3.5) imply $\|a\|_{M^p(S^d)} \lesssim \|a\|_{R^{s,p}[0,\infty)}$, which completes the proof. $\square$

Appendix 7

Elementary Theorems

In this appendix, we list several results that occur frequently in the harmonic analysis literature. The first result, Euler's homogeneous function theorem, provides a way to relate a homogeneous function and its derivatives.

Theorem 7.1 (Euler's Homogeneous Function Theorem). *Suppose $f : \mathbb{R}^p \rightarrow \mathbb{C}$ is a smooth, homogeneous function of order s . Then for any $x \in \mathbb{R}^p$, $x \cdot \nabla f(x) = sf(x)$.*

Proof. Differentiate the relation $f(ax) = a^s f(x)$ with respect to a , and then set $a = 1$. $\square$

The Sobolev embedding theorem allows us to trade smoothness for integrability.

Theorem 7.2 (The Sobolev Embedding Theorem). *Suppose X is a d -dimensional manifold, or $X = \mathbb{R}^d$. Fix $1 \leq p \leq q < \infty$ and let $r = d(1/p - 1/q)$. Then for any $s \in \mathbb{R}$, $W^{s,q}(X) \subset W^{s-r,p}(X)$, and if $s > d/p$, $W^{s,p}(X) \subset C_b(X)$, the space of continuous, bounded functions, where these inclusions are continuous.*

Proof. For the case where $X = \mathbb{R}^d$, see Theorem 6.2.4 of [13]. To obtain the result on a general compact manifold, apply the Sobolev embedding theorem in each coordinate system. $\square$

Interpolation is essential for reducing bounds to Lebesgue norms in which the boundedness of the operators is simpler to understand. If X and Y are measure spaces, recall that an operator T mapping elements of $L^p(X)$ to measurable functions on Y is said to be of *restricted weak type* (p, q) if it has the property that for any measurable set $E \subset X$,

$$\|T\mathbb{I}_E\|_{L^{q,\infty}(Y)} \lesssim \|\mathbb{I}_E\|_{L^p(X)} = |E|^{1/p}. \quad (7.0.1)$$

where $\mathbb{I}_E$ is the indicator function of E , and the implicit constant is uniform in E .

Theorem 7.3 (The Marcinkiewicz Interpolation Theorem). *Let X and Y be measure spaces, and fix exponents $1 \leq p_0, p_1 < \infty$ and $1 \leq q_0, q_1 < \infty$. For $\theta \in [0, 1]$, define p_θ and q_θ so that*

$$1/p_\theta = \theta/p_1 + (1 - \theta)/p_0 \quad \text{and} \quad 1/q_\theta = \theta/q_1 + (1 - \theta)/q_0.$$

Let T be a linear operator mapping elements of $L^{p_0}(X) + L^{p_1}(X)$ to measurable functions on Y . Then if T is of restricted weak type (p_0, q_0) and (p_1, q_1) , then for any $\theta \in (0, 1)$, the operator T is bounded from $L^{p_\theta}(X)$ to $L^{q_\theta}(Y)$.

Proof. See Theorem 1.4.19 of [12]. □

Schur's Test for integral kernels is a simple, but often optimal method, for obtaining the boundedness of integral operators from an L^1 norm, or into an L^∞ norm. Interpolation can then be used to obtain certain bounds for operators with respect to other norms.

Theorem 7.4 (Schur's Test for Integral Kernels). *Let X and Y be measure spaces.*

- *If $K_T \in L^\infty(Y)L^p(X)$ and $f \in L^1(Y)$, then for almost every x , the integral*

$$Tf(x) = \int K_T(x, y)f(y) dy$$

is absolutely integrable, and defines an operator from $L^1(Y)$ to $L^p(X)$ such that

$$\|T\|_{L^1(Y) \rightarrow L^p(X)} \leq \|K_T\|_{L^\infty(X)L^p(Y)}.$$

- If $K_T \in L^\infty(X)L^{p'}(Y)$, then for each $f \in L^p(Y)$, and for almost every x , the integral

$$Tf(x) = \int K_T(x, y)f(y) dy$$

is absolutely integrable, defining an operator T from $L^p(Y)$ to $L^\infty(X)$, such that

$$\|T\|_{L^p(Y) \rightarrow L^\infty(X)} \leq \|K_T\|_{L^\infty(Y)L^{p'}(X)}.$$

Proof. For any measurable $f : Y \rightarrow \mathbb{C}$, define $T^\dagger f : X \rightarrow [0, \infty]$ by setting

$$T^\dagger f(x) = \int |K_T(x, y)|f(y) dy. \quad (7.0.2)$$

Applying Minkowski's inequality and Hölder's inequality gives the bounds

$$\|T^\dagger f\|_{L^p(X)} \leq \|K_T\|_{L^\infty(X)L^p(Y)} \|f\|_{L^1(Y)} \quad (7.0.3)$$

and

$$\|T^\dagger f\|_{L^\infty(X)} \leq \|K_T\|_{L^\infty(Y)L^{p'}(X)} \|f\|_{L^p(Y)}. \quad (7.0.4)$$

These bounds immediately imply the required results for the integral operator T . $\square$

We will rely on the following ' L^p orthogonality' result.

Lemma 7.5. *Let X be a measure space. Suppose $\{T_j\}$ are operators on X which satisfy*

$$\|T_j\|_{L^1(X) \rightarrow L^{1,\infty}(X)} \lesssim 1,$$

uniformly in j , and suppose that the bound

$$\left\| \sum T_j f_j \right\|_{L^2(X) \rightarrow L^2(X)} \lesssim \left(\sum_j \|f_j\|_{L^2(X)}^2 \right)^{1/2}$$

holds. Then for $1 < p \leq 2$,

$$\left\| \sum T_j f_j \right\|_{L^p(X)} \lesssim \left(\sum \|f_j\|_{L^p(X)}^p \right)^{1/p}.$$

Proof. Define a vector-valued operator T from sequences of functions on X to functions on X , such that for a sequence $f = \{f_j\}$, $Tf = \sum T_j f_j$. Then by assumption,

$$\|Tf\|_{L^2(X)} \lesssim \|f\|_{l^2(\mathbb{N})L^2(X)}, \quad (7.0.5)$$

and by the triangle inequality,

$$\|Tf\|_{L^{1,\infty}(X)} \lesssim \|f\|_{l^1(\mathbb{N})L^1(X)}. \quad (7.0.6)$$

Real interpolation (i.e. the Marcinkiewicz interpolation theorem) thus tells us that for $1 < p < \infty$,

$$\|Tf\|_{L^p(X)} \lesssim \|f\|_{l^p(\mathbb{N})L^p(X)}, \quad (7.0.7)$$

which implies the required result. $\square$

Finally, we will often use a helpful interpolation lemma.

Lemma 7.6 (A Real Interpolation Lemma). *Consider a family of functions $\{f_H\}$, where H ranges over the dyadic numbers, and fix $0 < p_0 < p_1 < \infty$. If for each H ,*

$$\|f_H\|_{L^{p_0}(X)} \lesssim HW_H^{1/p_0} \quad \text{and} \quad \|f_H\|_{L^{p_1}(X)} \lesssim HW_H^{1/p_1},$$

with implicit constants uniform in H , then for any $p_0 < p < p_1$,

$$\left\| \sum_H f_H \right\|_{L^p(X)} \lesssim \left(\sum_H H^p W_H \right)^{1/p}.$$

Proof. See Lemma 2.2 of [16]. □

In an application of Lemma 7.6, the quantities H normally stand for the ‘height’ of a given input, and the quantities $\{W_H\}$ the ‘width’, and so this interpolation result shows that if we are not proving results at an endpoint, it suffices to prove bounds for ‘a fixed height scale’.

Appendix 8

Distributions Defined by Oscillatory Distributions

The kernels of many of the Schwartz operators T we study in this thesis have kernels which are defined by certain formal expression of the form

$$K_T(x, y) = \int_{\mathbb{R}^p} a(x, y, \theta) e^{i\phi(x, y, \theta)} d\theta, \quad (8.0.1)$$

where a is a symbol, and ϕ is smooth and homogeneous of order one. In this appendix we show how these formal expressions can be interpreted in a way that ensures these expressions are well defined as distributions, provided that $\nabla_{x, y}\phi$ and $\nabla_{\theta}\phi$ have no common zeros. This ensures that, when K_T is integrated against a pair of test functions on $\mathbb{R}^m$ and $\mathbb{R}^n$, the phase becomes rapidly oscillatory at high frequencies, which ensures convergence of appropriate integrals. We call a smooth, homogeneous function ϕ such that $\nabla_{x, y, \theta}\phi$ is non-vanishing a *phase function*.

Theorem 8.1. *Let Ω be an open subset of $\mathbb{R}^n$. Let $a : \Omega \times \mathring{\mathbb{R}}^p \rightarrow \mathbb{C}$ be a symbol, and let $\phi : W \times \mathring{\mathbb{R}}^p \rightarrow \mathbb{R}$ be a phase function. Fix $\chi \in C_c^\infty(\mathbb{R}^p)$, equal to one in a neighborhood*

of the origin. The smooth functions

$$I_R^{a,\theta} = \int_{\mathbb{R}^p} a(x, \theta) \chi(\theta/R) e^{2\pi i \phi(x, \theta)} d\theta$$

converge in the weak * topology to a distribution I , independent of the choice of χ .

Proof. If $\mu < -p$, then

$$\int_{\mathbb{R}^p} a(x, \theta) e^{2\pi i \phi(x, \theta)} d\theta \quad (8.0.2)$$

is absolutely integrable, and the result follows from the dominated convergence theorem. For $\mu \geq -p$, fix $f \in C_c^\infty(\mathbb{R}^d)$ supported on a compact set $K \subset \Omega$. Our goal is to show the quantities

$$\langle I_R^{a,\theta}, f \rangle = \int a(x, \theta) \chi(\theta/R) e^{2\pi i \phi(x, \theta)} d\theta \quad (8.0.3)$$

converges as $R \rightarrow \infty$ to a quantity independent of the choice of χ , and that moreover, there exists $N > 0$ such that

$$|\langle I_R^{a,\theta}, f \rangle| \lesssim \|f\|_{C^M(\mathbb{R}^d)}, \quad (8.0.4)$$

uniformly in R . Write

$$a_R(x, \theta) = a(x, \theta) \chi(\theta/R). \quad (8.0.5)$$

We will assume without loss of generality that $a(x, \theta) = 0$ for $|\theta| \leq 1$, since the quantity

$$\int a(x, \theta) \mathbb{I}(|\theta| \leq 1) e^{2\pi i \phi(x, \theta)} d\theta \quad (8.0.6)$$

defines a function in $C^\infty(\mathbb{R}^d)$. We now apply the principle of non-stationary phase, i.e. integrating by parts. Consider the *homogenized gradient*

$$(\nabla^H \phi)(x, \theta) = (\nabla_x \phi, |\theta| \nabla_\theta \phi(x, \theta)). \quad (8.0.7)$$

Then $\nabla_{x,\theta}^H \phi$ is homogeneous of degree one in the θ variable, and $|\nabla_{x,\theta}^H \phi| > 0$. We note that

$$(\nabla_{x,\theta}^H e^{2\pi i \phi}) = (2\pi i) e^{2\pi i \phi} \nabla_{x,\theta}^H \phi. \quad (8.0.8)$$

Note also that the *formal transpose* of $\nabla_{x,\theta}^H$ is the differential operator L given for pairs of functions $F_1 : \mathbb{R}^d \times \mathbb{R}^p \rightarrow \mathbb{R}^d$ and $F_2 : \mathbb{R}^d \times \mathbb{R}^p \rightarrow \mathbb{R}^p$ by the formula

$$\begin{aligned} L(F_1, F_2) &= -(\nabla_x \cdot F_1 + \nabla_\theta \cdot (|\theta| F_2)) \\ &= -\left(\nabla_x \cdot F_1 + |\theta|^{-1} (\theta \cdot F_2) + |\theta| (\nabla_\theta \cdot F_2) \right), \end{aligned} \quad (8.0.9)$$

i.e. so that for smooth, compactly supported functions F_1 , F_2 , and G ,

$$\int_{\mathbb{R}^d \times \mathbb{R}^p} (F_1, F_2) \cdot \nabla_{x,\theta}^H G = \int_{\mathbb{R}^d \times \mathbb{R}^p} L(F_1, F_2) \cdot G. \quad (8.0.10)$$

Thus we conclude that

$$\begin{aligned} \langle I_R^{a,\theta}, f \rangle &= \frac{1}{2\pi i} \int \frac{a \cdot f}{|\nabla_{x,\theta}^H \phi|^2} (\nabla_{x,\theta}^H e^{2\pi i \phi}) \cdot (\nabla_{x,\theta}^H \phi) \\ &= \frac{1}{2\pi i} \int L \left\{ \frac{a \cdot f}{|\nabla_{x,\theta}^H \phi|^2} \nabla_{x,\theta}^H \phi \right\} e^{2\pi i \phi}. \end{aligned} \quad (8.0.11)$$

Expanding out the differential operator L , we see we can write

$$\langle I_R^{a,\phi}, f \rangle = \sum_{|\alpha| \leq 1} \langle I_R^{a_\alpha, \phi}, \partial^\alpha f \rangle, \quad (8.0.12)$$

where a_α is a symbol of order at most $\mu - 1$. Iterating this argument, we find that for any $N > 0$, we can write

$$\langle I_R^{a,\theta}, f \rangle = \sum_{|\alpha| \leq N} \langle I_R^{a_\alpha, \phi}, \partial^\alpha f \rangle, \quad (8.0.13)$$

where a_α is a symbol of order at most $\mu - N$. If $N > \mu + p$, then we can apply the

dominated convergence theorem to each term on the right hand side, from which the result follows. $\square$

Appendix 9

Pseudo-Differential Operators

In this appendix we briefly introduce the theory of pseudo-differential operators relevant to this thesis. Recall that a *pseudo-differential operator* on $\mathbb{R}^d$ of order $s \in \mathbb{R}$ is a bounded Schwartz operator T whose kernel is an oscillatory integral distribution of the form

$$K_T(x, y) = \int_{\mathbb{R}^d \times \mathbb{R}^d} a(x, \xi) e^{2\pi i \xi \cdot (x-y)} d\xi, \quad (9.0.1)$$

where $a : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{C}$ is a fixed symbol of order s . The operator T is often written $a(x, D)$, because if a is a polynomial in ξ , i.e.

$$a(x, \xi) = \sum_{\alpha} c_{\alpha}(x) \xi^{\alpha}, \quad (9.0.2)$$

then the pseudo-differential operator $a(x, D)$ is really the differential operator $\sum c_{\alpha}(x) D^{\alpha}$, where $D_j = (2\pi i)^{-1} \partial_j$ is a self-adjoint normalization of the partial derivative operator.

An important property of pseudo-differential operators is that they are *pseudo-local*, in the sense that the Schwartz kernel K_T is a smooth function away from the diagonal $\Delta_{\mathbb{R}^d} = \{(x, x) : x \in \mathbb{R}^d\}$, and K_T and all of its derivatives rapidly decay away from the

diagonal*, i.e. if we write $k_T(x, z) = K_T(x, x + z)$, then

$$|\partial_x^\alpha \partial_z^\beta k_T(x, z)| \lesssim_{\alpha, \beta, N} |z|^{-d-s-|\beta|-N} \quad \text{for all } N > 0. \quad (9.0.3)$$

Using duality, one can extend the definition of the operator T to the space of tempered distributions, and the estimates in (9.0.3) imply that if f is a distribution smooth in a neighborhood of a given point, then Tf is smooth in a neighborhood of this point.

A symbol a of order s is *classical* if it has an expansion of the form

$$a \sim \sum_{k=0}^{\infty} a_{s-k}, \quad (9.0.4)$$

where $a_{s-k} : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{C}$ is a smooth function which is homogeneous of order $s - k$ in the second variable. A pseudo-differential operator $T = a(x, D)$ is then *classical* if the symbol a is classical. We then call a_s the *principal symbol* of the operator T . A classical pseudo-differential operator is *elliptic* if its principal symbol is non-vanishing.

All this generalizes to the setting of compact manifolds by working in coordinates. Let X be a compact manifold, and consider a cover of X by open sets $\{U_\alpha\}$, where the closure of U_α is contained in an open set V_α which is diffeomorphic to a pre-compact subset W_α of $\mathbb{R}^d$ by a diffeomorphism $F_\alpha : V_\alpha \rightarrow W_\alpha$. Consider a partition of unity $\{\psi_\alpha\}$ of X , subordinate to the cover $\{U_\alpha\}$, and also consider a smooth function ϕ_α on $\mathbb{R}^d$, equal to one on $F_\alpha(U_\alpha)$ but vanishing outside of W_α . A Schwartz operator T on a compact d -dimensional manifold X is a *pseudo-differential operator* of order $s > 0$ if its Schwartz kernel K_T , defined with respect to any fixed smooth volume density on X , agrees with a smooth function away from the diagonal $\Delta_X = \{(x, x) : x \in X\}$, and if for each α , the

*See e.g. Proposition 1 of Chapter 5, Section 4 of [47] for a proof of this rapid decay.

operators T_α on $\mathbb{R}^d$ given by

$$(T_\alpha f)(x) = \phi_\alpha(x) T\{\psi_\alpha \cdot (f \circ F_\alpha)\}(F_\alpha^{-1}(x)) \quad (9.0.5)$$

are pseudo-differential operators on $\mathbb{R}^d$. The operator T is classical if each of the operators T_α is classical, and if T_α has principal symbol $p_\alpha(x, \xi)$ we can then define the principal symbol of T to be the function $p : T^*X - \{0\} \rightarrow \mathbb{C}$ defined by

$$p(x, \xi) = \sum \psi_\alpha(x) p_\alpha(x, DF_\alpha(x)^{-T} \xi). \quad (9.0.6)$$

One verifies using the change-of-variable formulas for pseudo-differential operators* that if T is a pseudo-differential operator for one choice of cover $\{U_\alpha\}$, then it is a pseudo-differential operator for *all* choices of covers, and that for a classical pseudo-differential operator the function p is invariant of this choice when it is interpreted as defined on the cotangent bundle of X . A classical pseudo-differential operator T on a compact manifold is then *elliptic* if its principal symbol is non-vanishing.

*See Theorem 18.1.17 of [21].

Bibliography

- [1] Sriram Alladi. “A Multiplier Theorem for Ultraspherical Polynomials”. PhD thesis. University of Wisconsin-Madison, 2015.
- [2] Vladimir I. Arnol’d. *Mathematical Methods of Classical Mechanics*. Second. Vol. 60. Graduate Texts in Mathematics. Translated from the Russian by K. Vogtmann and A. Weinstein. Springer-Verlag, New York, 1989.
- [3] D. Bao, Chern S.-S., and Shen Z. *An Introduction to Riemann-Finsler Geometry*. Vol. 200. Graduate Texts in Mathematics. Springer-Verlag, New York, 2000.
- [4] Arthur L. Besse. *Manifolds All of Whose Geodesics Are Closed*. Vol. 93. Ergebnisse der Mathematik und ihrer Grenzgebiete. With appendices by D. B. A. Epstein, J.-P. Bourguignon, L. Bérard-Bergery, M. Berger and J. L. Kazdan. Springer-Verlag, Berlin - New York, 1978.
- [5] Aline Bonami and Jean-Louis Clerc. “Sommes de Cesàro et multiplicateurs des développements en harmoniques sphériques”. In: *Trans. Amer. Math. Soc.* 183 (1973), pp. 223–263.
- [6] F. M. Christ and C. D. Sogge. “The weak type L^1 convergence of eigenfunction expansions for pseudodifferential operators”. In: *Invent. Math.* 94.2 (1988), pp. 421–453.
- [7] Laura Cladek. “Radial Fourier Multipliers in $\mathbb{R}^3$ and $\mathbb{R}^4$ ”. In: *Anal. PDE* 11.2 (2018), pp. 467–498.
- [8] Song Dai, Liuwei Gong, and Ruixiang Zhang. “Oscillatory Integral Operators on Manifolds and Related Kakeya and Nikodym problems”. In: *Camb. J. Math.* 12.4 (2024), pp. 937–1015.
- [9] Johannes J. Duistermaat. “On the Morse index in variational calculus”. In: *Advances in Math.* 21.2 (1976), pp. 173–195.
- [10] Gerald Folland. *A Course in Abstract Harmonic Analysis*. Studies in Advanced Mathematics. CRC Press, Boca Raton, FL, 1995.
- [11] Gustavo Garrigós and Andreas Seeger. “Characterizations of Hankel Multipliers”. In: *Math. Ann.* 342.1 (2008), pp. 31–68.
- [12] Loukas Grafakos. *Classical Fourier Analysis*. Second. Vol. 249. Graduate Texts in Mathematics. Springer, New York, 2008.

- [13] Loukas Grafakos. *Modern Fourier Analysis*. Vol. 250. Graduate Texts in Mathematics. Springer, New York, 2009.
- [14] Shaoming Guo, Hong Wang, and Ruixiang Zhang. “A Dichotomy for Hörmander-Type Oscillatory Integral Operators”. In: *Invent. Math.* 238.2 (2024), pp. 503–584.
- [15] Yaryong Heo, Fedor Nazarov, and Andreas Seeger. “On Radial and Conical Fourier Multipliers”. In: *J. Geom. Anal.* 21.1 (2011), pp. 96–117.
- [16] Yaryong Heo, Fedor Nazarov, and Andreas Seeger. “Radial Fourier Multipliers in High Dimensions”. In: *Acta Math.* 206.1 (2011), pp. 55–92.
- [17] Heinz Hopf. *Differential Geometry In The Large*. Vol. 1000. Lectures Notes in Mathematics. Springer-Verlag, Berlin, 1983.
- [18] Lars Hörmander. “Estimates For Translation Invariant Operators In L^p Spaces”. In: *Acta Math.* 104.1-2 (1960), pp. 93–140.
- [19] Lars Hörmander. “On the Riesz means of spectral functions and eigenfunction expansions for elliptic differential operators”. In: *Some Recent Advances in the Basic Sciences, Vol. 2 (Proc. Annual Sci. Conf., Belfer Grad. School Sci., Yeshiva Univ., New York, 1965-1966)*. Yeshiva University, Belfer Graduate School of Science, New York, 1969, pp. 155–202.
- [20] Lars Hörmander. *The Analysis of Linear Partial Differential Operators. I: Distribution Theory and Fourier Analysis*. Vol. 256. Grundlehren der mathematischen Wissenschaften. Springer-Verlag, Berlin, 1990.
- [21] Lars Hörmander. *The Analysis of Linear Partial Differential Operators. III: Pseudo-Differential Operators*. Vol. 274. Grundlehren der mathematischen Wissenschaften. Springer-Verlag, Berlin, 1985.
- [22] Lars Hörmander. *The Analysis of Linear Partial Differential Operators. IV: Fourier Integral Operators*. Vol. 275. Grundlehren der mathematischen Wissenschaften. Springer-Verlag, Berlin, 1985.
- [23] Carlos E. Kenig, Robert J. Stanton, and Peter A. Tomas. “Divergence of Eigenfunction Expansions”. In: *J. Funct. Anal.* 46.1 (1982), pp. 28–44.
- [24] Jongchon Kim. “Endpoint Bounds for a Class of Spectral Multipliers on Compact Manifolds”. In: *Indiana Univ. Math. J.* 67.2 (2018), pp. 937–969.
- [25] Jongchon Kim. “Endpoint Bounds for Quasiradial Fourier Multipliers”. In: *Ann. Mat. Pura Appl. (4)* 196 (2017), pp. 773–789.
- [26] Sanghyuk Lee and Andreas Seeger. “Lebesgue Space Estimates for a Class of Fourier Integral Operators Associated with Wave Propagation”. In: *Math. Nachr.* 286.7 (2013), pp. 745–755.
- [27] Sanghyuk Lee and Andreas Seeger. “On Radial Fourier Multipliers and Almost Everywhere Convergence”. In: *J. Lond. Math. Soc.* 91.2 (2015), pp. 105–126.

- [28] Karel de Leeuw. “On L_p Multipliers”. In: *Ann. of Math. (2)* 81.2 (1965), pp. 364–379.
- [29] Makoto Matsumoto. “A Slope of a Mountain is a Finsler Surface With Respect to a Time Measure”. In: *J. Math. Kyoto Univ.* 29 (1989), pp. 17–25.
- [30] William P. Minicozzi and Christopher D. Sogge. “Negative Results for Nikodym Maximal Functions and Related Oscillatory Integrals in Curved Spaces”. In: *Math. Res. Lett.* 4.2 (2000), pp. 221–237.
- [31] Boris S. Mitjagin. “Divergenz von Spektralentwicklungen in L_p -Räumen”. In: *Internat. Ser. Numer. Math.* 25 (1974), pp. 521–530.
- [32] Jaak Peetre. “On Spaces of Triebel-Lizorkin Type”. In: *Ark. Math.* 13 (1975), pp. 123–130.
- [33] Jaak Peetre. “Une caractérisation abstraite des opérateurs différentiels”. In: *Math. Scand.* 7 (1959), pp. 211–218.
- [34] Walter Rudin. *Functional analysis*. McGraw-Hill Series in Higher Mathematics. McGraw-Hill Book Co., New York-Düsseldorf-Johannesburg, 1973.
- [35] A. Seeger and C. D. Sogge. “Bounds for eigenfunctions of differential operators”. In: *Indiana Univ. Math. J.* 38.3 (1989), pp. 669–682.
- [36] Andreas Seeger. “Endpoint Estimates for Multiplier Transformations on Compact Manifolds”. In: *Indiana Univ. Math. J.* 40.2 (1991), pp. 471–533.
- [37] Andreas Seeger. “Remarks on Singular Convolution Operators”. In: *Studia Math.* 97.2 (1990), pp. 91–114.
- [38] Andreas Seeger. “Some Inequalities for Singular Convolution Operators in L^p -Spaces”. In: *Trans. Amer. Math. Soc.* 308.1 (1988), pp. 259–272.
- [39] Robert Thomas Seeley. “Complex Powers Of An Elliptic Operator”. In: *Proc. Symp. Pure Math.* 10 (1967), pp. 288–307.
- [40] Mark R. Sepanski. *Compact Lie Groups*. Vol. 235. Graduate Texts in Mathematics. Springer, New York, 2007.
- [41] Christopher D. Sogge. “Concerning the L^p norm of spectral clusters for second-order elliptic operators on compact manifolds”. In: *J. Funct. Anal.* 77.1 (1988), pp. 123–138.
- [42] Christopher D. Sogge. *Fourier Integrals in Classical Analysis*. Second. Vol. 210. Cambridge Tracts in Mathematics. Cambridge University Press, Cambridge, 2017.
- [43] Christopher D. Sogge. *Hangzhou Lectures on Eigenfunctions of the Laplacian*. Vol. 188. Annals of Mathematics Studies. Princeton University Press, Princeton, NJ, 2014.
- [44] Christopher D. Sogge. “On the Convergence of Riesz Means on Compact Manifolds”. In: *Ann. of Math. (2)* 126.3 (1987), pp. 439–447.

- [45] Christopher D. Sogge. “Oscillatory integrals and spherical harmonics”. In: *Duke Math. J.* 53.1 (1986), pp. 43–65.
- [46] Michael Spivak. *A comprehensive introduction to differential geometry. Vol. IV.* Second. Publish or Perish, Inc., Wilmington, DE, 1979.
- [47] Elias M. Stein. *Harmonic analysis: real-variable methods, orthogonality, and oscillatory integrals.* Vol. 43. Princeton Mathematical Series. Princeton University Press, Princeton, NJ, 1993.
- [48] Terence Tao. *Nonlinear Dispersive Equations.* Vol. 106. CBMS Regional Conference Series in Mathematics. Conference Board of the Mathematical Sciences, Washington, DC; by the American Mathematical Society, Providence, RI, 2006.
- [49] Terence Tao. “The Bochner-Riesz Conjecture Implies the Restriction Conjecture”. In: *Duke Math. J.* 96.1 (1999), pp. 363–375.
- [50] Michael E. Taylor. *Partial Differential Equations.* Vol. 23. Texts in Applied Mathematics. Springer-Verlag, New York, 1996.
- [51] François Trèves. *Introduction to Pseudodifferential and Fourier Integral Operators. Vol. 2.* University Series in Mathematics. Plenum Press, New York - London, 1980.